

Reductive groups: splittings, subgroups, quotient groups

SGA 3, Exposé XXII

English draft

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About this draft

This is an unofficial English translation of Exposé XXII, “Reductive groups: splittings, subgroups, quotient groups” from *Schémas en groupes* (SGA 3). The source is the recomposition edited by P. Gille and P. Polo, PDF `Exp22-13oct24.pdf` (68 pages), from <https://webusers.imj-prg.fr/~patrick.polo/SGA3/>. The seminar was directed by M. Demazure and A. Grothendieck (IHÉS, 1962–1964).

Statement numbers follow that PDF. A clear typographical slip is corrected in the English and marked with a `% typo:` comment; such slips are listed in the README. A slip that would change the mathematics is left as printed and marked `% typo?:`. Scholarly proofreading remains outstanding.

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Exposé XXII. Reductive groups: splittings, subgroups, quotient groups

by M. Demazure

⁽⁰⁾This exposé comprises two parts. The first (1 to 5.5) collects the technical results necessary for the proof of the uniqueness and existence theorems. The second (5.6 to the end) will not be used in this proof; the end of no. 5 will be used in particular in exposé XXVI devoted to parabolic subgroups; no. 6 establishes in the framework of schemes the classical results on the derived group of a reductive group.

1. Roots and coroots. Split groups and root data

Theorem 1.1. *Let S be a scheme, G a reductive S -group, T a maximal torus of G , α a root of G with respect to T .*

(i) *There exists a unique morphism of groups with operator group T*

$$\exp_\alpha : W(\mathfrak{g}^\alpha) \longrightarrow G$$

which induces on the Lie algebras the canonical morphism $\mathfrak{g}^\alpha \longrightarrow \mathfrak{g}$. This morphism is a closed immersion. The corresponding morphism

$$T \cdot_\alpha W(\mathfrak{g}^\alpha) \longrightarrow G$$

is also a closed immersion.

If $p_\alpha : \mathbf{G}_{a,S} \longrightarrow G$ is a monomorphism normalized by T with multiplier α , there exists a unique $X_\alpha \in \Gamma(S, \mathfrak{g}^\alpha)^{\times(1)}$ such that $p_\alpha(x) = \exp_\alpha(xX_\alpha)$; one has $\text{Lie}(p_\alpha)(1) = X_\alpha$ and the two preceding formulas establish a bijective correspondence between $\Gamma(S, \mathfrak{g}^\alpha)^\times$ and the set of monomorphisms $\mathbf{G}_{a,S} \longrightarrow G$ normalized by T with multiplier α .

(ii) *There exists a unique duality (denoted $(X, Y) \longmapsto XY$)*

$$\mathfrak{g}^\alpha \otimes_{\mathcal{O}_S} \mathfrak{g}^{-\alpha} \xrightarrow{\sim} \mathcal{O}_S,$$

and a unique morphism of groups

$$\alpha^* : \mathbf{G}_{m,S} \longrightarrow T,$$

such that one has formula (F) of Exp. XX 2.1. One has

$$\alpha \circ \alpha^* = 2, \quad (-\alpha)^* = -\alpha^*,$$

and α^ is given by the formula of Exp. XX 2.7.*

Indeed, a morphism normalized by T with multiplier α necessarily factors through the closed subgroup $Z_\alpha = \text{Centr}_G(T_\alpha)$ of G (cf. Exp. XIX 3.9). Now (Z_α, T, α) is an elementary S -system (Exp. XX 1.4), and one is reduced to the results of exposé XX (1.5, 2.1 and 5.9).

Remark 1.2. Part (i) of theorem 1.1 remains valid if one assumes only that α is a character of T , nontrivial on each fiber. Indeed, one then has a decomposition $S = S' \amalg S''$, such that

⁽⁰⁾**N.D.E.** Version of 13/10/2024

⁽¹⁾**N.D.E.** The set $\Gamma(S, \mathfrak{g}^\alpha)^\times$ is defined in XIX 4.4.1.

$\alpha|_{S'}$ is a root of $G_{S'}$ with respect to $T_{S'}$ and $\mathfrak{g}^\alpha|_{S''} = 0$. If $S = S'$, one is reduced to 1.1; if $S = S''$ the result is trivial; the general case follows at once.

Notations 1.3. As in exposé XX, one denotes by U_α the image of $W(\mathfrak{g}^\alpha)$; it is a closed subgroup of G , canonically endowed with a vector structure. One will say that it is the *vector group associated with the root α* . One says that α^* is the *coroot associated with α* . Sections $X_\alpha \in \Gamma(S, \mathfrak{g}^\alpha)$ and $X_{-\alpha} \in \Gamma(S, \mathfrak{g}^{-\alpha})$ are said to be *paired* if $X_\alpha X_{-\alpha} = 1$. Then $X_\alpha \in \Gamma(S, \mathfrak{g}^\alpha)^\times$ and likewise for $X_{-\alpha}$. The corresponding morphisms p_α and $p_{-\alpha}$ are contragredient to one another in the sense of XX, 1.5⁽²⁾ and one has

$$p_\alpha(x) p_{-\alpha}(y) = p_{-\alpha} \left(\frac{y}{1+xy} \right) \alpha^*(1+xy) p_\alpha \left(\frac{x}{1+xy} \right).$$

Proposition 1.4. *Under the conditions of 1.1, let $w \in \text{Norm}_G(T)(S)$. Then $\beta = \alpha \circ \text{int}(w)^{-1} : T \rightarrow \mathbf{G}_{m,S}$ is a root of G with respect to T , $\beta^* = \text{int}(w) \circ \alpha^*$ is the corresponding coroot, and the following diagram is commutative:*

$$\begin{array}{ccc} W(\mathfrak{g}^\alpha) & \xrightarrow{\exp_\alpha} & G \\ \text{Ad}(w) \downarrow & & \downarrow \text{int}(w) \\ W(\mathfrak{g}^\beta) & \xrightarrow{\exp_\beta} & G. \end{array}$$

Trivial: transport of structure.

Definitions 1.5. (a) Under the conditions of 1.1, one denotes by s_α the automorphism of T defined by

$$s_\alpha(t) = t \cdot \alpha^*(\alpha(t))^{-1}.$$

One denotes by $(\ , \)$ the canonical pairing:

$$\begin{aligned} & \underline{\text{Hom}}_{S\text{-gr.}}(\mathbf{G}_{m,S}, T) \times \underline{\text{Hom}}_{S\text{-gr.}}(T, \mathbf{G}_{m,S}) \\ & \longrightarrow \underline{\text{Hom}}_{S\text{-gr.}}(\mathbf{G}_{m,S}, \mathbf{G}_{m,S}) = \mathbf{Z}_S. \end{aligned}$$

Then s_α acts on $\underline{\text{Hom}}_{S\text{-gr.}}(T, \mathbf{G}_{m,S})$, resp. $\underline{\text{Hom}}_{S\text{-gr.}}(\mathbf{G}_{m,S}, T)$, by the following formulas, where χ (resp. u) denotes an arbitrary section of $\underline{\text{Hom}}_{S\text{-gr.}}(T, \mathbf{G}_{m,S})$ (resp. of $\underline{\text{Hom}}_{S\text{-gr.}}(\mathbf{G}_{m,S}, T)$):

$$s_\alpha(\chi) = \chi - (\alpha^*, \chi)\alpha, \quad s_\alpha(u) = u - (u, \alpha)\alpha^*.$$

One has $s_\alpha \circ s_\alpha = \text{id}$ and $s_{-\alpha} = s_\alpha$.

(b) If $X \in \Gamma(S, \mathfrak{g}^\alpha)^\times$, then the inner automorphism $w_\alpha(X)$ of T defined by

$$w_\alpha(X) = \exp_\alpha(X) \exp_{-\alpha}(-X^{-1}) \exp_\alpha(X)$$

(cf. Exp. XX 3.1) coincides with s_α (*loc. cit.*). One then concludes from 1.4:

Corollary 1.6. *Let S be a scheme, G a reductive S -group, T a maximal torus of G , α and β two roots of G with respect to T . Then*

$$s_\alpha(\beta) = \beta - (\alpha^*, \beta)\alpha$$

is a root of G with respect to T , the corresponding coroot being

$$s_\alpha(\beta)^* = s_\alpha(\beta^*) = \beta^* - (\beta^*, \alpha)\alpha^*.$$

Corollary 1.7. *Under the preceding conditions, $\alpha^* = \beta^*$ implies $\alpha = \beta$.*

Indeed, if $\alpha^* = \beta^*$, one has cf. XXI.1.4

$$s_\beta(\alpha) = \alpha - 2\beta, \quad s_\alpha(\beta) = \beta - 2\alpha,$$

⁽²⁾**N.D.E.** That is, if one denotes by $\langle \ , \ \rangle$ the duality between $\mathfrak{g}^\alpha$ and $\mathfrak{g}^{-\alpha}$ and identifies $\mathfrak{g}^\alpha$ with U_α via $p_\alpha(x) = \exp(xX_\alpha)$, and likewise for $-\alpha$, then $\langle p_\alpha(x), p_{-\alpha}(y) \rangle = xy$.

and one deduces at once

$$(s_\beta s_\alpha)^n(\alpha) = \alpha + 2n(\beta - \alpha).$$

If $\beta \neq \alpha$, there exists an $s \in S$ such that $\alpha_s \neq \beta_s$. But then the preceding formula shows that there exist infinitely many distinct roots of G_s with respect to T_s , which is impossible.

Definitions 1.8.0. ⁽³⁾If $u : \mathbf{G}_{m,S} \rightarrow T$ is a morphism of groups, one will say that u is a *coroot* of G with respect to T if there exists a root α of G with respect to T such that $\alpha^* = u$. Consider the functor $\mathcal{R}^*$ of coroots of G with respect to T defined as follows:

$$\mathcal{R}^*(S') = \text{set of coroots of } G_{S'} \text{ with respect to } T_{S'}.$$

If $\mathcal{R}$ is the functor of roots of G with respect to T (Exp. XIX 3.8.), one has a canonical morphism $\mathcal{R} \rightarrow \mathcal{R}^*$. By 1.7 and Exp. XIX 3.8, one has:

Corollary 1.8. *The canonical morphism $\mathcal{R} \rightarrow \mathcal{R}^*$ is an isomorphism. In particular, $\mathcal{R}^*$ is representable by a twisted constant finite S -scheme which is an open and closed subscheme of $\underline{\text{Hom}}_{S\text{-gr.}}(\mathbf{G}_{m,S}, T)$.*

This leads one to give the following definition:

Definition 1.9. Let S be a scheme, T an S -torus. One calls a *twisted root datum* in T the datum:

- (i) of a finite subscheme $\mathcal{R}$ of $\underline{\text{Hom}}_{S\text{-gr.}}(T, \mathbf{G}_{m,S})$,
- (ii) of a finite subscheme $\mathcal{R}^*$ of $\underline{\text{Hom}}_{S\text{-gr.}}(\mathbf{G}_{m,S}, T)$,
- (iii) of an isomorphism $\mathcal{R} \xrightarrow{\sim} \mathcal{R}^*$ denoted $\alpha \mapsto \alpha^*$,

satisfying the following conditions:

- (DR 1) For every $S' \rightarrow S$ and every $\alpha \in \mathcal{R}(S')$, one has $\alpha \circ \alpha^* = 2$.
- (DR 2) For every $S' \rightarrow S$ and all $\alpha, \beta \in \mathcal{R}(S')$, one has

$$\alpha - (\beta^*, \alpha)\beta \in \mathcal{R}(S'), \quad \alpha^* - (\alpha^*, \beta)\beta^* \in \mathcal{R}^*(S').$$

Moreover, if $\alpha \in \mathcal{R}(S')$ ($S' \neq \emptyset$) implies $2\alpha \notin \mathcal{R}(S')$, one says that the root datum is *reduced*.

Proposition 1.10. *Let S be a scheme, G a reductive S -group, T a maximal torus of G , $\mathcal{R}$ (resp. $\mathcal{R}^*$) the scheme of roots (resp. of coroots) of G with respect to T . Then $(\mathcal{R}, \mathcal{R}^*)$ is a reduced twisted root datum in T .*

The only fact which remains to be verified is that this twisted root datum is reduced. This is what was done in Exp. XIX 3.10.

1.11. — Let $T = D_S(M)$ be a trivialized torus. If one denotes by M^* the dual abelian group of M , one has canonical isomorphisms (cf. Exp. VIII 1.5):

$$\begin{aligned} \underline{\text{Hom}}_{S\text{-gr.}}(T, \mathbf{G}_{m,S}) &\xrightarrow{\sim} M_S, \\ \underline{\text{Hom}}_{S\text{-gr.}}(\mathbf{G}_{m,S}, T) &\xrightarrow{\sim} M_S^*, \end{aligned}$$

hence isomorphisms of groups:

$$\begin{aligned} \text{Hom}_{S\text{-gr.}}(T, \mathbf{G}_{m,S}) &\longrightarrow \text{Hom}_{\text{loc.const.}}(S, M), \\ \text{Hom}_{S\text{-gr.}}(\mathbf{G}_{m,S}, T) &\longrightarrow \text{Hom}_{\text{loc.const.}}(S, M^*). \end{aligned}$$

A character of T (resp. a morphism of groups $\mathbf{G}_{m,S} \rightarrow T$) will be called *constant* (relative to the given trivialization) if the preceding isomorphism transforms it into a constant map from S to M (resp. M^*).

1.12. — With the same notations, let (M, M^*, R, R^*) be a root datum (Exp. XXI). Then (R_S, R_S^*) is a twisted root datum in T . Conversely, if $(\mathcal{R}, \mathcal{R}^*)$ is a twisted root datum in a

⁽³⁾**N.D.E.** The numbering 1.8.0 has been added to bring out these definitions.

torus T , one will call a *splitting* of this root datum the datum of an ordinary root datum (M, M^*, R, R^*) and of an isomorphism $T \simeq D_S(M)$ which transforms $(\mathcal{R}, \mathcal{R}^*)$ into (R_S, R_S^*) .

Definition 1.13. Let S be a scheme, G a reductive S -group, T a maximal torus of G . One calls a *splitting of G relative to T* the datum

- (i) of an abelian group M and an isomorphism $T \simeq D_S(M)$,
- (ii) of a system of roots R of G with respect to T (Exp. XIX 3.6),

satisfying the following two conditions:

(D₁) S is nonempty and the roots $\alpha \in R$ (resp. the corresponding coroots) are identified with constant functions from S to M (resp. M^*).

(D₂) The $\mathfrak{g}^\alpha$ ($\alpha \in R$) are free $\mathcal{O}_S$ -modules.

One says that G is *splittable relative to T* if there exists a splitting of G relative to T . One calls a *splitting of G* the datum of a maximal torus T of G and of a splitting of G with respect to T . One says that G is *splittable* if there exists a splitting of G . One calls a *split S -group* a reductive S -group endowed with a splitting; one will denote it by a symbol of the form (G, T, M, R) , or simply G if no confusion is possible.

Condition (D 1) implies that R (resp. R^*) is canonically identified with a subset of M (resp. M^*).

Proposition 1.14. Let S be a (nonempty) scheme, (G, T, M, R) a split S -group; then

$$\mathcal{R}(G, T, M, R) = (M, M^*, R, R^*)$$

is a reduced root datum (Exp. XXI 1.1 and 2.1.3); it is a splitting of the twisted root datum of 1.10.

This is a trivial consequence of 1.10 and of Exp. XIX 3.7.

One will sometimes write for simplicity $\mathcal{R}(G, T, M, R) = \mathcal{R}(G)$. One will systematically use the notations V , $\mathcal{V}(R)$, W , ... of Exp. XXI.

Remark 1.15. a) If S is connected and nonempty (resp. if $\text{Pic}(S) = 0$), condition (D 1) (resp. (D 2)) is automatically satisfied.

b) If (G, T, M, R) is a split S -group, then for every $S' \rightarrow S$, $S' \neq \emptyset$, $(G_{S'}, T_{S'}, M, R)$ is a split S' -group and $\mathcal{R}(G, T, M, R) = \mathcal{R}(G_{S'}, T_{S'}, M, R)$.

1.16. — Let $T = D_S(M)$ be a trivialized torus. The Lie algebra $\mathfrak{t}$ of T is canonically identified (Exp. II 5.1.1) with

$$\mathfrak{t} \simeq M^* \otimes \mathcal{O}_S.$$

For every morphism of groups $u : T \rightarrow \mathbf{G}_{m,S}$, $\underline{\text{Lie}}(u)$ is a linear form

$$\underline{\text{Lie}}(u) : \mathfrak{t} \rightarrow \mathcal{O}_S = \underline{\text{Lie}}(\mathbf{G}_{m,S}/S).$$

In particular, if u is defined by an element $\alpha \in M$, then $\underline{\text{Lie}}(u)$ is the linear form $\bar{\alpha}$ on $M^* \otimes \mathcal{O}_S$ defined by α :

$$\bar{\alpha}(m \otimes x) = (m, \alpha)x.$$

Symmetrically, for every morphism of groups $h : \mathbf{G}_{m,S} \rightarrow T$, $\underline{\text{Lie}}(h)$ is an $\mathcal{O}_S$ -morphism $\mathcal{O}_S = \underline{\text{Lie}}(\mathbf{G}_{m,S}/S) \rightarrow \mathfrak{t}$, canonically defined by the section

$$H = \underline{\text{Lie}}(h)(1) \in \Gamma(S, \mathfrak{t}).$$

In particular, if h is defined by an element $m \in M^*$, one has

$$H = \underline{\text{Lie}}(h)(1) = m \otimes 1.$$

Comparing the two definitions, one finds in particular

$$\bar{\alpha}(H) = (h, \alpha) \cdot 1 \in \Gamma(S, \mathcal{O}_S).$$

1.17. — These definitions apply in particular to the case where T is the maximal torus of a split group. Every root $\alpha \in R$ defines an *infinitesimal root* $\bar{\alpha} \in \text{Hom}_{\mathcal{O}_S}(\mathfrak{t}, \mathcal{O}_S)$ with

$$\bar{\alpha}(m \otimes x) = (m, \alpha)x.$$

Each coroot $\alpha \in R$ defines an *infinitesimal coroot*

$$H_\alpha \in \Gamma(S, \mathfrak{t}), \quad H_\alpha = \alpha^* \otimes 1.$$

For $\alpha, \beta \in R$, one has the relation

$$\bar{\alpha}(H_\beta) = (\beta^*, \alpha) \cdot 1,$$

and in particular

$$\bar{\alpha}(H_\alpha) = 2.$$

In particular, if 2 is invertible on S , then $\bar{\alpha}$ and H_α are nonzero on each fiber.

2. Existence of a splitting. Type of a reductive group

Proposition 2.1. *Let S be a scheme, G a reductive S -group, T a maximal torus of G . Assume T split. Then G is locally splittable with respect to T : for every $s_0 \in S$, there exists an open neighborhood U of s_0 such that the U -group G_U is splittable relative to T_U .*

Indeed, write $T \simeq D_S(M)$ and

$$\mathfrak{g} = \coprod_{m \in M} \mathfrak{g}^m.$$

Let $R = \{m \in M \mid m \neq 0, \mathfrak{g}^m(s_0) \neq 0\}$. Restricting S and replacing it by an open neighborhood of s_0 , one may assume the $\mathfrak{g}^\alpha$, $\alpha \in R$, free, and the $\mathfrak{g}^m$, $m \neq 0$, $m \notin R$, zero. One then has

$$\mathfrak{g} = \mathfrak{t} \oplus \coprod_{\alpha \in R} \mathfrak{g}^\alpha,$$

the $\mathfrak{g}^\alpha$ being free of rank 1. It follows that R is a system of roots of G with respect to T (Exp. XIX 3.6). The coroots α^* corresponding to the $\alpha \in R$ are then identified with locally constant functions on S with values in M^* . Restricting S further, one may assume them constant and one is done.

Note that the proof gives at once:

Proposition 2.2. *Let S be a nonempty connected scheme such that $\text{Pic}(S) = 0$, for example $\text{Spec}(\mathbf{Z})$ or a local scheme (in particular the spectrum of a field). If G is a reductive S -group possessing a split maximal torus T , then G is splittable relative to T .*

One deduces at once from 2.1 and the fact that a reductive group possesses maximal tori locally for the étale topology (Exp. XIX 2.5):

Corollary 2.3. *Let S be a scheme, G a reductive S -group (resp. and T a maximal torus of G). Then G is locally splittable (resp. locally splittable relative to T) for the étale topology on S .*

Corollary 2.4. *Let k be a field, G a reductive k -group. There exists a finite separable extension K/k such that G_K is splittable.*

Remark 2.5. Using 2.1 and the remark Exp. XIX 2.9, one proves at once the following result: let $G = (G, T, M, R)$ be a split S -group; there exists a covering of S by open subsets U_i such that each split group G_{U_i} comes by base change from a split group over a noetherian ring (and in fact a $\mathbf{Z}$ -algebra of finite type). We will moreover prove that every split group over S already comes from a split $\mathbf{Z}$ -group (Exp. XXV).

2.6. — Let k be an algebraically closed field and G a reductive k -group. One knows (2.4 for example) that splittings of G exist. Let (G, T, M, R) and (G, T', M', R') be two splittings of G ; the root data $\mathcal{R}(G, T, M, R)$ and $\mathcal{R}(G, T', M', R')$ are then isomorphic.

Indeed, one sees first that one may reduce to the case where $T = T'$ (for there exists $g \in G(k)$ such that $T' = \text{int}(g)T$, and one easily verifies that if one transports a splitting by an automorphism of G , one obtains a root datum isomorphic to the initial datum); but since $S = \text{Spec}(k)$ is connected, the isomorphism $D_k(M) \xrightarrow{\sim} T \xrightarrow{\sim} D_k(M')$ comes from a unique isomorphism $M \simeq M'$; for the same reason, there exists at most one system of roots of G with respect to T .

Definition 2.6.1. ⁽⁴⁾If G is a reductive k -group (k an algebraically closed field), one will call the *type of G* the isomorphism class of the root datum defined by any splitting of G ; if G is a torus, of type M in the sense of Exp. IX 1.4, then the type of G as a reductive group is given by the trivial root datum $(M, M^*, \emptyset, \emptyset)$.

By 1.15 b)⁽⁵⁾, the type is invariant under (algebraically closed) extension of the base field.

Definition 2.7. If G is a reductive S -group and if $s \in S$, one calls the *type of G at s* the type of the reductive $\bar{s}$ -group $G_{\bar{s}}$.

For every $S' \rightarrow S$ and every $s' \in S'$ projecting to $s \in S$, the type of $G_{S'}$ at s' is equal to the type of G at s .

If G is splittable, and if (G, T, M, R) is a splitting of G , then the type of G at s is the isomorphism class of $\mathcal{R}(G, T, M, R)$ by 1.15 b)⁽⁵⁾. One then obtains at once from 2.3 the

Proposition 2.8. *Let G be a reductive S -group ($S \neq \emptyset$). The function*

$$s \longmapsto \text{type of } G \text{ at } s$$

is locally constant on S . In particular, there exists a partition of S into nonempty open subschemes such that on each of them G has constant type. More precisely, let E be the set of types of the fibers of G ; for every $t \in E$, let S_t be the set of points $s \in S$ where G has type t ; then $(S_t)_{t \in E}$ is a partition of S and each S_t is open and closed (and nonempty).

3. The Weyl group

3.1. — Let S be a scheme, G a reductive S -group, T a maximal torus of G . Then

$$W_G(T) = \underline{\text{Norm}}_G(T)/T$$

is a finite étale S -group (Exp. XIX 2.5). The morphism $n \mapsto \text{int}(n)$ induces on passing to the quotient a canonical monomorphism (which is moreover an open immersion):

$$W_G(T) \longrightarrow \underline{\text{Aut}}_{S\text{-gr.}}(T).$$

3.2. — Assume now that G is splittable relative to T . Choose a splitting, say (G, T, M, R) . One then has a canonical isomorphism (Exp. VIII 1.5)

$$\underline{\text{Aut}}_{S\text{-gr.}}(T) \simeq (\text{Aut}_{\text{gr.}}(M))_S.$$

In particular, if W is the Weyl group of the root datum $\mathcal{R}(G)$ (Exp. XXI 1.1.8), one has a monomorphism

$$W_S \longrightarrow \underline{\text{Aut}}_{S\text{-gr.}}(T).$$

3.3. — For each root $\alpha \in R$, the reflection $s_\alpha \in W$ acts on M by

$$s_\alpha(x) = x - (\alpha^*, x)\alpha,$$

⁽⁴⁾**N.D.E.** The number 2.6.1 has been added, for later references.

⁽⁵⁾**N.D.E.** The original, which referred to 1.17, has been corrected.

hence on T (by the preceding morphism), by

$$s_\alpha(t) = t \cdot \alpha^*(\alpha(t))^{-1}.$$

On the other hand, since $\mathfrak{g}^\alpha$ is assumed free, there exists an $X \in \Gamma(S, \mathfrak{g}^\alpha)^\times$. Consider then $w_\alpha(X) \in \underline{\text{Norm}}_G(T)(S)$ (Exp. XX 3.1). One has (*loc. cit.*)

$$\text{int}(w_\alpha(X))(t) = s_\alpha(t).$$

Since W is generated by the s_α , $\alpha \in R$, it follows from the preceding remarks that if one considers W and $\underline{\text{Norm}}_G(T)(S)/T(S)$ as groups of automorphisms of T , one has

$$W \subset \underline{\text{Norm}}_G(T)(S)/T(S) \subset W_G(T)(S).$$

By definition of the constant group W_S associated with W (cf. I 1.8), one therefore has a commutative diagram

$$\begin{array}{ccc} W_S & \xrightarrow{\quad} & W_G(T) \\ & \searrow & \swarrow \\ & \underline{\text{Aut}}_{S\text{-gr.}}(T) & \end{array}$$

Proposition 3.4. *Let S be a scheme, (G, T, M, R) a split S -group, W the Weyl group of the root datum $\mathcal{R}(G)$. Then the canonical monomorphism*

$$W_S \longrightarrow W_G(T) = \underline{\text{Norm}}_G(T)/T$$

is an isomorphism.

These are indeed étale groups over S ; it therefore suffices to verify that for every $s \in S$, $W_S(\bar{s}) \longrightarrow W_G(T)(\bar{s})$ is an isomorphism.⁽⁶⁾ Now this last assertion follows, for example, from *Bible*, § 11.3, th. 2.

Remark 3.5. Using 2.3, the preceding proposition gives a new proof of the fact that the Weyl group of a maximal torus of a reductive S -group G is finite over S (Exp. XIX 2.5 (ii)).⁽⁷⁾

3.6. — Under the conditions of 3.1, for every $w \in W_G(T)(S)$, one denotes by N_w ⁽⁸⁾ the fibered product of the following diagram:

$$\begin{array}{ccc} N_w & \longrightarrow & \underline{\text{Norm}}_G(T) \\ \downarrow & & \downarrow \\ S & \xrightarrow{\quad w \quad} & W_G(T). \end{array}$$

It is an open and closed subscheme of $\underline{\text{Norm}}_G(T)$, which is a principal homogeneous bundle under T on the left (resp. on the right) by the law $(t, q) \mapsto tq$ (resp. $(q, t) \mapsto qt$). If $n \in N_w(S)$, one has

$$N_{ww'} = n \cdot N_{w'}, \quad N_{w'w} = N_{w'} \cdot n.$$

3.7. — In particular, if α is a root of G with respect to T , N_{s_α} is none other than what had been denoted by $N^\times$ in Exp. XX 3.0. If $\mathfrak{g}^\alpha$ is free over S , one therefore has $N_{s_\alpha}(S) \neq \emptyset$.

By 3.4 and condition (D 2) of the splitting, one deduces the

⁽⁶⁾**N.D.E.** Indeed, since W_S and $W_G(T) = \underline{\text{Norm}}_G(T)/T$ are étale over S , the morphism $f : W_S \longrightarrow W_G(T) = \underline{\text{Norm}}_G(T)/T$ is étale (EGA IV₄, 17.3.4); if moreover each $f_{\bar{s}}$ is an isomorphism then, according to *loc. cit.*, 17.9.1, f will be a surjective open immersion, hence an isomorphism.

⁽⁷⁾**N.D.E.** Indeed, let T be a maximal torus of G . The fact that $W_G(T)$ is finite over S is local for the (fpqc) topology (EGA IV₂, 2.7.1), hence a fortiori for the étale topology. By 2.3, one may therefore assume that G is split relative to T , in which case the assertion follows from 3.4.

⁽⁸⁾**N.D.E.** Q_w has been replaced by N_w , just as in XX 3.0 Q had been replaced by $N^\times$.

Corollary 3.8. *Under the conditions of 3.4, the morphism*

$$\underline{\text{Norm}}_G(T)(S) \longrightarrow W_G(T)(S) = \text{Hom}_{\text{loc.cons.}}(S, W)$$

is surjective. In particular, for every $w \in W$, there exists an $n_w \in \underline{\text{Norm}}_G(T)(S)$ such that $\text{int}(n_w)|_T = w$.

4. Homomorphisms of split groups

4.1. The “big cell”

4.1.1. — Let (G, T, M, R) be a split reductive S -group. Choose a *system of positive roots* (Exp. XXI 3.2.1) R_+ of the root datum $\mathcal{R}(G)$. Put $R_- = -R_+$.

Choose a total order on R_+ (resp. R_-) and consider the morphism induced by the product in G

$$u : \prod_{\alpha \in R_-} U_\alpha \times_S T \times_S \prod_{\alpha \in R_+} U_\alpha \longrightarrow G.$$

It is an *open immersion*. Indeed, since both sides are flat and of finite presentation over S , it suffices to verify this on each geometric fiber (SGA 1, I 5.7 and VIII 5.4); one is therefore reduced to the case where S is the spectrum of an algebraically closed field; but, by *Bible*, § 13.4, cor. 2 to th. 3, u is radicial and dominant; since the tangent map to u at the origin is an isomorphism (definition of a system of roots), u is birational; but since G is normal, one may apply Zariski’s “Main Theorem” (EGA III₁, 4.4.9) and u is an open immersion.

Let us show that the image Ω of this open immersion is independent of the order chosen on R_+ (resp. R_-). Since this amounts to comparing open subsets of G , one is reduced to proving that they have the same geometric points, so one may again assume that S is the spectrum of an algebraically closed field. But then the assertion is none other than *Bible*, § 13, prop. 1 (c) and th. 1 (a).

One has therefore proved:

Proposition 4.1.2. *Let (G, T, M, R) be a split S -group. Let R_+ be a system of positive roots of R . There exists an open subset Ω_{R_+} of G such that for every total order on R_+ (resp. R_-), the morphism induced by the product in G*

$$\prod_{\alpha \in R_-} U_\alpha \times_S T \times_S \prod_{\alpha \in R_+} U_\alpha \longrightarrow G$$

is an open immersion with image Ω_{R_+} .

Remark 4.1.3. One may express 4.1.2 as follows: choose for every $\alpha \in R$ an isomorphism of vector groups $p_\alpha : \mathbf{G}_{a,S} \xrightarrow{\sim} U_\alpha$ (cf. 1.19); then the morphism (one puts $N = \text{Card}(R_+) = \text{Card}(R_-)$)

$$\mathbf{G}_{a,S}^N \times_S T \times_S \mathbf{G}_{a,S}^N \longrightarrow G$$

defined set-theoretically by

$$((x_\alpha)_{\alpha \in R_-}, t, (x_\alpha)_{\alpha \in R_+}) \longmapsto \prod_{\alpha \in R_-} p_\alpha(x_\alpha) \cdot t \cdot \prod_{\alpha \in R_+} p_\alpha(x_\alpha)$$

is an open immersion whose image depends only on R_+ (and not on the choice of the p_α and of the orders on R_+ and R_-).

Notation 4.1.4. One denotes $\Omega_{R_+} = \prod_{\alpha \in R_-} U_\alpha \cdot T \cdot \prod_{\alpha \in R_+} U_\alpha$.⁽⁹⁾

Proposition 4.1.5. *The scheme Ω_{R_+} is of finite presentation over S (hence retrocompact in G) and is universally schematically dense in G relative to S (cf. Exp. XVIII 1).*

⁽⁹⁾**N.D.E.** And one calls it the “big cell” corresponding to R_+ .

The first assertion is trivial. Then,⁽¹⁰⁾ Ω_{R_+} is flat and of finite presentation over S , and contains the identity section, hence meets each fiber of G in a nonempty, hence dense, open subset; the second assertion therefore follows from Exp. XVIII 1.3⁽¹¹⁾.

Corollary 4.1.6. *Let (G, T, M, R) be a split reductive S -group. Then*

$$\underline{\text{Centr}}(G) = \bigcap_{\alpha \in R} \text{Ker}(\alpha).$$

⁽¹²⁾ *Consequently, $\underline{\text{Centr}}(G)$ is representable by a diagonalizable closed subgroup of G .*

The second assertion follows at once from the first. To prove the latter, one may invoke Exp. XII 4.8 and 4.11; one may also proceed directly as follows.⁽¹³⁾ Let $S' \rightarrow S$. If $t \in T(S')$ and if $\alpha(t) = 1$ for every $\alpha \in R$, then $\text{int}(t)$ induces the identity on $T_{S'}$ and on each $(U_\alpha)_{S'}$, $\alpha \in R$, hence also on $(\Omega_{R_+})_{S'}$, hence on $G_{S'}$ by schematic density, whence $t \in \underline{\text{Centr}}(G)(S')$.

Conversely, since $\underline{\text{Centr}}_{G_{S'}}(T_{S'}) = T_{S'}$ (cf. Exp. XIX 2.8), if $g \in G(S')$ centralizes $T_{S'}$ and the $(U_\alpha)_{S'}$, it is a section of $T_{S'}$ which annihilates the $\alpha \in R$.

Corollary 4.1.7. *Let S be a scheme, G a reductive S -group. Then the center of G is representable by a closed subgroup of G , of multiplicative type; it is also “the intersection of the maximal tori of G ” in the following sense: for every $S' \rightarrow S$, $\underline{\text{Centr}}(G)(S')$ is the set of $g \in G(S')$ whose inverse image in $G(S'')$, for every $S'' \rightarrow S'$, is contained in all the $T(S'')$, where T ranges over the set of maximal tori of $G_{S''}$.*

Taking 2.3 into account, the first assertion follows from 4.1.6 by descent.⁽¹⁴⁾ Let us prove the second assertion. Let H be “the intersection of the maximal tori of G ” in the preceding sense. One clearly has $\underline{\text{Centr}}(G) \subset H$.⁽¹⁵⁾ Then, by descent, it suffices to prove $\underline{\text{Centr}}(G) = H$ in the case where G is split. Since H is contained in the intersection of the maximal tori of G in the usual sense, this then follows from the following remark: if (G, T, M, R) is a splitting, $\alpha \in R$ and $X \in \Gamma(S, \mathfrak{g}^\alpha)^\times$, then $\text{int}(\exp_\alpha(X))(T) \cap T = \text{Ker}(\alpha)$, as a trivial calculation shows. (Cf. also Exp. XII 8.6 and 8.8 for a more general statement).

Remark 4.1.8. In what follows, in the split case we will systematically identify T with $D_S(M)$. Then $\underline{\text{Centr}}(G)$ is none other than $D_S(M/\Gamma_0(R))$, where $\Gamma_0(R)$ is the subgroup of M generated by R (cf. Exp. XXI 1.1.6). If $\{\alpha_1, \alpha_2, \dots, \alpha_n\}$ is a system of simple roots of R , one has at once (cf. Exp. XX 1.19):

$$\underline{\text{Centr}}(G) = \bigcap \text{Ker}(\alpha_i) = \bigcap \underline{\text{Centr}}(Z_{\alpha_i}).$$

Proposition 4.1.9. *Let S be a scheme, (G, T, M, R) a split S -group, Q an S -torus, α_0 a character of Q , $\mathcal{L}$ an invertible $\mathcal{O}_S$ -module,*

$$f : Q \rightarrow T, \quad p : W(\mathcal{L}) \rightarrow G$$

morphisms of groups satisfying the set-theoretic relation

$$p(\alpha_0(q)x) = \text{int}(f(q)) \cdot p(x),$$

⁽¹⁰⁾**N.D.E.** The original has been expanded in what follows.

⁽¹¹⁾**N.D.E.** See also EGA IV₃, 11.10.10.

⁽¹²⁾**N.D.E.** The following sentence has been added.

⁽¹³⁾**N.D.E.** The original has been expanded in what follows.

⁽¹⁴⁾**N.D.E.** Indeed, representability of the center by a closed subscheme of G is local for the (fpqc) topology (SGA 1, VIII 5.2 and 5.4), hence a fortiori for the étale topology, and the same holds for the property “of multiplicative type”.

⁽¹⁵⁾**N.D.E.** The original has been expanded in what follows.

for all $q \in Q(S')$, $x \in W(\mathcal{L})(S')$, $S' \rightarrow S$. Assume that f separates the elements of R in the following sense: if $\alpha, \alpha' \in R$ and if $m, m' \in \mathbf{Z}$, then $m\alpha \circ f = m'\alpha' \circ f$ implies $m\alpha = m'\alpha'$.⁽¹⁶⁾ Finally let $s \in S$ be such that $(\alpha_0)_{\bar{s}} \neq e$ and $p_{\bar{s}} \neq e$.

There then exist an open subset U of S containing s , an integer $q > 0$ such that $x \mapsto x^q$ is an endomorphism of $\mathbf{G}_{a,U}$, a root $\alpha \in R$ and an isomorphism of $\mathcal{O}_U$ -modules

$$h : (\mathcal{L}|_U)^{\otimes q} \xrightarrow{\sim} \mathfrak{g}^\alpha|_U$$

such that

- (i) $(\alpha \circ f)_U = (q\alpha_0)_U$,
- (ii) $p(X) = \exp_\alpha(h(X^q))$ for every $X \in W(\mathcal{L})(S')$, $S' \rightarrow U$.

Moreover, once U has been chosen, q , α and h are uniquely determined.

Restricting S , we may assume that α_0 is nonzero on each fiber of S . Choose a system of positive roots R_+ of R and let $V = p^{-1}(\Omega_{R_+})$. It is an open subset of $W(\mathcal{L})$ containing the zero section and stable under multiplication by every $\alpha_0(q)$, $q \in Q(S')$, $S' \rightarrow S$. Since α_0 is nontrivial on each fiber, it follows immediately that $V = W(\mathcal{L})$, hence that p factors through Ω_{R_+} . Choose any order on R_+ and R_- ; all products will be assumed taken in this order. One therefore has unique morphisms

$$a_\alpha : W(\mathcal{L}) \rightarrow U_\alpha, \quad \alpha \in R, \quad b : W(\mathcal{L}) \rightarrow T$$

such that

$$p(x) = \prod_{\alpha \in R_-} a_\alpha(x) \cdot b(x) \cdot \prod_{\alpha \in R_+} a_\alpha(x).$$

Writing the covariance condition under Q , one obtains at once

$$\begin{aligned} a_\alpha(\alpha_0(q)x) &= \alpha(f(q))a_\alpha(x), & \alpha \in R, \\ b(\alpha_0(q)x) &= b(x) \end{aligned}$$

for all $x \in W(\mathcal{L})(S')$, $q \in Q(S')$, $S' \rightarrow S$. The second condition gives at once $b = e$.

Let now $\alpha \in R$ be such that $(a_\alpha)_{\bar{s}} \neq e$ (we know that such an α exists, since $p_{\bar{s}}$ is assumed $\neq e$). Applying Exp. XIX 4.12 (a), one deduces that there exists an integer $n > 0$ such that $(\alpha \circ f)_{\bar{s}} = (n\alpha_0)_{\bar{s}}$. Restricting S , one may assume $\alpha \circ f = n\alpha_0$ (Exp. IX 5.3). But then, for every $\alpha' \in R$, $\alpha' \neq \alpha$, one has $(\alpha' \circ f)_{\bar{s}} \neq m\alpha_0$ for every integer $m > 0$ by the hypothesis on f (and the fact that the only roots proportional to α are α and $-\alpha$). Applying Exp. XIX 4.12 (a) again, this time to $a_{\alpha'}$, one deduces that $a_{\alpha'}$ is zero in a neighborhood of S ; since R is finite, one may, restricting S further, assume the $a_{\alpha'}$ zero for $\alpha' \in R$, $\alpha' \neq \alpha$. One then has $p = a_\alpha$, and one may apply to it Exp. XIX 4.12 (b), then (c), which gives the stated result (the uniqueness assertions are obvious).

Remark 4.1.10. The condition imposed on f in 4.1.9 holds in particular if f is surjective (= faithfully flat).

Proposition 4.1.11. Let (G, T, M, R) be a split S -group, R_+ a system of positive roots of R , Ω_{R_+} the corresponding “big cell”.

- (i) Let H be an S -group functor, separated⁽¹⁷⁾ for (fppf). If $f, g : G \rightrightarrows H$ are two morphisms of groups which coincide on Ω_{R_+} , then $f = g$.

⁽¹⁶⁾**N.D.E.** Note that this is equivalent to the hypothesis: if $\alpha, \alpha' \in R$ and $m, m' \in \mathbf{Z}$ and if $(m\alpha \circ f)_{\bar{s}} = (m'\alpha' \circ f)_{\bar{s}}$ for every geometric point $\bar{s}$ of S , then $m\alpha = m'\alpha'$. In particular, this separation hypothesis is stable under base change.

⁽¹⁷⁾**N.D.E.** Recall (cf. Exp. IV, 4.3.5) that an S -presheaf H is separated for a topology $\mathcal{T}$ if for every $S' \rightarrow S$ and every family of S -morphisms $(S'_i \rightarrow S')_{i \in I}$ covering for $\mathcal{T}$, the map $H(S') \rightarrow \prod_i H(S'_i)$ is injective.

- (ii) Let H be an S -sheaf of groups for (fppf) and $f : \Omega_{R_+} \rightarrow H$ an S -morphism satisfying the following condition: for every $S' \rightarrow S$ and all $x, y \in \Omega_{R_+}(S')$ such that $xy \in \Omega_{R_+}(S')$, one has $f(xy) = f(x)f(y)$. There then exists a (unique, by (i)) morphism of groups $\bar{f} : G \rightarrow H$ extending f .

Indeed, by 4.1.5, (i) (resp. (ii)) follows at once from Exp. XVIII 2.2 (resp. 2.3 and 2.4).

Remark 4.1.12. If $\alpha \in R_+$, one has

$$(\dagger) \quad \Omega_{R_+} \cap Z_\alpha = U_{-\alpha} \cdot T \cdot U_\alpha.$$

⁽¹⁸⁾Indeed, for every $S' \rightarrow S$, if $g = \prod_{\beta \in R_-} p_\beta(x_\beta) \cdot t \cdot \prod_{\beta \in R_+} p_\beta(x_\beta)$ is an element of $\Omega_{R_+}(S')$ and if $t' \in T_\alpha(S'')$, then

$$t'gt'^{-1} = \prod_{\beta \in R_-} p_\beta(\beta(t')x_\beta) \cdot t \cdot \prod_{\beta \in R_+} p_\beta(\beta(t')x_\beta)$$

and since α and $-\alpha$ are the only two elements of R which take the value 1 on T_α , one obtains that g belongs to $Z_\alpha = \text{Centr}(T_\alpha)$ if and only if $x_\beta = 0$ for $\beta \neq \pm\alpha$.

By $(\dagger)$, one deduces from XX 2.1 that if $X \in \Gamma(X, \mathfrak{g}^\alpha)$ and $Y \in \Gamma(S, \mathfrak{g}^{-\alpha})$, one has:

$$\exp_\alpha(X) \exp_{-\alpha}(Y) \in \Omega_{R_+}(S) \iff 1 + XY \text{ invertible.}$$

4.2. Morphisms of split groups

Definition 4.2.1. Let S be a (nonempty) scheme, (G, T, M, R) and (G', T', M', R') two split S -groups. One says that the morphism of S -groups $f : G \rightarrow G'$ is *compatible with the splittings*, or defines a *morphism of split groups*, if the restriction of f to T factors as a morphism $f_T : T \rightarrow T'$ of the form $f_T = D_S(h)$, where $h : M' \rightarrow M$ is a morphism of groups satisfying the following condition:

there exists a bijection $d : R \xrightarrow{\sim} R'^{(19)}$ and for each $\alpha \in R$ an integer $q(\alpha) > 0$ such that $x \mapsto x^{q(\alpha)}$ is an endomorphism of $\mathbf{G}_{a,S}$ and such that

$$h(d(\alpha)) = q(\alpha)\alpha, \quad {}^th(\alpha^*) = q(\alpha)d(\alpha)^*.$$

Remark 4.2.2. It is immediate that $h, d, q(\alpha)$ for $\alpha \in R$ are uniquely determined by f . One writes $h = \mathcal{R}(f)$. The $q(\alpha)$ are the *radicial exponents* of f (or of h).

Let p be the prime number (if it exists) which is zero on S ; put $p = 1$ if no prime number is zero on S . Then $\mathcal{R}(f)$ is a p -morphism of reduced root data in the sense of Exp. XXI 6.8. One has therefore defined a functor $\mathcal{R}$ from the category of split S -groups to that of reduced root data (endowed with the p -morphisms).

Proposition 4.2.3. *Under the conditions of 4.2.1, one has the following properties:*

- (i) For every $\alpha \in R$, there exists a unique isomorphism of $\mathcal{O}_S$ -modules

$$f_\alpha : (\mathfrak{g}^\alpha)^{\otimes q(\alpha)} \xrightarrow{\sim} \mathfrak{g}'^{d(\alpha)}$$

such that

$$f(\exp_\alpha(X)) = \exp_{d(\alpha)}(f_\alpha(X^{q(\alpha)}))$$

for every $X \in W(\mathfrak{g}^\alpha)(S')$, $S' \rightarrow S$.

- (ii) For every $\alpha \in R$, one has $q(-\alpha) = q(\alpha)$ and f_α and $f_{-\alpha}$ are contragredient to one another.

- (iii) For every $\alpha \in R$, every $Z \in W(\mathfrak{g}^\alpha)^*(S')$, $S' \rightarrow S$, one has

$$f(w_\alpha(Z)) = w_{d(\alpha)}(Z^{q(\alpha)}).$$

⁽¹⁸⁾**N.D.E.** The original has been expanded in what follows.

⁽¹⁹⁾**N.D.E.** The notation $u : R \rightarrow R'$ has been replaced by $d : R \rightarrow R'$.

By hypothesis the diagram

$$\begin{array}{ccccc} \mathbf{G}_{m,S} & \xrightarrow{\alpha^*} & T & \xrightarrow{\alpha} & \mathbf{G}_{m,S} \\ q(\alpha) \downarrow & & \downarrow f_T & & \downarrow q(\alpha) \\ \mathbf{G}_{m,S} & \xrightarrow{d(\alpha)^*} & T' & \xrightarrow{d(\alpha)} & \mathbf{G}_{m,S} \end{array}$$

is commutative. It follows that f maps $\text{Ker}(\alpha)$ into $\text{Ker}(d(\alpha))$, hence T_α into $T'_{d(\alpha)}$, hence Z_α into $Z'_{d(\alpha)}$. One then has only to apply Exp. XXI 3.10 and 3.11 to the groups Z_α and $Z'_{d(\alpha)}$.

Proposition 4.2.4. *The morphism f induces a morphism f_N from $\underline{\text{Norm}}(T)$ to $\underline{\text{Norm}}_{G'}(T')$, hence a morphism f_W from $W_G(T)$ to $W_{G'}(T')$; the latter is an isomorphism. More precisely, if one denotes by $\bar{d} : W(\mathcal{R}(G)) = W \rightarrow W' = W(\mathcal{R}(G'))$ the isomorphism extending $s_\alpha \mapsto s_{d(\alpha)}$ (Exp. XXI 6.8.4), one has a commutative diagram of isomorphisms:*

$$\begin{array}{ccc} W_G(T) & \xrightarrow[\sim]{f_W} & W_{G'}(T') \\ \sim \uparrow & & \uparrow \sim \\ W_S & \xrightarrow[\bar{d}_S]{\sim} & W'_S. \end{array}$$

This follows at once from 3.4, Exp. XXI 6.8.4, and (iii) above.

Remark 4.2.5. With the notations of 4.2.3, the restriction of f to Ω_{R_+} (for a system of positive roots R_+) is written explicitly: it maps Ω_{R_+} into $\Omega'_{d(R_+)}$ ($d(R_+)$ is a system of positive roots of R' by Exp. XXI 6.8.7) and is given by the set-theoretic formula:

$$\begin{aligned} & f \left(\prod_{\alpha \in R_-} \exp_\alpha(X_\alpha) \cdot t \cdot \prod_{\alpha \in R_+} \exp_\alpha(X_\alpha) \right) \\ &= \prod_{\alpha \in R_-} \exp_{d(\alpha)}(f_\alpha(X_\alpha^{q(\alpha)})) \cdot f_T(t) \cdot \prod_{\alpha \in R_+} \exp_{d(\alpha)}(f_\alpha(X_\alpha^{q(\alpha)})). \end{aligned}$$

Proposition 4.2.6.

- (i) f is surjective (= faithfully flat in the present case, cf. VIB 3.11) if and only if f_T is.
- (ii) One has $\text{Ker}(f) \subset \Omega_{R_+}$.

Let us prove (i): if f is surjective, then $f_T(T) = f(T)$ is a maximal torus of G' (indeed $f(T)$ is a subtorus of a maximal torus T' (Exp. IX 6.8); to verify that $f(T) = T'$, one is reduced to the case of an algebraically closed field, where this is *Bible*, § 7.3, th. 3 (a)).

If f_T is surjective, then the preceding formula shows that f induces a surjection from $\Omega = \Omega_{R_+}$ onto $\Omega' = \Omega'_{d(R_+)}$.⁽²⁰⁾ Since the fibers of G' are connected, it follows (cf. Exp. VIA, 0.5) that f is surjective.

Let us prove (ii), and for this admit a result which will be proved below (5.7.4): choose for each $w \in W$ an $n_w \in \underline{\text{Norm}}_G(T)(S)$ representing it; then the open subsets $n_w \Omega$ ($w \in W$) form a covering of G . It then suffices to prove that $\text{Ker}(f) \cap n_w \Omega \neq \emptyset$ implies $w = 1$. If $x \in \Omega(S')$, $S' \rightarrow S$ and $f(n_w x) = 1$, one has $f(x) = f(n_w)^{-1}$; now $f(x) \in \Omega'(S')$ and $f(n_w)^{-1} \in \underline{\text{Norm}}_{G'}(T')(S')$. By 4.2.4, one is reduced to proving:

Lemma 4.2.7. *Under the conditions of 4.1.2, one has $\Omega \cap \underline{\text{Norm}}_G(T) = T$.*

⁽²⁰⁾**N.D.E.** The reference to Exp. VI has been expanded in what follows.

Let

$$x = \prod_{\alpha \in R_-} p_\alpha(x_\alpha) \cdot t \cdot \prod_{\alpha \in R_+} p_\alpha(x_\alpha) = vt u \in \Omega(S').$$

If x normalizes $T_{S'}$, one has for every $t' \in T(S')$,

$$xt'x^{-1} = t'' \in T(S'),$$

that is, $xt' = t''x$, which is written

$$v(tt')(t'^{-1}ut') = (t''vt''^{-1})(t''t)u,$$

which gives $t'^{-1}ut' = u$, hence $u \in \underline{\text{Centr}}_G(T)(S') = T(S')$, that is $u = 1$. Likewise $v = 1$.

Corollary 4.2.8. *One has*

$$\text{Ker}(f) = \prod_{\alpha \in R_-} K_\alpha \cdot \text{Ker}(f_T) \cdot \prod_{\alpha \in R_+} K_\alpha,$$

where for each $\alpha \in R$, K_α denotes the finite S -group

$$K_\alpha = \text{Ker}(U_\alpha \longrightarrow U_\alpha^{\otimes q(\alpha)}) \simeq \alpha_{q(\alpha), S}.$$

To apply this corollary, put:

Definition 4.2.9. Let S be a scheme, G and G' two reductive S -groups. A faithfully flat finite morphism of S -groups $f : G \longrightarrow G'$ (i.e. surjective and with kernel finite over S) is called an *isogeny*. If moreover $\text{Ker}(f)$ is a central subgroup of G , one says that f is a *central isogeny*.

Proposition 4.2.10. *Let $f : G \longrightarrow G'$ be a morphism of split groups. For f to be an isogeny (resp. a central isogeny), it is necessary and sufficient that f_T be an isogeny, i.e. that $\mathcal{R}(f)$ be injective with finite cokernel (resp. and that for every $\alpha \in R$, one have $q(\alpha) = 1$).*

Indeed, by 4.2.8, $\text{Ker}(f)$ is finite over S if and only if $\text{Ker}(f_T)$ is finite over S , and $\text{Ker}(f) \subset T$ if and only if each $q(\alpha)$ is 1 ($\text{Ker}(f)$ is then central since it is of multiplicative type and invariant, cf. Exp. IX 5.5).

Remark 4.2.11. a) One therefore sees that $f : G \longrightarrow G'$ is a central isogeny if and only if $\mathcal{R}(f) : \mathcal{R}(G') \longrightarrow \mathcal{R}(G)$ is an isogeny in the sense of Exp. XXI 6.2; moreover one has in this case (with the notations of *loc. cit.*):

$$\text{Ker}(f) = D_S(K(\mathcal{R}(f))), \quad K(\mathcal{R}(f)) = \text{Coker}(\mathcal{R}(f)).$$

b) If G and G' are semisimple, every morphism of split groups $f : G \longrightarrow G'$ is an isogeny⁽²¹⁾.

c) If $f : G \longrightarrow G'$ is faithfully flat and finite and if G is reductive (resp. semisimple), then G' is also. It is indeed of finite presentation over S (Exp. V 9.1), affine over S (EGA II 6.7.1), smooth over S (Exp. VI 9.2), with connected reductive (resp. semisimple) geometric fibers by Exp. XIX 1.7.

Definition 4.2.1 may seem arbitrary. It is justified by the following proposition (which we will state, for simplicity, for semisimple groups).

Let us say that a morphism $f : G \longrightarrow G'$ of reductive S -groups is *splittable* if there exist splittings of G and G' with which f is compatible. One then has the

Proposition 4.2.12. *Let S be a scheme, G and G' two semisimple S -groups, $f : G \longrightarrow G'$ a morphism of groups. Let $s \in S$. The following conditions are equivalent:*

- (i) $\text{Ker } f_{\bar{s}}$ is finite ($\Leftrightarrow e$ is isolated in $\text{Ker } f(\bar{s})$) and $f_{\bar{s}}$ is surjective, i.e. $f_{\bar{s}}$ is an isogeny.
- (ii) $f_{\bar{s}}$ is splittable.

⁽²¹⁾**N.D.E.** Indeed, since G and G' are semisimple, f_T is surjective with finite kernel, hence f is faithfully flat with finite kernel by 4.2.6 (i) and 4.2.8.

- (iii) *There exists an étale morphism $S' \rightarrow S$ covering s such that $f_{S'} : G_{S'} \rightarrow G'_{S'}$ is splittable.*

One clearly has (iii) $\Leftrightarrow$ (ii); (ii) $\Rightarrow$ (i) follows from 4.2.11 (b) (it is here that the hypothesis that G and G' are semisimple enters—the other implications hold for reductive groups).

Let us now prove (i) $\Rightarrow$ (iii). One may assume G and G' split in such a way that f induces a morphism $f_T : T \rightarrow T'$ (2.3 and Exp. XIX 2.8); restricting S , one may assume that $f_T = D_S(h)$, where h is a morphism of groups $M' \rightarrow M$. Let $\alpha \in R$, consider the composite morphism

$$p : W(\mathfrak{g}^\alpha) \xrightarrow{\exp_\alpha} G \xrightarrow{f} G'.$$

Since $\text{Ker}(p_{\bar{s}})$ is finite, $p_{\bar{s}} \neq e$. On the other hand $f_{T_{\bar{s}}}$ is surjective; one may therefore apply 4.1.9 and there exist an open subset V_α of S containing s , a root $\alpha' \in R'$, an integer $q(\alpha)$ such that $x \mapsto x^{q(\alpha)}$ is an endomorphism of $\mathbf{G}_{a, V_\alpha}$, and an isomorphism of $\mathcal{O}_{V_\alpha}$ -modules

$$f_\alpha : (\mathfrak{g}^\alpha)^{\otimes q(\alpha)}|_{V_\alpha} \rightarrow \mathfrak{g}'^{\alpha'}|_{V_\alpha}$$

such that $f(\exp_\alpha(X_\alpha)) = \exp_{d(\alpha)}(f_\alpha(X_\alpha^{q(\alpha)}))$ and $\alpha' \circ f_T = h(\alpha') = q(\alpha)\alpha$. One may replace S by the intersection of the V_α , for $\alpha \in R$. Put $\alpha' = d(\alpha)$. It is clear that $d : R \rightarrow R'$ is a bijection, for the kernel of h is finite ($f_{T_{\bar{s}}}$ being surjective). It only remains to prove that $f_T \circ \alpha^* = q(\alpha)\alpha'^*$, which is done by a trivial modification of the argument used in Exp. XX 3.11.

In any case, as was seen in the course of the proof, one has (i) $\Rightarrow$ (iii). One therefore has:

Corollary 4.2.13. *Let S be a scheme, $f : G \rightarrow G'$ an isogeny of reductive groups. Then f is locally splittable for the étale topology.*

4.3. Central quotients of reductive groups

Consider first a particular case.

Proposition 4.3.1. *Let S be a scheme, (G, T, M, R) a split S -group, N a subgroup of M containing R , $Q = D_S(M/N) \subset \text{Centr}(G)$. Then:*

- (i) $G' = G/Q$ is a reductive S -group, $T' = T/Q$ is a maximal torus of it;
- (ii) if one identifies T' with $D_S(N)$, then $R \subset N$ is a system of roots of G' with respect to T' , (G', T', N, R) is a splitting of G , and $\mathcal{R}(G')$ is canonically identified with the induced root datum (Exp. XXI 6.5) $\mathcal{R}(G)_N$;
- (iii) the canonical morphism $G \rightarrow G'$ is compatible with the splittings, with radicial exponents 1, and gives by functoriality the canonical morphism (loc. cit.) $\mathcal{R}(G)_N \rightarrow \mathcal{R}(G)$.

One knows that $G' = G/Q$ is representable by a group scheme affine over S (Exp. VIII 5.7), smooth over S (Exp. VIB 9.2), with connected reductive geometric fibers (as quotients of reductive groups, cf. Exp. XIX 1.7); G' is therefore a reductive S -group.

It is clear that $T' = T/Q \simeq D_S(N)$ is a maximal torus of it. Note next that on choosing a system of positive roots R_+ of R , the open subset Ω_{R_+} of 4.2 is stable under Q and that one has a canonical isomorphism

$$\Omega_{R_+}/Q \simeq \prod_{\alpha \in R_-} U_\alpha \times_S (T/Q) \times_S \prod_{\alpha \in R_+} U_\alpha,$$

and that Ω_{R_+}/Q is an open subset of G' , containing the identity section (cf. Exp. IV, 4.7.2 and 6.4.1).

It follows that if one denotes by $\mathfrak{g}'$ the Lie algebra of G' and by α the character of T/Q induced by α (or, equivalently, defined by $\alpha \in N$ in the identification $T/Q = D_S(N)$), the canonical morphism $\mathfrak{g} \rightarrow \mathfrak{g}'$ induces for each $\alpha \in R$ an isomorphism

$$\mathfrak{g}^\alpha \xrightarrow{\sim} \mathfrak{g}'^{\alpha'}.$$

One has therefore proved that R is a system of roots of G' with respect to T' , and one finishes the proof without difficulty.

Corollary 4.3.2. *Let S be a scheme, G a reductive S -group, Q an invariant subgroup of multiplicative type of G . Then Q is central in G , the quotient G/Q is representable by a reductive S -group, and the canonical morphism $G \rightarrow G/Q$ is locally splittable for the étale topology (with radicial exponents equal to 1).*

The first assertion follows from Exp. IX 5.5; the others are local for the étale topology and one is reduced to 4.3.1.

Definition 4.3.3. Let G be a reductive S -group. One says that G is *adjoint* (resp. *simply connected*) if for every $s \in S$, the type of G at s is given by an adjoint (resp. simply connected) root datum, i.e. (Exp. XXI 6.2.6) such that M is generated by R (resp. M^* generated by R^*).

Proposition 4.3.4.

- (i) *An adjoint (resp. simply connected) reductive group is semisimple.*
- (ii) *If T is a maximal torus of the adjoint (resp. simply connected) reductive group G and if α is a root of G with respect to T , then the infinitesimal root $\bar{\alpha}$ is nonzero on each fiber (resp. α^* is a monomorphism and the infinitesimal coroot H_α is nonzero on each fiber).*

Indeed, (i) is trivial; (ii) is verified on the geometric fibers and follows at once from Exp. XXI 6.2.8.

Proposition 4.3.5.

- (i) *For the reductive group G to be adjoint, it is necessary and sufficient that $\underline{\text{Centr}}(G) = \{e\}_S$.*
- (ii) *For every reductive group G , the quotient group $G/\underline{\text{Centr}}(G)$ is an adjoint reductive group.*

Indeed, one may assume G split; then (i) is trivial (for $\underline{\text{Centr}}(G) = D_S(M/\Gamma_0(R))$), and (ii) follows from 4.3.1.

Definition 4.3.6. Let G be a reductive S -group. One calls the *adjoint group* of G , and denotes by $\text{ad}(G)$, the group $G/\underline{\text{Centr}}(G)$. One calls the *radical* of G , and denotes by $\text{rad}(G)$, the maximal torus (unique by Exp. XII 1.12) of $\underline{\text{Centr}}(G)$. One calls the *semisimple group associated with G* the quotient $G/\text{rad}(G)$.

The preceding definitions are compatible with base change. If $s \in S$, $\text{rad}(G)_{\bar{s}}$ is indeed the radical of $G_{\bar{s}}$ in the usual sense (Exp. XIX 1.6).

4.3.7. — If (G, T, M, R) is a split group, then $\text{rad}(G) = D_S(M/N)$, where $N = M \cap \mathcal{V}(R)$, hence the semisimple group associated with G (as indeed the adjoint group of G) is endowed with a canonical splitting (4.3.1) and one has a diagram of split groups

$$G \rightarrow \text{ss}(G) \rightarrow \text{ad}(G)$$

corresponding to the canonical diagram of root data (Exp. XXI 6.5.5)

$$\text{ad}(\mathcal{R}(G)) \rightarrow \text{ss}(\mathcal{R}(G)) \rightarrow \mathcal{R}(G).$$

Remark 4.3.8. Let (G, T, M, R) be an adjoint (resp. simply connected) split S -group, Δ a system of simple roots of R . Then the family $\{\alpha\}_{\alpha \in \Delta}$, resp. $\{\alpha^*\}_{\alpha \in \Delta}$, induces an isomorphism

$$T \xrightarrow{\sim} (\mathbf{G}_{m,S})^\Delta, \quad \text{resp.} \quad (\mathbf{G}_{m,S})^\Delta \xrightarrow{\sim} T.$$

Indeed, $M = \Gamma_0(R)$ (resp. ...) and Δ is a basis of the free abelian group $\Gamma_0(R)$ (Exp. XXI 3.1.8).

Remark 4.3.9. The radical of a reductive group is a *characteristic* subgroup (i.e. stable under $\underline{\mathrm{Aut}}_{S\text{-gr.}}(G)$), in view of its definition.

5. Subgroups of type (R)

We are particularly interested in reductive groups, but some of the results we are going to establish hold more generally for a larger class of groups: groups of type (RR).

5.1. Groups of type (RR)

Definition 5.1.1. Let S be a scheme, G an S -group scheme. One says that G is of *type (RR)* if it satisfies the following conditions:

- (i) G is smooth and of finite presentation over S and has connected fibers.
- (ii) G possesses maximal tori locally for the (fpqc) topology.
- (iii) For every $s \in S$, every maximal torus T of $G_{\bar{s}}$ and every root of $G_{\bar{s}}$ with respect to $T_{\bar{s}}$ (Exp. XIX 1.10), $\underline{\mathrm{Lie}}(G_{\bar{s}})^{\alpha}$ has rank 1 (as a vector space over $\kappa(\bar{s})$).
- (iv) For every $s \in S$ and every maximal torus T of $G_{\bar{s}}$, denote by R the set of roots of $G_{\bar{s}}$ with respect to T and by $\Gamma_0(R)$ the subgroup of $\mathrm{Hom}_{\bar{s}\text{-gr.}}(T, \mathbf{G}_{m, \bar{s}})$ generated by R ; then the content⁽²²⁾ of every root $\alpha \in R$ in the free abelian group $\Gamma_0(R)$ (which is thus an integer > 0) is invertible on S .

Recall 5.1.1.1. ⁽²³⁾Recall that if G is a smooth connected algebraic group over an algebraically closed field k , a *Cartan subgroup* of G is the centralizer of a maximal torus of G (XII 1.0), and such a subgroup is *smooth and connected*: for this, as well as for other characterizations of Cartan subgroups, see *Bible*, § 7.1, Th. 1 in the case G affine and Exp. XII Th. 6.6 in the general case. If S is an arbitrary scheme and G a smooth S -group of finite type, one calls a *Cartan subgroup* of G a smooth S -subgroup C of G such that, for every $s \in S$, $C_{\bar{s}}$ is a Cartan subgroup of $G_{\bar{s}}$ (XII Def. 3.1).

Remark 5.1.2. a) By Exp. XII 7.1 (where the hypothesis G separated holds here since G has connected fibers, cf. Exp. VIB 5.5), (i) and (ii) imply that G possesses maximal tori (resp. Cartan subgroups) locally for the étale topology, conjugate locally for the étale topology.

b) The Cartan subgroups of G have connected fibers (Exp. XII 6.6).

c) If G is affine over S , (i) and (ii) are equivalent respectively to

(i') G is smooth over S and has connected fibers.

(ii') The reductive rank of the fibers of G is locally constant (Exp. XII

1.7).

d) By Exp. XII 8.8 (c) and (d), G possesses a reductive center Z and for every $s \in S$ one has, with the notations of (iv),⁽²⁴⁾ $Z_{\bar{s}} = \bigcap_{\alpha \in R} \mathrm{Ker}(\alpha)$, whence

$$\mathrm{Hom}_{\bar{s}\text{-gr.}}((T/Z)_{\bar{s}}, \mathbf{G}_{m, \bar{s}}) \simeq \Gamma_0(R).$$

e) Condition (iv) holds in particular in the following two cases:

(1) S has characteristic 0;

(2) every root is an indivisible element of the group generated by the roots.

f) Being of type (RR) is stable under base change and is local for the (fpqc) topology.

From remarks (c) and (e), one obtains at once the

⁽²²⁾**N.D.E.** The content of the root α is the positive generator of the ideal $\{f(\alpha), f \in \Gamma_0(R)^*\}$ of $\mathbf{Z}$; it is the largest integer $c > 0$ such that $\alpha/c \in \Gamma_0(R)$.

⁽²³⁾**N.D.E.** This reminder has been added.

⁽²⁴⁾**N.D.E.** The original indicated “under the conditions of (iv)”, but the last condition of (iv) does not seem to be used here.

Proposition 5.1.3. *A reductive group is of type (RR).*

Proposition 5.1.4. *Let S be a scheme, G an S -group of type (RR), Q a central subgroup of G of finite presentation over S such that the quotient G/Q is representable (for example G affine over S and Q of multiplicative type (IX 2.3), or S artinian (VIA 3.3.2)); then G/Q is an S -group of type (RR).*

Indeed G/Q is smooth over S (Exp. VIB 9.2), of finite presentation over S (Exp. V 9.1) and has connected fibers, so condition (i) holds. On the other hand, condition (ii) follows from Exp. XII 7.6; it remains to verify conditions (iii) and (iv).

Write $G' = G/Q$, let $u : G \rightarrow G'$ be the canonical morphism, $T' = u(T)$ the maximal torus of G' which is the image of T (cf. Exp. XII 7.1 (e)); for each $\alpha \in R$, denote again by α the character of T' defined by α (one has $Q \cap T \subset \bigcap_{\alpha \in R} \text{Ker}(\alpha)$ by 5.1.2 (d)). Let us first prove:

Lemma 5.1.5. *Under the conditions of 5.1.4, let $T = D_S(M)$ be a trivialized maximal torus of G , assume that the decomposition of $\mathfrak{g} = \underline{\text{Lie}}(G)$ under $\text{Ad}(T)$ is of the form*

$$\mathfrak{g} = \mathfrak{g}^0 \oplus \coprod_{\alpha \in R} \mathfrak{g}^\alpha, \quad R \subset M - \{0\},$$

where for every $s \in S$, $\mathfrak{g}^\alpha(s) \neq 0$ for $\alpha \in R$ (so $\mathfrak{g}^\alpha$ is an invertible $\mathcal{O}_S$ -module for every $\alpha \in R$ and R is the set of roots of $G_{\bar{s}}$ with respect to $T_{\bar{s}}$ for every $s \in S$).

Then the Lie algebra $\mathfrak{g}'$ of G' decomposes under $\text{Ad}(T')$ as follows:

$$\mathfrak{g}' = \mathfrak{g}'^0 \oplus \coprod_{\alpha \in R} \mathfrak{g}'^\alpha$$

and $\underline{\text{Lie}}(u)$ induces an isomorphism of $\mathfrak{g}^\alpha$ onto $\mathfrak{g}'^\alpha$.

Indeed, let $p = \underline{\text{Lie}}(u) : \mathfrak{g} \rightarrow \mathfrak{g}'$. One has at once $p(\mathfrak{g}^\alpha) \subset \mathfrak{g}'^\alpha$ for every $\alpha \in R$, and $p(\mathfrak{g}^0) \subset \mathfrak{g}'^0$. Since

$$\text{Ker}(p) = \underline{\text{Lie}}(Q) \subset \underline{\text{Lie}}(\text{Centr}_G(T)) = \mathfrak{g}^0,$$

p induces a monomorphism of $\mathfrak{g}^\alpha$ into $\mathfrak{g}'^\alpha$, for every $\alpha \in R$.

To prove the lemma, it suffices to do so when S is the spectrum of an algebraically closed field, and by the preceding remarks it then suffices to prove that $\text{rg}(\mathfrak{g}') = \text{rg}(\mathfrak{g}'^0) + \text{Card}(R)$. Now put $C = \underline{\text{Centr}}_G(T)$, $C' = \underline{\text{Centr}}_{G'}(T')$; by Exp. XII 7.1 (e), u induces a faithfully flat morphism $C \rightarrow C'$ with kernel Q . One therefore has

$$\dim C' + \dim Q = \dim C.$$

But G , G' , C and C' are smooth, hence

$$\begin{aligned} \dim G &= \text{rg}(\mathfrak{g}) = \text{rg}(\mathfrak{g}^0) + \text{Card}(R) = \dim C + \text{Card}(R) \\ &= \dim Q + \dim C' + \text{Card}(R) \\ &= \dim Q + \text{rg}(\mathfrak{g}'^0) + \text{Card}(R), \\ \text{rg}(\mathfrak{g}') &= \dim G' = \dim G - \dim Q, \end{aligned}$$

which implies

$$\text{rg}(\mathfrak{g}') = \text{rg}(\mathfrak{g}'^0) + \text{Card}(R),$$

that is, the desired relation.

Let us return to the proof of 5.1.4; one may assume that S is the spectrum of an algebraically closed field. Let T be a maximal torus of G ; applying 5.1.5, one has at once (iii) and (iv) for G/Q .

To use the preceding proposition, let us introduce a definition:

Definition 5.1.6. One says that the S -group scheme G is of type (RA) if it is of type (RR) and satisfies in addition condition (iv') (stronger than (iv)):

(iv') For every $s \in S$ and every maximal torus T of $G_{\bar{s}}$, every root of $G_{\bar{s}}$ with respect to T has a content in $\text{Hom}_{\bar{s}\text{-gr.}}(T, \mathbf{G}_{m, \bar{s}})$ which is invertible on S .

Remarks 5.1.7. a) An adjoint reductive S -group is of type (RA).

b) If S has characteristic 0, every group of type (RR) is of type (RA).

c) Being of type (RA) is stable under base change and is local for the (fpqc) topology.

The preceding remark (a) generalizes as:

Proposition 5.1.8. *Let S be a scheme, G an S -group of type (RR), Z its reductive center; assume the quotient G/Z representable (for example G affine over S , or S artinian); then G/Z is of type (RA).*

Indeed, this follows at once from 5.1.4, 5.1.5 and 5.1.2 (d).

Remark 5.1.9. If G is of type (RR) and if T is a maximal torus of G , one may apply Exp. XIX 6.3. In particular $W_G(T)$ is étale, quasi-finite and separated over S .

5.2. Subgroups of type (R)

Definition 5.2.1. Let S be a scheme, G a smooth S -group scheme of finite presentation with connected fibers,⁽²⁵⁾ H a group subscheme of G . One says that H is of type (R) if:

- (i) H is smooth, of finite presentation over S and has connected fibers.⁽²⁵⁾
- (ii) For every $s \in S$, $H_{\bar{s}}$ contains a Cartan subgroup of $G_{\bar{s}}$.

This notion is stable under base change and local for the (fpqc) topology.

Recall 5.2.2. (cf. Exp. XII 7.9) Under the preceding conditions:

- a) $H = \underline{\text{Norm}}_G(H)^0$.
- b) If G contains Cartan subgroups (resp. maximal tori) locally for the étale topology, the same holds for H , and the Cartan subgroups (resp. maximal tori) of H are Cartan subgroups (resp. maximal tori) of G .

Examples 5.2.3. a) *Borel subgroups:* a Borel subgroup of G is a subgroup H of type (R) whose geometric fibers are Borel subgroups of those of G .⁽²⁶⁾

b) *Parabolic subgroups:* a parabolic subgroup of G is a subgroup of type (R) whose geometric fibers contain Borel subgroups.

Other examples are given by the following propositions.

Proposition 5.2.4. *Let G be as in 5.2.1, $K \subset H$ two group subschemes of G , H being assumed of type (R). Then K is a subgroup of type (R) of H if and only if it is a subgroup of type (R) of G .*

⁽²⁷⁾Indeed, let $s \in S$. Since H is of type (R), every maximal torus of $H_{\bar{s}}$ is a maximal torus of $G_{\bar{s}}$, and therefore the same holds for Cartan subgroups.

Proposition 5.2.5. *Let G be as in 5.2.1, T a maximal torus of G , Q a subtorus of T , $Z = \underline{\text{Centr}}_G(Q)$. If H is a subgroup of type (R) of G containing T , then $H \cap Z$ is a subgroup of type (R) of Z .*

⁽²⁵⁾**N.D.E.** The hypothesis that G (resp. H) is of finite presentation over S holds automatically since G (resp. H), being smooth over S and with connected fibers, is quasi-compact and separated over S (VIB 5.5), hence of finite presentation over S .

⁽²⁶⁾**N.D.E.** It amounts to the same to say that H is a smooth subgroup of G , each of whose geometric fibers $H_{\bar{s}}$ is a Borel subgroup of $G_{\bar{s}}$ (since every Borel subgroup of $G_{\bar{s}}$ is connected and contains a Cartan subgroup of $G_{\bar{s}}$).

⁽²⁷⁾**N.D.E.** What follows has been expanded.

Recall first that Z is a closed group subscheme of G , of finite presentation (Exp. XI 6.11), with connected fibers (Exp. XII 6.6), smooth over S (Exp. XI 2.4), hence satisfies the conditions imposed in definition 5.2.1. Likewise, $H \cap Z$ is of finite presentation, smooth and with connected fibers (for $H \cap Z = \underline{\text{Centr}}_H(Q)$); moreover $H \cap Z \supset \underline{\text{Centr}}_G(T)$.

Proposition 5.2.6. *Let S be a scheme, G an S -group of type (RR) (resp. (RA)), H a subgroup of type (R) of G . Then H is an S -group of type (RR) (resp. (RA)).*

Indeed (i) is clear, (ii) follows from 5.2.2 (b), (iii) and (iv) (resp. (iv')) are to be verified when S is the spectrum of an algebraically closed field. Then H contains a maximal torus T of G (and hence also $C = \underline{\text{Centr}}_G(T)^{(28)}$) and the assertions to be proved follow at once from:

Lemma 5.2.7. *Let S be a scheme, G an S -group of type (RR), T a maximal torus of G endowed with a trivialization $T \simeq \text{D}_S(M)$; assume that*

$$\mathfrak{g} = \mathfrak{g}^0 \oplus \coprod_{\alpha \in R} \mathfrak{g}^\alpha$$

(the $\mathfrak{g}^\alpha$ being then invertible $\mathcal{O}_S$ -modules).

Let H be a subgroup of type (R) containing $C = \underline{\text{Centr}}_G(T)$ (i.e. containing T). Then $\mathfrak{h} = \underline{\text{Lie}}(H/S)$ is locally over S of the form

$$\mathfrak{g}^0 + \coprod_{\alpha \in R'} \mathfrak{g}^\alpha = \mathfrak{g}_{R'};$$

more precisely, let for each $s \in S$, $R'(s) = \{\alpha \in R \mid \mathfrak{g}^\alpha(s) \subset \mathfrak{h}(s)\}$. Then $R'(s)$ is a locally constant function of s ; if U is an open subset of S on which $R'(s) = R'$, one has

$$\mathfrak{h}_U = \mathfrak{g}_U^0 \oplus \coprod_{\alpha \in R'} \mathfrak{g}_U^\alpha.$$

Indeed, $\mathfrak{h}$ is a submodule of $\mathfrak{g}$, locally a direct summand, containing $\mathfrak{g}^0$ and stable under T .

5.3. Strict transporter of two subgroups of type (R). Applications

Recall 5.3.0. ⁽²⁹⁾Let S be a scheme, G a smooth S -group, $\mathfrak{g} = \underline{\text{Lie}}(G/S)$ and $\mathfrak{h}$ an $\mathcal{O}_S$ -submodule of $\mathfrak{g}$ locally a direct summand. The $\mathcal{O}_S$ -algebra $\mathcal{A} = \text{Sym}(\omega_{G/S}^1)$ is locally free, hence the S -scheme $\underline{\text{Lie}}(G/S) = W(\mathfrak{g}) = \text{Spec}(\mathcal{A})$ is essentially free in the sense of Exp. VIII, 6.1. Since $W(\mathfrak{h})$ is a closed subscheme of $\underline{\text{Lie}}(G/S)$, of finite presentation over $\underline{\text{Lie}}(G/S)$, $N = \underline{\text{Norm}}_G(\mathfrak{h})$ is representable by a closed group subscheme of G , of finite presentation over G , by Exp. VIII, 6.5 (a). (See also additions 6.2.3 and 6.2.4 (a) in Exp. VIB.) On the other hand, by Exp. II 5.3.1, one has $\underline{\text{Lie}}(N/S) = \underline{\text{Norm}}_{\underline{\text{Lie}}(G/S)}(\mathfrak{h})$.

Finally, by Exp. VIB 3.10, if N is smooth over S at the points of the identity section, then the subgroup functor N^0 (defined in VIB 3.1) is represented by an open group subscheme of N , smooth over S .

Proposition 5.3.1. *Let S be a scheme, G an S -group of type (RA) (5.1.6), H a subgroup of type (R) of G , $\mathfrak{g} \supset \mathfrak{h}$ their Lie algebras.*

Then $\underline{\text{Norm}}_G(\mathfrak{h})$ (which is representable by a closed subscheme of G of finite presentation over S by 5.3.0) is smooth over S at every point of the identity section and one has

$$\underline{\text{Norm}}_G(\mathfrak{h})^0 = H.$$

⁽²⁸⁾**N.D.E.** By hypothesis, H contains $\underline{\text{Centr}}_G(T')$ for some maximal torus T' of G ; then T and T' are conjugate in H , hence H also contains $C = \underline{\text{Centr}}_G(T)$.

⁽²⁹⁾**N.D.E.** This reminder, which is used in the proof of 5.3.1 and 5.3.4, has been added.

PROOF. ⁽³⁰⁾Put $N = \underline{\text{Norm}}_G(\mathfrak{h})$ and $\mathfrak{n} = \underline{\text{Lie}}(N/S)$. One has $H \subset N$ and, by Exp. II 5.3.1, one has for every $s \in S$

$$\mathfrak{h}(s) \subset \mathfrak{n}(s) = \underline{\text{Norm}}_{\mathfrak{g}(s)}(\mathfrak{h}(s)).$$

Now, by 5.3.2 below, one has $\mathfrak{h}(s) = \underline{\text{Norm}}_{\mathfrak{g}(s)}(\mathfrak{h}(s))$, and since H is smooth over S , one has $\dim_{\kappa(s)} \mathfrak{h}(s) = \dim H_s$ (cf. [DG70], § II.5, Th. 2.1). One therefore obtains

$$\dim_{\kappa(s)} \mathfrak{n}(s) = \dim_{\kappa(s)} \mathfrak{h}(s) = \dim H_s \leq \dim N_s$$

whence $N_s^0 = H_s^0 = H_s$ (H having connected fibers). It follows that the subgroup functor N^0 (defined in VIB, 3.1) is represented by the smooth S -group H . This proves 5.3.1, modulo the following lemma: $\square$

Lemma 5.3.2. *Under the conditions of 5.2.7, if G is of type (RA), one has, for every $s \in S$,*

$$\underline{\text{Norm}}_{\mathfrak{g}(s)}(\mathfrak{h}(s)) = \mathfrak{h}(s).$$

Indeed, one is reduced to the case where S is the spectrum of a field, hence where $\mathfrak{h} = \mathfrak{g}_{R'}$ for some $R' \subset R$. But one already has

$$\underline{\text{Transp}}_{\mathfrak{g}}(\mathfrak{t}, \mathfrak{h}) = \mathfrak{h}.$$

Indeed, if $H \in \mathfrak{t}$ and $X \in \mathfrak{g}^\alpha$, one has $[H, X] = \bar{\alpha}(H)X$, where $\bar{\alpha} : \mathfrak{t} \rightarrow \mathcal{O}_S$ is the derivative morphism of α . Now condition (iv') says precisely that $\bar{\alpha} \neq 0$ for every $\alpha \in R$.

Corollary 5.3.3. *Let S be a scheme, G an S -group of type (RA), H and H' two subgroups of type (R) of G , $\mathfrak{h}$ and $\mathfrak{h}'$ their Lie algebras. Then*

$$H = H' \iff \mathfrak{h} = \mathfrak{h}'.$$

Corollary 5.3.4. *Under the conditions of 5.2.7, with G of type (RA), the maps*

$$H \mapsto \underline{\text{Lie}}(H/S), \quad \mathfrak{h} \mapsto \underline{\text{Norm}}_G(\mathfrak{h})^0$$

establish a bijective correspondence between the set of subgroups of type (R) of G containing T , and the set of Lie subalgebras of $\mathfrak{g}$ containing $\mathfrak{g}^0$, stable under T , and whose normalizer in G is smooth over S at every point of the identity section.

⁽³¹⁾Indeed, let $\mathfrak{h}$ be a Lie subalgebra of $\mathfrak{g}$ satisfying the above properties. By 5.3.0, $H = \underline{\text{Norm}}_G(\mathfrak{h})^0$ is a smooth S -group scheme. Moreover, since $C = \underline{\text{Centr}}_G(T)$ stabilizes each $\mathfrak{g}^\alpha$ and has connected fibers (XII 6.6), one has $C \subset H$. Hence H is a subgroup of G of type (R). By Exp. II 5.3.1, one has $\underline{\text{Lie}}(H) = \underline{\text{Norm}}_{\mathfrak{g}}(\mathfrak{h})$. Finally, by the proof of 5.3.2, one has $\underline{\text{Norm}}_{\mathfrak{g}}(\mathfrak{h}) = \mathfrak{h}$.

Corollary 5.3.5. *Let S be a scheme, G an S -group of type (RR) (5.1.1), T a maximal torus of G , H and H' two subgroups of type (R) of G , containing T . Then*

$$H = H' \iff \mathfrak{h} = \mathfrak{h}'.$$

By the finite presentation hypotheses, one reduces as usual (cf. EGA IV₃, § 8 and Exp. VIB § 10) to the case where S is noetherian; it then suffices to verify that $\mathfrak{h} = \mathfrak{h}'$ implies $H_{S'} = H'_{S'}$ for every S' which is the spectrum of an artinian quotient of a local ring of S ; ⁽³²⁾one is therefore reduced to the case where S is artinian and where one may apply 5.1.8. Let then $u : G \rightarrow G' = G/Z$ be the canonical morphism and $T' = T/Z$ the maximal torus of G' corresponding to T . By Exp. XII 7.12, there exist subgroups of type (R) H_1 and H'_1 of G' ,

⁽³⁰⁾N.D.E. The original has been expanded in what follows.

⁽³¹⁾N.D.E. The following proof has been added.

⁽³²⁾N.D.E. Indeed, let $g \in G$, s its image in S , $\mathfrak{m}$ the maximal ideal of $\mathcal{O}_{G,g}$, $\mathfrak{n}$ that of $\mathcal{O}_{S,s}$, and I (resp. I') the kernel of the morphism from $\mathcal{O}_{G,g}$ to $\mathcal{O}_{H,g}$ (resp. $\mathcal{O}_{H',g}$) (the latter being the zero ring if $g \notin H$, resp. $g \notin H'$). Since $\mathcal{O}_{G,g}$ is noetherian, I and I' are closed for the $\mathfrak{m}$ -adic topology, hence a fortiori for the $\mathfrak{n}$ -adic topology, so it suffices to show that $I + \mathfrak{n}^n \mathcal{O}_{G,g} = I' + \mathfrak{n}^n \mathcal{O}_{G,g}$ for every $n \in \mathbb{N}$.

containing T' , such that $H = u^{-1}(H_1)$ and $H' = u^{-1}(H'_1)$. It suffices to prove that $H_1 = H'_1$. But by 5.2.7 and 5.1.5, one has

$$\underline{\mathrm{Lie}}(H_1) = \underline{\mathrm{Lie}}(H'_1),$$

and one is reduced to 5.3.3.

Remark 5.3.6. The fact that H and H' contain the same maximal torus is essential for the validity of 5.3.5 when G is not of type (RA). Example: maximal tori of $\mathrm{SL}_{2,k}$, for k of characteristic 2.⁽³³⁾

Corollary 5.3.7. *Let S be a scheme, G an S -group of type (RR), T a maximal torus of G , H and H' two subgroups of type (R) of G , containing T . The set U of $s \in S$ such that $H_s = H'_s$ is open and closed in S and $H_U = H'_U$.*

Indeed, this follows at once from 5.3.5 and 5.2.7.

Corollary 5.3.8. *The “functor of subgroups of type (R) containing T ”, where T is a given maximal torus in a group G of type (RR), is formally unramified (Exp. XI 1.1).*

Theorem 5.3.9. *Let G be an S -group of type (RR) (5.1.1), H and H' two subgroups of type (R) (5.2.1). Let $\underline{\mathrm{Transt}}_G(H, H')$ be the strict transporter of H into H' defined by*

$$\underline{\mathrm{Transt}}_G(H, H')(S') = \{g \in G(S') \mid \mathrm{int}(g)H_{S'} = H'_{S'}\}.$$

Then $\underline{\mathrm{Transt}}_G(H, H')$ is representable by a closed subscheme of G , which is smooth and of finite presentation over S .

The fact that $\underline{\mathrm{Transt}}_G(H, H')$ is representable by a closed subscheme of G , of finite presentation over S , follows from Exp. XI 6.11 (a). To prove that it is smooth over S , one must prove that if S is affine and if S_0 is the closed subscheme defined by a nilpotent ideal J , and if $g_0 \in G(S_0)$ and $\mathrm{int}(g_0)H_0 = H'_0$, there exists a $g \in G(S)$ projecting to g_0 and such that $\mathrm{int}(g)H = H'$. Since the question of smoothness is local for the étale topology, one may assume that H contains a maximal torus T of G .

Then T_0 is a maximal torus of H_0 , hence $\mathrm{int}(g_0)T_0$ a maximal torus of H'_0 . By Exp. IX 3.6 bis, there exists a torus T' of H' such that $T'_0 = \mathrm{int}(g_0)T_0$; by Exp. IX 3.3 bis, there therefore exists a $g \in G(S)$ projecting to g_0 and such that $\mathrm{int}(g)T = T'$. Replacing H by $\mathrm{int}(g)H$, one may therefore assume that H and H' contain the same maximal torus T and that $H_0 = H'_0$. But then $H = H'$ by 5.3.7. Q.E.D.

Corollary 5.3.10. *Let G be an S -group of type (RR), H a subgroup of type (R) of G . Then $\underline{\mathrm{Norm}}_G(H)$ is representable by a closed group subscheme of G , of finite presentation and smooth over S .*

Using now the argument which served, in Exp. XI, to deduce 5.4 bis from 5.2 bis, one obtains:

Corollary 5.3.11. *Under the hypotheses of 5.3.9, the following conditions are equivalent:*

- (i) H and H' are conjugate locally in G for the étale topology.
- (i bis) idem for the (fpqc) topology.
- (ii) For every $s \in S$, H_s and H'_s are conjugate by an element of $G(\bar{s})$.
- (ii bis) The structural morphism $\underline{\mathrm{Transt}}_G(H, H') \rightarrow S$ is surjective.
- (iii) $\underline{\mathrm{Transt}}_G(H, H')$ is a principal homogeneous bundle under the action of the smooth S -group scheme of finite presentation $\underline{\mathrm{Norm}}_G(H)$.

Let us simply note that the nontrivial assertion (iii) $\Rightarrow$ (i) is Hensel's lemma.

⁽³³⁾**N.D.E.** Indeed, if k is algebraically closed, all maximal tori of $G = \mathrm{SL}_{2,k}$ are conjugate under $G(k)$, and all have as Lie algebra the line $k \cdot \mathrm{id} \subset M_2(k)$ (which is invariant under the adjoint action).

Using now *Bible*, § 6.4, th. 4 (= [Ch05], § 6.5 th. 5) and § 9.3, th. 1, one obtains by 5.3.10 and 5.3.11:

Corollary 5.3.12. *Let G be an S -group of type (RR) . The Borel subgroups of G are closed in G , equal to their normalizer, and conjugate locally for the étale topology.*

Definition 5.3.13. Let S be a scheme, G a smooth S -group scheme of finite presentation and with connected fibers.⁽³⁴⁾ One calls a *Killing pair* of G a pair $T \subset B$, where T is a maximal torus of G and B a Borel subgroup of G containing T .

Using now the conjugacy of maximal tori in B (cf. 5.1.2 (a) and 5.2.6, for example), one has:

Corollary 5.3.14. *Let G be an S -group of type (RR) . The Killing pairs of G are conjugate locally for the étale topology.*

Corollary 5.3.15. *Let G be an S -group of type (RR) . Let T be a maximal torus of G , $W_G(T) = \underline{\text{Norm}}_G(T)/\underline{\text{Centr}}_G(T)$ the corresponding Weyl group (Exp. XIX 6.3). The “functor of Borel subgroups of G containing T ” is formally principal homogeneous under $W_G(T)$.*

This follows at once from 5.3.14 and the fact that if B is a Borel subgroup of G containing T , one has

$$\underline{\text{Norm}}_G(T) \cap B = \underline{\text{Centr}}_G(T),$$

cf. Exp. XIV 4.4.

Proposition 5.3.16. *Let G be an S -group of type (RR) , H a subgroup of type (R) , $N = \underline{\text{Norm}}_G(H)$ its normalizer (5.3.10). Let T be a maximal torus of H , $W_H(T)$ and $W_N(T)$ the corresponding Weyl groups (étale quasi-finite separated by Exp. XIX 6.3). One has the following exact sequence of sheaves (for the étale topology) (the morphisms are induced by the morphisms $\underline{\text{Norm}}_H(T) \longrightarrow \underline{\text{Norm}}_N(T) \longrightarrow N/H$):*

$$1 \longrightarrow W_H(T) \longrightarrow W_N(T) \longrightarrow N/H \longrightarrow 1.$$

The only nontrivial point is the fact that the last arrow is an epimorphism. Let therefore $n \in N(S')$, $S' \longrightarrow S$. The two maximal tori T and $\text{int}(n)T$ of H are conjugate in H locally for the étale topology. There therefore exist a covering family $\{S'_i \longrightarrow S'\}$ and for each i an $h_i \in H(S'_i)$ such that $\text{int}(h_i)T = \text{int}(n)T$. One therefore has $nh_i^{-1} \in \underline{\text{Norm}}_N(T)$, which implies the desired result.

Remark 5.3.17. One may describe $W_N(T)$ as follows: assume we have reduced to the situation of 5.2.7, with $\mathfrak{h} = \mathfrak{g}_{R'}$. Then $W_N(T)$ equals $\underline{\text{Norm}}_W(R')$, the sheaf of sections of $W = W_G(T)$ which, acting on R , normalize R' . Indeed, by 5.3.5, one has

$$\underline{\text{Norm}}_N(T) = \underline{\text{Norm}}_G(H) \cap \underline{\text{Norm}}_G(T) = \underline{\text{Norm}}_G(\mathfrak{h}) \cap \underline{\text{Norm}}_G(T).$$

Corollary 5.3.18. *Let G be an S -group of type (RR) , H a subgroup of type (R) . Assume “the Weyl groups of G finite”, i.e. that for every $S' \longrightarrow S$ and every maximal torus T of $G_{S'}$, the étale S' -scheme $\underline{\text{Norm}}_{G_{S'}}(T)/\underline{\text{Centr}}_{G_{S'}}(T)$ is finite (cf. Exp. XIX 6.3 (iii)). The following conditions are equivalent:*

- (i) H is closed in G .
- (ii) $\underline{\text{Norm}}_G(H)/H$ is representable by a finite étale S -scheme.
- (iii) “the Weyl groups of H are finite”.

Indeed, one may assume that H possesses a maximal torus T . By 5.3.10, $N = \underline{\text{Norm}}_G(H)$ is closed in G , hence $W_N(T)$ is closed in $W_G(T)$ and therefore finite over S . One clearly has

⁽³⁴⁾**N.D.E.** cf. N.D.E. (25).

(i) $\Rightarrow$ (iii), and one has (iii) $\Rightarrow$ (ii) by the exact sequence of 5.3.16. Finally, one has (ii) $\Rightarrow$ (i) for if N/H is finite, it is separated, hence H is closed in N , hence in G .

Remark 5.3.19. When G is reductive, the preceding conditions on H seem always to hold. We prove them below in most cases.

5.4. Subgroups of type (R) of a split reductive group (generalities)

5.4.1. — If H is a subgroup of type (R) of the reductive group G , then H contains a maximal torus of G locally for the étale topology (5.2.2). By 2.3, one may, locally for the étale topology, assume G split with respect to this torus. Let therefore (G, T, M, R) be a split S -group, H a subgroup of type (R) of G containing T . By 5.3.5 such a subgroup is characterized by its Lie algebra, which (5.2.7) is locally over S of the form $\mathfrak{g}_{R'}$:

$$\mathfrak{g}_{R'} = \mathfrak{t} \oplus \coprod_{\alpha \in R'} \mathfrak{g}^\alpha.$$

Definition 5.4.2. Let (G, T, M, R) be a split S -group. One will say that the subset R' of R is of type (R) if $\mathfrak{g}_{R'}$ is the Lie algebra of a subgroup of type (R) of G containing T . This subgroup, uniquely determined by R' , is denoted $H_{R'}$.

Lemma 5.4.3. Under the preceding conditions, one has the following equivalences:

$$\begin{aligned} H \cap Z_\alpha = T &\iff \alpha \notin R', \quad -\alpha \notin R', \\ H \supset U_\alpha &\iff \alpha \in R', \\ H \cap U_\alpha = e &\iff \alpha \notin R', \\ H \supset Z_\alpha &\iff \alpha \in R', \quad -\alpha \in R'. \end{aligned}$$

Indeed, $H \cap Z_\alpha$ is a subgroup of type (R) of Z_α , by 5.2.5; but a subgroup of type (R) of Z_α , containing T , is locally equal to one of the following subgroups: T , $T \cdot U_\alpha$, $T \cdot U_{-\alpha}$, Z_α , by 5.3.5.

Lemma 5.4.4. Under the conditions of 5.4.2, let R_+ be a system of positive roots; choose orders on $R' \cap R_+$ and $R' \cap -R_+$. The morphism

$$\Omega_{R_+, R'} = \prod_{\alpha \in R' \cap -R_+} U_\alpha \times_S T \times_S \prod_{\alpha \in R' \cap R_+} U_\alpha \longrightarrow G$$

induced by the product in G induces an open immersion

$$\Omega_{R_+, R'} \longrightarrow H_{R'}.$$

Indeed, by 5.4.3, this morphism factors through $H_{R'}$ and therefore induces an immersion $\Omega_{R_+, R'} \longrightarrow H_{R'}$. One then argues as in 4.1.1.

Proposition 5.4.5. Let (G, T, M, R) be a split S -group. Let R' and R'' be two subsets of R , of type (R).

(i) $H_{R'} \cap H_{R''}$ is smooth at every point of the identity section, $R' \cap R''$ is of type (R), and one has

$$(H_{R'} \cap H_{R''})^0 = H_{R' \cap R''}.$$

(ii) One has the equivalence

$$H_{R'} \subset H_{R''} \iff R' \subset R''.$$

Indeed, (ii) follows at once from (i). To prove (i), it suffices to prove that $H_{R'} \cap H_{R''}$ is smooth at every point of the identity section: its identity component (cf. Exp. VIB 3.10) will then be a subgroup of type (R) containing T , hence equal to $H_{R' \cap R''}$; but $\Omega_{R_+, R'} \cap \Omega_{R_+, R''} = \Omega_{R_+, R' \cap R''}$ is an open subset of $H_{R'} \cap H_{R''}$ containing the identity section and smooth over S .

Corollary 5.4.6. *Let S be a scheme, G a reductive S -group, T a maximal torus of G , s a point of S . If H and H' are two subgroups of type (R) of G containing T such that $H_s \subset H'_s$, there exists an open subset U of S containing s such that $H_U \subset H'_U$.*

One may indeed assume G split with respect to T . The assertion then follows at once from 5.4.5 (ii).

One is led to ask which are the subsets R' of type (R) of R . One may assume the group adjoint; one must then verify that $\mathfrak{g}_{R'}$ is a Lie algebra and that its normalizer is smooth at every point of the identity section. The most important case is given by the

Theorem 5.4.7. *Every closed subset R' of R is of type (R). (Recall, cf. Exp. XXI 3.1.4, that $R' \subset R$ is said to be closed if $\alpha, \beta \in R'$, $\alpha + \beta \in R$ implies $\alpha + \beta \in R'$).*

Remark 5.4.8. We will see later (Exp. XXIII 6.6) that if $6 \cdot 1_S \neq 0^{(35)}$ (for example, if S possesses a residue characteristic distinct from 2 and 3), the fact that $\mathfrak{g}_{R'}$ is a Lie algebra already implies that R' is closed, hence R' is of type (R) if and only if it is closed. Theorem 5.4.7 therefore gives all subsets of type (R) “independent of the characteristic”.

Let us first prove:

Lemma 5.4.9. *Choose for each $\alpha \in R$ an $X_\alpha \in \Gamma(S, \mathfrak{g}^\alpha)^\times$. Let $\alpha, \beta \in R$, with $\alpha + \beta \neq 0$, and let q be the largest integer i such that $\alpha + i\beta \in R$. There exist uniquely determined sections $M_{\alpha, \beta, i} \in \Gamma(S, \mathcal{O}_S)$ such that*

$$\mathrm{Ad}(\exp_\alpha(xX_\alpha))(X_\beta) = X_\beta + \sum_{i=1}^q M_{\alpha, \beta, i} x^i X_{\beta+i\alpha},$$

for every $x \in \mathbf{G}_a(S')$, $S' \rightarrow S$.

Indeed, $x \mapsto \mathrm{Ad}(\exp_\alpha(xX_\alpha))(X_\beta)$ defines a morphism $\mathbf{G}_{a,S} \rightarrow \mathrm{W}(\mathfrak{g}) \simeq \mathbf{G}_{a,S}^m$. There therefore exist uniquely determined sections $Y_n \in \Gamma(S, \mathfrak{g})$ such that

$$\mathrm{Ad}(\exp_\alpha(xX_\alpha))(X_\beta) = \sum_{n \geq 0} x^n Y_n.$$

Applying the inner automorphism defined by a section t of T , one finds at once

$$\mathrm{Ad}(t)(Y_n) = \beta(t)\alpha(t)^n Y_n,$$

which implies $Y_n \in \Gamma(S, \mathfrak{g}^{\beta+n\alpha})$. Since α and β are not proportional, none of the $\beta + n\alpha$ is zero; one therefore has $Y_n = 0$ for $n > q$, $Y_n = M_{\alpha, \beta, n} X_{\beta+n\alpha}$ for $0 \leq n \leq q$, where $M_{\alpha, \beta, n} \in \mathbf{G}_a(S)$ is uniquely determined. Putting $x = 0$ in the formula obtained, one finds $Y_0 = X_\beta$, which completes the proof.

Remark 5.4.10. Differentiating the preceding formula at $x = 0$, one finds

$$[X_\alpha, X_\beta] = \begin{cases} N_{\alpha, \beta} X_{\alpha+\beta}, & \text{where } N_{\alpha, \beta} = M_{\alpha, \beta, 1}, & \text{if } \alpha + \beta \in R, \\ 0 & & \text{if } \alpha + \beta \notin R, \quad \alpha + \beta \neq 0. \end{cases}$$

Let us now prove 5.4.7. If R' is a closed subset of R , then $\mathfrak{g}_{R'}$ is a Lie subalgebra of $\mathfrak{g}$, by 5.4.10 and Exp. XX 2.10, formula (3). By 5.4.9 and Exp. XX 2.10, formula (2), U_α normalizes $\mathfrak{g}_{R'}$ for each $\alpha \in R'$. Choose a system of positive roots R_+ and consider the open subset Ω_{R_+} of 4.1.2; let $\Omega_{R_+, R'}$ be the closed subscheme of Ω_{R_+} defined as follows:

$$\Omega_{R_+, R'} = \left(\prod_{\alpha \in R' \cap -R_+} U_\alpha \right) \cdot T \cdot \left(\prod_{\alpha \in R' \cap R_+} U_\alpha \right).$$

The canonical immersion $\Omega_{R_+, R'} \rightarrow G$ factors through $i : \Omega_{R_+, R'} \rightarrow \underline{\mathrm{Norm}}_G(\mathfrak{g}_{R'})$. Assume G adjoint; the tangent map to i at the points of the identity section is bijective by 5.3.2; in

⁽³⁵⁾**N.D.E.** In fact, it suffices (cf. *loc. cit.*) that 2 and 3 be nonzero on S .

particular, the morphism i is étale at every point of the identity section, hence is a local immersion⁽³⁶⁾ at every point of the identity section, hence $\underline{\text{Norm}}_G(\mathfrak{g}_{R'})$ is smooth at every point of the identity section, as was to be proved.

5.5. Borel subgroups of a split reductive group

Proposition 5.5.1. *Let (G, T, M, R) be a split S -group. For every system of positive roots R_+ of R , H_{R_+} (which exists by 5.4.7) is a Borel subgroup of G and, for every order on R_+ , the morphism induced by the product in G*

$$T \times_S \prod_{\alpha \in R_+} U_\alpha \longrightarrow G$$

is a closed immersion with image H_{R_+} . One writes $B_{R_+} = H_{R_+}$.

By definition of Borel subgroups, the first assertion may be verified on replacing S by the spectrum of an algebraically closed field. Let then B be the Borel subgroup of G containing T and corresponding to the system of positive roots R_+ (Bible, § 10.4, prop. 9); the Lie algebra of B is $\mathfrak{g}_{R_+}$; one therefore has $B = H_{R_+}$ by 5.3.5.

Let us prove the second assertion: the morphism of the statement induces an open immersion $i : T \times_S \prod_{\alpha \in R_+} U_\alpha \longrightarrow H_{R_+}$ (5.4.4). Now i is surjective (Bible, § 15.1, cor. 1 to prop. 1).

Corollary 5.5.2. *Choose any order on R_+ and for each $\alpha \in R_+$ an $X_\alpha \in \Gamma(S, \mathfrak{g}^\alpha)^\times$. Let $\alpha, \beta \in R_+$. For each pair $(i, j) \in \mathbf{N}^* \times \mathbf{N}^*$ such that $i\alpha + j\beta \in R$, there exists a unique section*

$$C_{i,j,\alpha,\beta} \in \Gamma(S, \mathcal{O}_S)$$

such that, for all $x, y \in \mathbf{G}_a(S')$, $S' \longrightarrow S$, one has

$$\begin{aligned} & \exp_\alpha(xX_\alpha) \exp_\beta(yX_\beta) \exp_\alpha(xX_\alpha)^{-1} = \\ & \exp_\beta(yX_\beta) \prod_{\substack{i,j \in \mathbf{N}^* \\ i\alpha + j\beta \in R}} \exp_{i\alpha + j\beta}(C_{i,j,\alpha,\beta} x^i y^j X_{i\alpha + j\beta}). \end{aligned}$$

If $\alpha = \beta$, the assertion is trivial. Assume therefore $\alpha \neq \beta$; then, by the proposition, there exist unique morphisms

$$F_0 : \mathbf{G}_{a,S}^2 \longrightarrow T, \quad F_\gamma : \mathbf{G}_{a,S}^2 \longrightarrow \mathbf{G}_{a,S} \quad (\gamma \in R_+)$$

such that one has

$$\exp(xX_\alpha) \exp(yX_\beta) \exp(xX_\alpha)^{-1} = F_0(x, y) \prod_{\gamma \in R_+} \exp(F_\gamma(x, y)X_\gamma).$$

Let $t \in T(S')$, $S' \longrightarrow S$. Apply $\text{int}(t)$ to this formula; one has at once the relations

$$(1) \quad F_0(\alpha(t)x, \beta(t)y) = F_0(x, y),$$

$$(2) \quad F_\gamma(\alpha(t)x, \beta(t)y) = \gamma(t)F_\gamma(x, y).$$

Since α and β are two linearly independent characters (over $\mathbf{Q}$) of T , one concludes as usual from the first relation that F_0 is constant, hence $F_0(x, y) = e$. Write next

$$F_\gamma(x, y) = \sum a_{ij} x^i y^j, \quad \text{with } a_{ij} \in \Gamma(S, \mathcal{O}_S).$$

Substituting in relation (2) and identifying the polynomials on both sides, one finds

$$a_{ij}(\alpha(t)^i \beta(t)^j - \gamma(t)) = 0.$$

⁽³⁶⁾**N.D.E.** By EGA IV₄, 17.11.2, i is étale at every point of the identity section (and $\underline{\text{Norm}}_G(\mathfrak{g}_{R'})$ is smooth at every point of the identity section). Moreover, let V be the largest open subset of $\Omega_{R_+, R'}$ on which i is étale; since i is a monomorphism, i_V is an open immersion (ibid., 17.9.1).

If $\gamma \neq i\alpha + j\beta$, one knows (Exp. XIX 4.13) that there exist a faithfully flat quasi-compact $S' \rightarrow S$ and a $t \in T(S')$ such that $\alpha(t)^i \beta(t)^j - \gamma(t) = 1$. One therefore has $a_{ij} = 0$ on S' , hence on S . If $\gamma = i\alpha + j\beta$, one puts $a_{ij} = C_{i,j,\alpha,\beta}$. Putting $x = 0$ (resp. $y = 0$), one finds $C_{0,1,\alpha,\beta} = 1$ (resp. $C_{1,0,\alpha,\beta} = 0$).

Remark 5.5.3. Differentiating at $y = 0$ and comparing with 5.4.9, one finds

$$C_{i,1,\alpha,\beta} = M_{\alpha,\beta,i}.$$

Corollary 5.5.4. *Let S be a scheme, G a reductive S -group, T a maximal torus of G , $\alpha \neq \beta$ two roots of G with respect to T such that $\alpha + \beta$ is nontrivial on each fiber. Order the set of $i\alpha + j\beta$ ($i, j \in \mathbf{N}^*$) in any way. For all $i, j \in \mathbf{N}^*$ such that $i\alpha + j\beta \in R$, there exists a unique morphism of $\mathcal{O}_S$ -modules*

$$f_{\alpha,\beta,i,j} : (\mathfrak{g}^\alpha)^{\otimes i} \otimes (\mathfrak{g}^\beta)^{\otimes j} \longrightarrow \mathfrak{g}^{i\alpha+j\beta}$$

such that for every $S' \rightarrow S$ and all $X \in W(\mathfrak{g}^\alpha)(S')$, $Y \in W(\mathfrak{g}^\beta)(S')$ one has (the exp on the right-hand side being taken in the sense of 1.2⁽³⁷⁾):

$$\begin{aligned} \exp_\alpha(X) \exp_\beta(Y) \exp_\alpha(-X) = \\ \exp_\beta(Y) \prod_{(i,j)} \exp_{i\alpha+j\beta}(f_{\alpha,\beta,i,j}(X^i \otimes Y^j)). \end{aligned}$$

The assertion is local for (fpqc). By § 2, one may therefore assume G split relative to T , α and β constant in the splitting. Since $\alpha + \beta \neq 0$, there exists a system of positive roots R_+ containing α and β (Exp. XXI 3.5.4) and one is reduced to 5.5.2.

Corollary 5.5.5. *Let S be a scheme, G a reductive S -group.*

- (i) *G possesses Borel subgroups locally for the étale topology. If T is a maximal torus of G , then G also possesses Borel subgroups containing T locally for the étale topology.*
- (ii) *If T is a maximal torus of G , the “functor of Borel subgroups of G containing T ” is representable by a principal homogeneous bundle under $W_G(T)$.*
- (iii) *If (G, T, M, R) is split, every Borel subgroup B of G containing T is locally over S of the form B_{R_+} , where R_+ is a system of positive roots of R .*
- (iv) *If $T \subset B$ is a Killing pair of G , there exists a covering family $\{S_i \rightarrow S\}$ for the étale topology, and for each i a splitting $(G_{S_i}, T_{S_i}, M_i, R_i)$ and a system of positive roots R_{i+} of R_i such that $B_{S_i} = B_{R_{i+}}$.*

Indeed, (i) follows from 2.3 and 5.5.1, (ii) from (i) and 5.3.15, (iii) from (ii) and 5.5.1, (iv) from (iii) and 2.3.

Lemma 5.5.6. *Choose on the group $\Gamma_0(R)$ generated by the roots a totally ordered group structure such that the roots > 0 are the elements of R_+ (cf. Exp. XXI 3.5.6)⁽³⁸⁾. Let $\alpha_1 < \dots < \alpha_N$ be the elements of R_+ . Consider the isomorphism*

$$f : T \times_S U_{\alpha_1} \times_S \dots \times_S U_{\alpha_N} \longrightarrow B_{R_+}$$

induced by the product in G . Put for $i = 1, \dots, N$,

$$U_{\geq i} = f(U_{\alpha_i} \times_S \dots \times_S U_{\alpha_N}).$$

- (i) *Each $U_{\geq i}$ is an invariant subgroup of B_{R_+} .*
- (ii) *For $1 \leq i \leq N-1$, $U_{\geq i}$ is identified with the semidirect product*

$$U_{\geq i} = U_{\alpha_i} \cdot U_{\geq i+1}.$$

⁽³⁷⁾**N.D.E.** That is, if $\mathfrak{g}^{i\alpha+j\beta} = 0$ on a connected component of S , the corresponding exponential is 1 on this component.

⁽³⁸⁾**N.D.E.** Note that such an order is necessarily compatible with ord_Δ , where $\Delta = \mathcal{S}(R_+)$ (cf. XXI 3.2.15).

(iii) B_{R_+} is identified with the semidirect product

$$B_{R_+} = T \cdot U_{\geq 1}.$$

(iv) For $1 \leq i \leq N-1$, the inner automorphisms of $U_{\geq 1}$ act trivially on $U_{\geq i}/U_{\geq i+1}$ (which is identified with U_{α_i} by (ii)).

Let us first prove by induction on i the following assertion:

$U_{\geq i}$ is an invariant subgroup of B_{R_+} , a semidirect product of U_{α_i} and $U_{\geq i+1}$.

The assertion is true for $i = N$; assume it true for $i+1$ and let us prove it for i . One has (as schemes)

$$U_{\geq i} = U_{\alpha_i} \cdot U_{\geq i+1};$$

it is first clear that $U_{\geq i}$ is stable under the inner automorphisms of B_{R_+} . This is clear for $\text{int}(t)$, $t \in T(S)$; it suffices to verify it for $\text{int}(x)$, $x \in U_{\alpha}(S)$, $\alpha \in R_+$. Now $U_{\geq i+1}$ is assumed invariant, so it suffices to see that $\text{int}(x)U_{\alpha_i} \subset U_{\geq i}$. By 5.5.2, if $y \in U_{\alpha_i}(S')$, one has $y^{-1}xyx^{-1} \in U_{\geq i+1}(S')$, which implies $\text{int}(x)(y) \in U_{\geq i}(S')$.

Let us now prove that $U_{\geq i}$ is a subgroup of B_{R_+} . If $x, y \in U_{\geq i}(S)$, one may write $x = px'$, $y = qy'$, with $p, q \in U_{\alpha_i}(S)$, and $x', y' \in U_{\geq i+1}(S)$. One has

$$xy = px'qy' = pq(q^{-1}x'q)y' \in U_{\alpha_i}(S')U_{\geq i+1}(S');$$

likewise $x^{-1} = p^{-1}(px'^{-1}p^{-1}) \in U_{\alpha_i}(S')U_{\geq i+1}(S')$. We have therefore proved (i) and (ii), as well as (iv) along the way. As for (iii), it is a trivial consequence of 5.5.1.

Lemma 5.5.7. *With the preceding notations, choose for each $1 \leq i \leq N$ an $X_i \in \Gamma(S, \mathfrak{g}^{\alpha_i})^\times$ and consider the isomorphism*

$$a : \mathbf{G}_{a,S}^N \longrightarrow U_{\geq 1}$$

defined set-theoretically by

$$a(x_1, \dots, x_N) = \exp_{\alpha_1}(x_1 X_1) \cdots \exp_{\alpha_N}(x_N X_N).$$

There exists a unique family (Q_i) , $i = 1, \dots, N$, of polynomials

$$Q_i = Q_i(x_1, \dots, x_N, y_1, \dots, y_N)$$

with coefficients in $\Gamma(S, \mathcal{O}_S)$ such that one has set-theoretically

$$a(x_1, \dots, x_N) a(y_1, \dots, y_N) = a(Q_1(x_1, \dots, y_N), \dots, Q_N(x_1, \dots, y_N)).$$

Moreover, the Q_i have coefficients in the subring of $\Gamma(S, \mathcal{O}_S)$ generated by the $C_{i,j,\alpha,\beta}$ of 5.5.2 ($\alpha, \beta \in R_+$, $i, j \in \mathbf{N}^*$) and each Q_i has the form

$$Q_i(x_1, \dots, y_N) = x_i + y_i + Q'_i(x_1, \dots, x_{i-1}, y_1, \dots, y_{i-1}).$$

The existence and uniqueness of the Q_i follow at once from the fact that a is an isomorphism of schemes. Denoting by z, z', z'' sections of $U_{\geq i+1}$, one has

$$\begin{aligned} a(x_1, \dots, x_N) a(y_1, \dots, y_N) &= a(x_1, \dots, x_{i-1}, 0, \dots, 0) \exp(x_i X_i) z \\ &\quad \cdot a(y_1, \dots, y_{i-1}, 0, \dots, 0) \exp(y_i X_i) z'; \end{aligned}$$

using 5.5.6 (i) and (iv), the right-hand term is written

$$\begin{aligned} &a(x_1, \dots, x_{i-1}, 0, \dots, 0) a(y_1, \dots, y_{i-1}, 0, \dots, 0) \\ &\quad \cdot \exp((x_i + y_i) X_i) z''; \end{aligned}$$

which gives, using 5.5.6 again,

$$Q_i(x_1, \dots, x_N, y_1, \dots, y_N) = x_i + y_i + Q'_i(x_1, \dots, x_{i-1}, y_1, \dots, y_{i-1}),$$

with

$$\begin{aligned} &Q'_i(x_1, \dots, x_{i-1}, y_1, \dots, y_{i-1}) \\ &= Q_i(x_i, \dots, x_{i-1}, 0, \dots, 0, y_1, \dots, y_{i-1}, \dots, 0); \end{aligned}$$

that is, the precise form requested.

Let us finally prove the assertion on the coefficients of the polynomials Q_i . Let A be the subring of $\Gamma(S, \mathcal{O}_S)$ generated by the $C_{i,j,\alpha,\beta}$ ($\alpha, \beta \in R_+$, $i, j \in \mathbf{N}^*$). Let us prove by descending induction on i that if $x_1 = \dots = x_{i-1} = 0$ and $y_1 = \dots = y_{i-1} = 0$, that is, if $a(x_1, \dots, x_N)$ and $a(y_1, \dots, y_N)$ are sections of $U_{\geq i}$, then the polynomials

$$R_j(x_i, \dots, x_N, y_i, \dots, y_N) = Q_j(x_1, \dots, x_N, y_1, \dots, y_N),$$

have coefficients in A . This is trivial for $i = N$ and also for $j < i$ (for $R_j = 0$ for $j < i$). Let $i < N$, assume the assertion holds for $i + 1$ and let us prove it for i (and $j \geq i$). One has

$$\begin{aligned} a(0, \dots, 0, x_i, \dots, x_N) &= \exp(x_i X_i) a(0, \dots, 0, x_{i+1}, \dots, x_N) \\ &= \exp(x_i X_i) Z_i. \end{aligned}$$

Write likewise

$$a(0, \dots, y_i, \dots, y_N) = \exp(y_i X_i) T_i.$$

One has

$$\begin{aligned} a(0, \dots, x_i, \dots, x_N) a(0, \dots, y_i, \dots, y_N) \\ = \exp((x_i + y_i) X_i) \text{int}(\exp(-y_i X_i))(Z_i) \cdot T_i. \end{aligned}$$

Now

$$\begin{aligned} &\text{int}(\exp(-y_i X_i))(Z_i) \\ &= \text{int}(\exp(-y_i X_i))(\exp(x_{i+1} X_{i+1}) \cdots \exp(x_N X_N)) \end{aligned}$$

is a product of $N - i - 1$ sections of $U_{\geq i+1}$ whose coefficients in the decomposition $U_{\geq i+1} = U_{\alpha_{i+1}} \cdots U_{\alpha_N}$ are polynomials in y_i and $x_{i+1}, \dots, x_N$ with coefficients in A (by 5.5.2). Applying the induction hypothesis, one deduces that the coefficients of

$$\text{int}(\exp(-y_i X_i))(Z_i) \cdot T_i$$

are also polynomials with coefficients in A , which completes the proof.

Note that the preceding induction gives at once a proof of the

Lemma 5.5.8. *With the notations of 5.5.6, let for each $i = 1, \dots, N$ a morphism of groups*

$$f_i : U_{\alpha_i} \longrightarrow H,$$

where H is an S -group functor. For the morphism

$$f : U_{\geq 1} \longrightarrow H$$

defined by

$$f(\exp(x_1 X_1) \cdots \exp(x_N X_N)) = f_1(\exp(x_1 X_1)) \cdots f_N(\exp(x_N X_N))$$

to be a morphism of groups, it is necessary and sufficient that for every pair $i < j$, one have

$$\begin{aligned} f_j(\exp(x_j X_j)) f_i(\exp(x_i X_i)) f_j(\exp(-x_j X_j)) = \\ f(\exp(x_j X_j) \exp(x_i X_i) \exp(-x_j X_j)). \end{aligned}$$

5.6. Subgroups of type (R) with solvable fibers

Proposition 5.6.1. *Let (G, T, M, R) be a split S -group, R' a subset of R of type (R) (5.4.2), $H_{R'}$ the corresponding subgroup of G . The following conditions are equivalent:*

- (i) $H_{R'}$ has solvable geometric fibers.
- (ii) There exists a system of positive roots R_+ such that $R' \subset R_+$, hence $H_{R'} \subset B_{R_+}$ (cf. 5.4.5).
- (iii) $R' \cap -R' = \emptyset$.
- (iv) For every order on R' , the morphism induced by the product in G

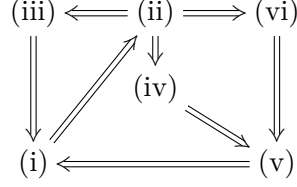
$$T \times_S \prod_{\alpha \in R'} U_{\alpha} \longrightarrow H_{R'}$$

is an isomorphism.

- (v) $H_{R'} \cap \underline{\text{Norm}}_G(T) = T$.
 (vi) For every subset R'' of R , of type (R), one has (cf. 5.4.5)

$$H_{R'} \cap \underline{\text{Norm}}_G(H_{R''}) = H_{R' \cap R''}.$$

We are going to prove these equivalences according to the logical diagram



One clearly has $(ii) \Rightarrow (iii)$ and $(vi) \Rightarrow (v)$ (take $R'' = \emptyset$). By 5.4.6, it suffices to verify $(i) \Rightarrow (ii)$ on the geometric fibers; now if S is the spectrum of an algebraically closed field, $H_{R'}$ is contained in a Borel group containing T , hence of the form H_{R_+} (5.5.5 (iii)).

Likewise $(iii) \Rightarrow (i)$ is verified on the geometric fibers; assume (iii) satisfied; if $H_{R'}$ were not solvable, there would exist a subtorus Q of T , of codimension 1 in T , such that $\underline{\text{Centr}}_{H_{R'}}(Q)$ is not solvable (*Bible*, § 10.4, prop. 8); now $\underline{\text{Centr}}_{H_{R'}}(Q)$ has as Lie algebra $\mathfrak{g}_{R''}$, where R'' is the set of roots of R' vanishing on Q , hence $R'' = \emptyset$ or $\{\alpha\}$ (by (iii)); hence $\underline{\text{Centr}}_{H_{R'}}(Q)$, which is a subgroup of type (R) of G , is T or $T \cdot U_\alpha$, hence solvable contrary to the hypothesis.

Likewise $(ii) \Rightarrow (iv)$ is verified on the geometric fibers (for these are flat S -schemes of finite presentation); by *Bible*, § 13.2, th. 1 d), the morphism in question is bijective; it induces an isomorphism on the tangent spaces at the origin, and one concludes as usual (cf. 4.1.1).

One has $(iv) \Rightarrow (v)$ by 4.2.7. To prove $(v) \Rightarrow (i)$, one is again reduced to the case where S is the spectrum of an algebraically closed field and one concludes by *Bible*, § 10.3, cor. to prop. 6 and § 9.3, cor. 3 to th. 1. It therefore only remains to prove assertion $(ii) \Rightarrow (vi)$. One may reduce to the case where G is adjoint. One then has, by 5.3.3

$$\underline{\text{Norm}}_G(H_{R''}) = \underline{\text{Norm}}_G(\mathfrak{g}_{R''}) \subset \underline{\text{Transp}}_G(\mathfrak{t}, \mathfrak{g}_{R''}).$$

⁽³⁹⁾By 5.4.5, it suffices to prove

$$(x) \quad H_{R'}(S) \cap \underline{\text{Transp}}_{G(S)}(\mathfrak{t}, \mathfrak{g}_{R''}) \subset H_{R' \cap R''}(S).$$

Let us first prove a lemma.

Lemma 5.6.2. *In the notations of 5.5.7, let*

$$u = \exp(x_1 X_1) \cdots \exp(x_N X_N)$$

where $x_i \in \mathbf{G}_a(S)$. Let m be an integer, $1 \leq m \leq N$, such that $x_i = 0$ for $i < m$.

a) If $H \in \Gamma(S, \mathfrak{t})$, the component of $\text{Ad}(u)H$ on $\mathfrak{g}^{\alpha_m}$ is

$$-\bar{\alpha}_m(H)x_m X_m.$$

b) If $Y \in \Gamma(S, \mathfrak{g}^{-\alpha_m})$, the component of $\text{Ad}(u)Y$ on $\mathfrak{t}$ is (with the notations of Exp. XX 2.6)

$$x_m \langle X_m, Y \rangle H_{\alpha_m}.$$

Denote indeed by $\mathfrak{u} = \mathfrak{g}^{\alpha_{m+1}} + \cdots + \mathfrak{g}^{\alpha_N}$. By 5.4.9, one has

$$\text{Ad}(\exp(x_i X_i))\mathfrak{u} \subset \mathfrak{u}, \quad \text{for } i > m.$$

By Exp. XX 2.10, one has

$$\text{Ad}(\exp(x_i X_i))H = H - \bar{\alpha}_i(H)x_i X_i.$$

This gives at once, modulo $\mathfrak{u}$,

$$\text{Ad}(u)H = \text{Ad}(\exp(x_m X_m))H = H - \bar{\alpha}_m(H)x_m X_m,$$

which implies the first result.

⁽³⁹⁾**N.D.E.** The inclusion follows from $\mathfrak{t} \subset \mathfrak{g}_{R''}$.

Likewise, denote⁽⁴⁰⁾ by $\mathfrak{n} = \mathfrak{g}^{\alpha_1} + \cdots + \mathfrak{g}^{\alpha_N}$ and $u_1 = \exp(x_{m+1}X_{m+1}) \cdots \exp(x_N X_N)$. For $i > m$, one has $\alpha_i > \alpha_m$, hence, by 5.4.9, one has, modulo $\mathfrak{n}$,

$$\mathrm{Ad}(u_1)Y \equiv Y \quad \text{whence} \quad \mathrm{Ad}(u)Y \equiv \mathrm{Ad}(\exp(x_m X_m))Y.$$

Applying Exp. XX 2.10, one therefore obtains, modulo $\mathfrak{n}$,

$$\mathrm{Ad}(u)Y - Y \equiv x_m \langle X_m, Y \rangle H_{\alpha_m}$$

whence the second result.

Let us return to the proof of inclusion (x). Assume there exists $h \in H_{R'}(S)$, $h \notin H_{R' \cap R''}(S)$, such that

$$\mathrm{Ad}(h)\mathfrak{t} \subset \mathfrak{g}_{R''}.$$

One may write

$$h = t \exp(x_1 X_1) \cdots \exp(x_N X_N).$$

Since $h \notin H_{R' \cap R''}(S)$, there exists a smallest m such that

$$\begin{aligned} t \exp(x_1 X_1) \cdots \exp(x_{m-1} X_{m-1}) &\in H_{R' \cap R''}(S) \\ \text{and } \alpha_m &\notin R'', \quad x_m \neq 0. \end{aligned}$$

Then $h' = \exp(x_m X_m) \cdots \exp(x_N X_N)$ also satisfies the conditions imposed on h above. But by 5.6.2, for every $H \in \Gamma(S, \mathfrak{t})$ the component of $\mathrm{Ad}(h')H$ on $\mathfrak{g}^{\alpha_m}$ is $-\bar{\alpha}_m(X)x_m X_m$. By the hypothesis on h and on m , one therefore has $\bar{\alpha}_m(H) = 0$ for every $H \in \Gamma(S, \mathfrak{t})$, which is impossible (for G is assumed adjoint and $\bar{\alpha}_m$ is therefore nonzero on each fiber).

Remark 5.6.2 bis. Let us return to the notations of 5.6.2. If $\mathrm{Ad}(u)$ is the identity on $\mathfrak{t}$ and on $\mathfrak{g}^{-\alpha_m}$, one has $x_m = 0$. Indeed, one has $x_m \bar{\alpha}_m = 0$ and $x_m H_{\alpha_m} = 0$; if $\alpha_m \notin 2M$, then $\bar{\alpha}_m$ is nonzero on each fiber; if $\alpha_m \in 2M$, then $\alpha_m^* \notin 2M^*$ and H_{α_m} is nonzero on each fiber; in each case, this implies $x_m = 0$. It follows that $u = e$ if $\mathrm{Ad}(u)$ acts trivially on $\mathfrak{g}$.

Remark 5.6.3. If H is a subgroup of type (R) of the reductive group G , with solvable geometric fibers, then H is closed in G and $\mathrm{Norm}_G(H)/H$ is representable by a separated finite étale S -scheme.

This follows from 5.3.18 and, as one chooses, 3.5 or Exp. XIX 2.5 (ii).

Corollary 5.6.4. *Let (G, T, M, R) be a split reductive group. If $R' \subset R$ is closed and if $R' \cap -R' = \emptyset$, then R' is contained in a system of positive roots.⁽⁴¹⁾*

Indeed, R' is of type (R) by 5.4.7, hence the result follows from 5.6.1.

Corollary 5.6.5. *Under the conditions of 5.6.1, the product in G induces an isomorphism*

$$\prod_{\alpha \in R'} U_{\alpha} \xrightarrow{\sim} U_{R'},$$

where $U_{R'}$ is a closed group subscheme of G , smooth over S , with connected unipotent geometric fibers, independent of the choice of the order on R' . Moreover, $H_{R'}$ is the semidirect product $T \cdot U_{R'}$ ($U_{R'}$ invariant).

Indeed, if $R' \subset R_+$, then $H_{R'} \cap U_{\geq 1}$ (notations of 5.5.6) is a closed subgroup of G of finite presentation, invariant in $H_{R'}$. By 5.6.1 (iv), one has $H_{R'} = T \cdot U_{R'}$, which implies the other assertions.

Remark 5.6.6. In particular, U_{R_+} is the group $U_{\geq 1}$ of 5.5.6.

Let us extract some corollaries from the preceding results concerning groups of the type $U_{R'}$.

⁽⁴⁰⁾**N.D.E.** The original has been corrected in what follows.

⁽⁴¹⁾**N.D.E.** This is in fact a statement on root systems, supplementing exposé XXI (cf. [BLie], VI § 1.7, Prop. 22), and proved here by an indirect route.

Corollary 5.6.7. *Let (G, T, M, R) be a split reductive group, R' and R'' two subsets of R of type (R) , with $R' \cap -R' = \emptyset$.*

(i) *One has*

$$U_{R'} \cap \underline{\text{Norm}}_G(H_{R''}) = U_{R' \cap R''}.$$

(ii) *Assume R' closed. If the conditions $\alpha \in R'$, $\beta \in R''$ and $\alpha + \beta \in R$ imply $\alpha + \beta \in R'$, then $H_{R''}$ normalizes $U_{R'}$.*

Indeed, (i) follows at once from 5.6.5 and 5.6.1 (vi). To prove (ii), it suffices, in view of 5.4.4, to show that T and each U_β , $\beta \in R''$, normalize $U_{R'}$.⁽⁴²⁾ For T , this is trivial; for U_β , this follows from 5.5.2 and Exp. XXI 3.1.2.⁽⁴³⁾

Corollary 5.6.8. *Let (G, T, M, R) be a split S -group, R_+ a system of positive roots, α a simple root of R_+ (i.e. an element of R_+ such that $R_+ - \{\alpha\}$ is closed). Write*

$$U_{\hat{\alpha}} = U_{R_+ - \{\alpha\}}.$$

Then

- (i) $U_{\hat{\alpha}}$ *is an invariant subgroup of B_{R_+} .*
- (ii) U_{R_+} *is the semidirect product of $U_{\hat{\alpha}}$ by U_α .*
- (iii) $U_{-\alpha}$ *normalizes $U_{\hat{\alpha}}$.*
- (iv) Z_α *normalizes $U_{\hat{\alpha}}$.*

If one similarly defines $U_{-\hat{\alpha}} = U_{R_- - \{-\alpha\}}$ (where $R_- = -R_+$), one has

$$\Omega_{R_+} = U_{-\hat{\alpha}} \cdot U_{-\alpha} \cdot T \cdot U_\alpha \cdot U_{\hat{\alpha}}.$$

Indeed, (ii) follows from 5.6.5, and (i) from 5.6.7 (ii). Likewise, (iii) follows from 5.5.2 (indeed, if $\beta \in R_+$, $\beta \neq \alpha$, no combination $i(-\alpha) + j\beta$, with $i, j > 0$, can be negative, for β contains at least one simple root $\neq \alpha$). Then (iv) follows from (i) and (iii), for $U_{-\alpha} \cdot T \cdot U_\alpha$ is schematically dense in Z_α . Finally, the last assertion follows from (ii) and its analogue for U_{R_-} .

Let us return to the general situation.

Proposition 5.6.9. *Let S be a scheme, G a reductive S -group, H a subgroup of type (R) , with solvable geometric fibers.*

- (i) $D_S(H) = \underline{\text{Hom}}_{S\text{-gr.}}(H, \mathbf{G}_{m,S})$ *is representable by a twisted constant S -group, whose type at $s \in S$ is $\mathbf{Z}^{\text{rgred}(G_s)}$. The biduality morphism (Exp. VIII § 1)*

$$f : H \longrightarrow D_S(D_S(H))$$

is smooth and surjective.

- (ii) *The kernel H^u of f is the largest invariant closed group subscheme of H , smooth over S , with connected unipotent geometric fibers. One says that it is the unipotent part of H and one also writes $H^u = \text{rad}^u(H)$.*

Then H^u is also the commutator sheaf of H : every morphism of groups from H into an S -presheaf of commutative groups, separated for (fppf), vanishes on H^u and therefore factors through $H/H^u = D_S(D_S(H))$.

- (iii) *If T is a maximal torus of H , the morphism $T \longrightarrow H$ induces isomorphisms $D_S(H) \xrightarrow{\sim} D_S(T)$ and $T \xrightarrow{\sim} D_S(D_S(H))$. Moreover, H is identified with the semidirect product of H^u by T .*

⁽⁴²⁾**N.D.E.** Indeed, G has connected fibers, hence is separated over S by VIB 5.5, so by XI 6.11 (see also addition 6.5.5 in VIB), $\underline{\text{Norm}}_G(U_{R'})$ is represented by a closed group subscheme N of G . If N contains T and the U_β , for $\beta \in R''$, it contains the big cell $\Omega_{R_+, R''}$; this is schematically dense in $H_{R''}$ by 5.4.4 and EGA IV₃, 11.10.10 (the fibers of $H_{R''}$ are integral and $\Omega_{R_+, R''}$ contains a nonempty open subset of each fiber). It follows that $H_{R''} \subset N$, hence $H_{R''}$ normalizes $U_{R'}$.

⁽⁴³⁾**N.D.E.** One must see that, under the hypotheses of (ii), if $\alpha \in R'$ and $\beta \in R''$, then all roots of the form $i\alpha + j\beta$ with $i, j \in \mathbf{N}^*$ belong to R' , and for this the reference XXI 2.3.5 has been replaced by XXI 3.1.2. This may also be seen directly by inspection of the root systems of rank 2.

(iv) In the situation of 5.6.1, if $H = H_{R'}$, then $H^u = U_{R'}$.

The assertions of the proposition are local for the étale topology (Exp. X 5.5). One may therefore reduce to the case of 5.6.1. One then knows (5.6.5) that $H_{R'}$ is the semidirect product of $U_{R'}$ by T . Let us show that $U_{R'}$ is the commutator sheaf of $H_{R'}$: since $H_{R'}/U_{R'} = T$ is commutative, it suffices to prove that every morphism of groups $\varphi : H_{R'} \rightarrow V$ as in (ii) vanishes on $U_{R'}$. It suffices to prove that φ vanishes on each U_α , $\alpha \in R'$. Now if $t \in T(S')$, $X \in \mathbf{W}(\mathfrak{g}^\alpha)(S')$, one has

$$1 = \varphi(t \exp_\alpha(X) t^{-1} \exp_\alpha(-X)) = \varphi(\exp_\alpha((\alpha(t) - 1)X)).$$

Since $\alpha : T \rightarrow \mathbf{G}_{m,S}$ is faithfully flat, one deduces at once that φ vanishes on $U_\alpha^\times$; but every section of U_α is locally a sum of two sections of $U_\alpha^\times$. One therefore has

$$\mathrm{Hom}_{S\text{-gr.}}(H, V) = \mathrm{Hom}_{S\text{-gr.}}(H/U_{R'}, V)$$

for every V as above. Applying this result to $V = \mathbf{G}_{m,S}$, one deduces at once (i) and (iii), then (iv) and the second assertion of (ii). It now suffices to prove the first assertion of (ii); the only nontrivial fact is that every invariant closed subgroup U of H , smooth over S with connected unipotent geometric fibers, is a subgroup of H^u . Now one first has:

Lemma 5.6.10. *Let G be a reductive S -group, T a maximal torus, U a group subscheme of G , smooth over S , with unipotent geometric fibers, normalized by T . Then $U \cap T = e$.*

Indeed, since $T = \mathrm{Centr}_G(T)$, one has $U \cap T = U^T$ (invariants under $\mathrm{int}(T)$). Applying Exp. XIX 1.4, one deduces that $U \cap T$ is smooth over S , but it is also radicial over S : for every $s \in S$, $U(\bar{s}) \cap T(\bar{s})$ consists of elements which are both unipotent and semisimple. This proves the lemma.

Let us return to the proof of 5.6.9 (ii). If U is an invariant subgroup of H as above, then the semidirect product $T \cdot U$ is a subgroup of type (R) of G , with solvable geometric fibers. One may therefore assume it of the form $H_{R''}$, with $R'' \subset R'$. It suffices to prove $U = U_{R''}$ and one is therefore reduced to the case where $H = T \cdot U$; but since the quotient H/U is commutative, U is a subsheaf of the commutator sheaf of H , which is H^u . Q.E.D.

Note that we have in fact just proved:

Proposition 5.6.11. *Let S be a scheme, G a reductive S -group, T a maximal torus of G . The maps*

$$H \mapsto H^u, \quad U \mapsto T \cdot U$$

are mutually inverse bijections between the set of subgroups H of type (R) of G , containing T and with solvable geometric fibers, and the set of subgroups U of G , smooth over S , normalized by T , with connected unipotent geometric fibers.⁽⁴⁴⁾

In particular, when (G, T, M, R) is split, the groups $H_{R'}$ and $U_{R'}$ correspond to one another.

Corollary 5.6.12. *Let S be a scheme, (G, T, M, R) a split S -group (resp. and R_+ a system of positive roots of R defining the Borel subgroup B).*

Every smooth group subscheme of G , with connected unipotent geometric fibers (resp. every smooth group subscheme of B^u), normalized by T , is locally over S of the form $U_{R'}$, where R' is a subset of R contained in a system of positive roots (resp. a subset of R_+) of type (R).

For the “resp.” case, it suffices to note that the geometric fibers of the given group are unipotent and connected by *Bible*, § 13.2, th. 1 (d).

⁽⁴⁴⁾**N.D.E.** The hypothesis that U is closed has been deleted, since it holds automatically. Indeed, for such a U , one has $U \cap T = e$ by 5.6.10, hence the semidirect product $H = T \cdot U$ is a subgroup of type (R) of G (cf. 5.2.1), with solvable geometric fibers. Hence, by 5.6.3, H is closed in G , and since U is closed in H , it is closed in G .

Proposition 5.6.9 has moreover the following corollary:

Corollary 5.6.13. *Let S be a scheme, G a reductive S -group, H a subgroup of type (R) with solvable geometric fibers, $\underline{\mathrm{Tor}}(H)$ the functor of maximal tori of H :*

$$\underline{\mathrm{Tor}}(H)(S') = \{\text{maximal tori of } H_{S'}\}.$$

Then $\underline{\mathrm{Tor}}(H)$ is representable by an affine smooth S -scheme, which is a principal homogeneous bundle under H^u for the law $(h, T) \mapsto \mathrm{int}(h)T$.⁽⁴⁵⁾

Indeed, if T and T' are two maximal tori of $H_{S'}$, there exists a unique section $h \in H^u(S')$ such that $\mathrm{int}(h)T = T'$. The uniqueness of h follows at once from the equality

$$\underline{\mathrm{Norm}}_G(T) \cap H^u = e$$

(cf. for example 5.6.1); it therefore suffices to prove the existence of h locally for the étale topology. By 5.2.6 and 5.1.2 (a), one may assume T and T' conjugate by a section of H , whence the desired conclusion since $H = H^u \cdot T$ by 5.6.9 (iii). It follows that $\underline{\mathrm{Tor}}(H)$ is a principal homogeneous sheaf under H^u , which is affine and smooth over S , which implies the statement at once⁽⁴⁶⁾.

5.7. Bruhat's theorem

Recall 5.7.1. Let k be an algebraically closed field, G a reductive k -group, B a Borel subgroup of G , T a maximal torus of B , $N = \underline{\mathrm{Norm}}_G(T)$. Then

$$G(k) = B(k)N(k)B(k);$$

this is Bruhat's theorem (*Bible*, § 13.4, cor. 1 to th. 3); more precisely, with the notations of 3.6, the sets

$$B(k)N_w(k)B(k) = B^u(k)N_w(k)B^u(k)$$

form, as w ranges over $(N/T)(k)$, a partition of $G(k)$. If B' is another Borel subgroup of G containing T , the sets $B'(k)N_w(k)B(k)$ also form a partition of $G(k)$. Indeed, if $y \in N(k)$ is such that $\mathrm{int}(y)B = B'$, one has

$$yB(k)N_w(k)B(k) = B'(k)N_{yw}(k)B(k).$$

Definition 5.7.2. Let (G, T, M, R) be a split S -group, R_- a system of positive roots of R , $B' = B_{R_-}$ the Borel group it defines. For $w \in W$, one writes (cf. 5.6.5):

$$R_-^w = R_- \cap w(R_-), \quad B'_w{}^u = U_{R_-^w} = \prod_{\alpha \in R_-^w} U_\alpha.$$

If $n_w \in \underline{\mathrm{Norm}}_G(T)(S)$ is a representative of w (3.8), one may also write

$$B'_w{}^u = B'^u \cap \mathrm{int}(n_w)B'^u.$$

Lemma 5.7.3. *Let (G, T, M, R) be a split S -group, R_+ a system of positive roots of R , $R_- = -R_+$, B (resp. B') the Borel subgroup of G defined by R_+ (resp. R_-). Let $w \in W$, N_w and $B'_w{}^u$ be the corresponding subschemes of G (3.8 and 5.7.2).*

(i) *The sheaf $B' \cdot N_w \cdot B$, image of the morphism*

$$B' \times_S N_w \times_S B \longrightarrow G$$

induced by the product in G , is representable by a subscheme of G (and in fact a closed subscheme of the open subset $n_w\Omega_{R_+}$).

⁽⁴⁵⁾**N.D.E.** This proves again and sharpens XII 1.10 (for G reductive).

⁽⁴⁶⁾**N.D.E.** By SGA 1, VIII 2.1 and EGA IV₄, 17.7.1.

(ii) *The morphism*

$$B'_w \times_S N_w \times_S B^u \longrightarrow G,$$

induced by the product in G , is an immersion with image the preceding subscheme.

Let us first show that the morphism of (ii) is an immersion. By definition, $\text{int}(n_w)^{-1}$ induces a closed immersion of B'^u_w into B'^u , hence the morphism

$$(u, b) \longmapsto n_w^{-1} u n_w b$$

induces a closed immersion

$$B'^u_w \times_S B \longrightarrow \Omega_{R_+}.$$

This implies immediately that the morphism of (ii) induces a closed immersion of the first side into the open subset $n_w \Omega_{R_+}$. To prove (i), it suffices to see that

$$B'(S)N_w(S)B(S) = B'^u_w(S)N_w(S)B^u(S).$$

Now, if $\alpha \in R$, one has $\text{int}(n_w)U_\alpha(S) = U_{w(\alpha)}(S)$, hence if $w^{-1}(\alpha) \in R_+$,

$$\begin{aligned} U_\alpha(S)N_w(S)B(S) &= U_\alpha(S)n_w T(S)B^u(S) \\ &= n_w U_{w^{-1}(\alpha)}(S)T(S)B^u(S) \\ &= n_w B(S) = N_w(S)B^u(S). \end{aligned}$$

This implies at once, in view of the definition of R_-^w , the desired assertion.

Theorem 5.7.4. *Let S be a scheme, (G, T, M, R) a split S -group, B the Borel subgroup defined by the system of positive roots R_+ , B' the Borel subgroup defined by $R_- = -R_+$.*

- (i) *(Bruhat's theorem) The schemes $B'^u_w \cdot N_w \cdot B$ form, as w ranges over W , a partition of the underlying set of G .*
- (ii) *For each $w \in W$, let n_w be a representative of w in $\text{Norm}_G(T)(S)$ (3.8); then the open subsets $n_w \Omega = n_w B'^u \cdot T \cdot B^u$ form, as w ranges over W , a covering of G .*

The two assertions are indeed verified on the geometric fibers, where one concludes by 5.7.1 and 5.7.3.

Remark 5.7.5. (i) implies that if S is the spectrum of a field, $G(S)$ is the disjoint union of the $B'^u_w(S) \cdot T(S) \cdot B^u(S)$. The corresponding assertion for arbitrary S (even local or artinian) is clearly false. Note however that (ii) implies that if S is local, $G(S)$ is indeed the union of the $n_w \Omega(S)$. In fact:

Corollary 5.7.6. *Let Δ be a system of simple roots of the split group G over the local scheme S .*

- (i) *Then $G(S)$ is generated by $T(S)$ and the $U_\alpha(S)$, $\alpha \in \Delta \cup -\Delta$.*
- (ii) *If G is simply connected (4.3.3), $G(S)$ is already generated by the $U_\alpha(S)$, $\alpha \in \Delta \cup -\Delta$.*

Indeed, let H be the subgroup of $G(S)$ generated by the $U_\alpha(S)$, $\alpha \in \Delta \cup -\Delta$. Note first that H contains a representative of each s_α ($\alpha \in \Delta$) in $\text{Norm}_G(T)(S)$ (Exp. XX 3.1), hence a representative n_w of each $w \in W$.

Since every $\alpha \in R$ is written $w(\alpha_0)$ with $w \in W$, $\alpha_0 \in \Delta$, one has

$$U_\alpha(S) = \text{int}(n_w)U_{\alpha_0}(S) \subset H.$$

The subgroup generated by H and $T(S)$ therefore contains $\Omega(S)$ and is therefore all of $G(S)$, by the remark made earlier.

If now G is simply connected, let us prove that $H \supset T(S)$. By Exp. XX 2.7, H contains $\alpha^*(\mathbf{G}_m(S))$ for every $\alpha \in \Delta$, and it suffices to apply 4.3.8.

Remark 5.7.6.1. Instead of taking, for each $\alpha \in \Delta$, $U_\alpha(S)$ and $U_{-\alpha}(S)$, one may content oneself with taking $U_\alpha(S)$ and a representative w_α of the reflection s_α , or $U_\alpha(S)$ and a section of $U_{-\alpha}^\times, \dots$

Corollary 5.7.7. *If G has semisimple rank 1, choose a $u_\alpha \in U_\alpha^\times(S)$. Then Ω and $u_\alpha\Omega$ form a covering of G .*

Indeed, if $u_{-\alpha}$ is the section of $U_{-\alpha}$ paired with u_α (cf. 1.3), one has, by 5.7.4 (ii),

$$G = \Omega \cup u_{-\alpha}^{-1} u_\alpha u_{-\alpha}^{-1} \Omega,$$

whence

$$G = u_{-\alpha} G = u_{-\alpha} \Omega \cup u_\alpha u_{-\alpha}^{-1} \Omega = \Omega \cup u_\alpha \Omega.$$

Corollary 5.7.8. *Let S be a scheme, G a reductive S -group. Then G is essentially free over S (Exp. VIII 6.1).⁽⁴⁷⁾*

Indeed, the assertion is local for the (fpqc) topology; one may assume G split. Then G admits a covering by open subsets isomorphic to $\mathbf{G}_{a,S}^N \times_S \mathbf{G}_{m,S}^n$, hence essentially free.

Lemma 5.7.9. *Under the conditions of 5.7.4, let α be a simple root of R_+ and $u_\alpha \in U_\alpha^\times(S)$. For every $v \in U_{-\alpha}(S)$, one has*

$$\Omega \cdot v \subset \Omega \cup u_\alpha \cdot \Omega.$$

One must compare two open subsets of G ; it suffices to do so when S is the spectrum of a field k . One must therefore prove

$$\Omega(k)v \subset \Omega(k) \cup u_\alpha \Omega(k).$$

Now

$$\begin{aligned} \Omega(k)v &= B'^u(k)T(k)B^u(k)v \\ &= U_{-\hat{\alpha}}(k)U_{-\alpha}(k)T(k)U_\alpha(k)U_{\hat{\alpha}}(k)v \\ &\subset U_{-\hat{\alpha}}(k)Z_\alpha(k)U_{\hat{\alpha}}(k)v. \end{aligned}$$

(One uses the decomposition of 5.6.8). Applying now 5.6.8 (iii) and using 5.7.7 for the group Z_α , one obtains

$$\begin{aligned} \Omega(k)v &\subset U_{-\hat{\alpha}}(k)Z_\alpha(k)vU_{\hat{\alpha}}(k) \subset U_{-\hat{\alpha}}(k)Z_\alpha(k)U_{\hat{\alpha}}(k) \\ &\subset U_{-\hat{\alpha}}(k)U_{-\alpha}(k)T(k)U_\alpha(k)U_{\hat{\alpha}}(k) \\ &\quad \cup U_{-\hat{\alpha}}(k)u_\alpha U_{-\alpha}(k)T(k)U_\alpha(k)U_{\hat{\alpha}}(k). \end{aligned}$$

Using 5.6.8 (iii) again (for R_- instead of R_+), one obtains the result.

Proposition 5.7.10. *Under the conditions of 5.7.4, choose for each simple root α a $u_\alpha \in U_\alpha^\times(S)$. Let U_1 be the submonoid of $B^u(S)$ generated by the u_α . The open subsets $u\Omega$, for $u \in U_1$, form a covering of G .*

Once again, one may assume that S is the spectrum of a field k ; by 5.7.6, it suffices to prove that $\bigcup_{u \in U_1} u\Omega(k)$ is stable under right multiplication by $T(k)$, $U_\alpha(k)$, $U_{-\alpha}(k)$ (for α simple). In the first two cases, this is trivial. In the last, this follows from the lemma.

Remark 5.7.11. Let us point out a particular case of 5.7.2. If $w = s_\alpha$ is the reflection with respect to the simple root α , then

$$R_- \cap s_\alpha(R_-) = R_- - \{-\alpha\}$$

(Exp. XXI 3.3.1), and, in the notations of 5.6.8, one therefore has

$$B'^u_{s_\alpha} = U_{-\hat{\alpha}}.$$

Remark 5.7.12. In fact, the proof of 5.7.10 gives at once the following statement: under the conditions of 5.7.10, let Γ be a submonoid of $G(S)$; for the open subsets $g\Omega$ ($g \in \Gamma$) to form a covering of G , it is necessary and sufficient that for every $s \in S$ and every simple root α , one have

$$(u_\alpha)_{\bar{s}} B'^u(\bar{s}) \subset \Gamma \cdot B'^u(\bar{s}) \cdot T(\bar{s}) \cdot B^u(\bar{s}).$$

⁽⁴⁷⁾**N.D.E.** See also the additions made in VIB, 6.2.1 to 6.2.6 and 6.5.2 to 6.5.5.

Remark 5.7.13. By 5.5.5 (iii), arguing as in 5.7.1, one obtains at once the following variant of 5.7.4: let (G, T, M, R) be a split S -group, B and B' two Borel subgroups of G containing T ; for every $w \in W$, the sheaf $B' \cdot N_w \cdot B$ is representable by a subscheme of G ; these subschemes form, for $w \in W$, a partition of the underlying set of G . One may also give the analogue of 5.7.3 (ii): one must put

$$B'^u_w = B'^u \cap \text{int}(n_w) \tilde{B}^u,$$

where $\tilde{B}$ is the Borel subgroup “opposite” to B relative to T (cf. 5.9.2).

Proposition 5.7.14. *Let S be a scheme, G a reductive S -group, and*

$$\text{Ad} : G \longrightarrow \text{GL}_{\mathcal{O}_S}(\mathfrak{g})$$

its adjoint representation. Then $\text{Ker}(\text{Ad}) = \underline{\text{Centr}}(G)$ (in other words, the canonical homomorphism deduced from Ad on passing to the quotient:

$$\overline{\text{Ad}} : G/\underline{\text{Centr}}(G) = \text{ad}(G) \longrightarrow \text{GL}_{\mathcal{O}_S}(\mathfrak{g})$$

is a monomorphism).⁽⁴⁸⁾

One may assume G split. Choose on $\Gamma_0(R)$ a total order structure compatible with the group structure and let R_+ be the set of positive roots. By 5.7.4 (ii) and 4.1.6, it suffices to prove that if n_w is a representative of the element w of W , if $u \in U(S)$, $t \in T(S)$, $v \in U_-(S)$, and if $\text{Ad}(n_w v t u) = \text{id}$, then $w = e$, $v = e$, $u = e$. For each $m \in R \cup \{0\}$, put

$$\mathfrak{g}^{>m} = \coprod_{n>m} \mathfrak{g}^n, \quad \mathfrak{g}^{<m} = \coprod_{n<m} \mathfrak{g}^n.$$

Let $X \in \Gamma(S, \mathfrak{g}^m)$; write $\text{Ad}(tu)X = \text{Ad}(v^{-1}n_w^{-1})X$. Now

$$\text{Ad}(t) \text{Ad}(u)X - m(t)X \in \Gamma(S, \mathfrak{g}^{>m}),$$

$$\text{Ad}(v^{-1}n_w^{-1})X - \text{Ad}(n_w^{-1})X \in \Gamma(S, \mathfrak{g}^{<w^{-1}(m)}).$$

If $w \neq e$, there exists an $\alpha \in R$ such that $w^{-1}(\alpha) < \alpha$, and putting $m = \alpha$, one obtains a contradiction, for

$$\text{Ad}(tu)X \in \Gamma(S, \mathfrak{g}^\alpha + \mathfrak{g}^{>\alpha}) \cap \Gamma(S, \mathfrak{g}^{w^{-1}(\alpha)} + \mathfrak{g}^{<w^{-1}(\alpha)}) = 0.$$

One therefore has $w = e$, and one may choose $n_w = e$; one then has

$$\text{Ad}(v^{-1})X - X \in \Gamma(S, \mathfrak{g}^{<m} \cap (\mathfrak{g}^m + \mathfrak{g}^{>m})) = 0,$$

whence $\text{Ad}(v)X = X$ for every $X \in \Gamma(S, \mathfrak{g}^m)$, hence $\text{Ad}(v) = \text{id}$. Likewise $\text{Ad}(u) = \text{id}$. One then concludes by 5.6.2 bis.

5.8. Schemes associated with a reductive group

Theorem 5.8.1. *Let S be a scheme, G a reductive S -group. Let $\mathcal{H}$ be the functor of subgroups of type (R) of G : for every $S' \rightarrow S$, $\mathcal{H}(S')$ is the set of subgroups of type (R) of $G_{S'}$ (cf. 5.2.1). Then $\mathcal{H}$ is representable by a quasi-projective S -scheme, of finite presentation over S .*

⁽⁴⁹⁾Let $G' = G/\underline{\text{Centr}}(G)$ be the adjoint group of G (4.3.6) and u the morphism $G \rightarrow G'$. By Exp. XII 7.12, the map $H' \mapsto u^{-1}(H')$ establishes a bijection between the set of subgroups of type (R) of G' and of G (and this remains valid after every base change). Hence, replacing G by G' , one may assume that G is adjoint. Consider then the morphism

$$u : \mathcal{H} \longrightarrow \underline{\text{Grass}}(\mathfrak{g})$$

⁽⁴⁸⁾**N.D.E.** It is even a closed immersion, by Exp. XVI 1.5 (a).

⁽⁴⁹⁾**N.D.E.** The original has been expanded in what follows.

which associates with each subgroup of type (R) its Lie algebra (which is a submodule locally a direct summand of $\mathfrak{g}^{(50)}$). Then u is a monomorphism by 5.3.3. It suffices to prove that it is representable by an immersion of finite presentation, in other words to prove the following assertion: for every $S_1 \rightarrow S$, given a submodule $\mathfrak{h}$ locally a direct summand of $\mathfrak{g}_{S_1}$, the $S' \rightarrow S_1$ such that $\mathfrak{h}_{S'}$ is the Lie algebra of a subgroup of type (R) of $G_{S'}$ are exactly those which factor through a certain subscheme Σ of finite presentation of S_1 . Replacing S_1 by S , one reduces to $S_1 = S$, and one may moreover assume S affine; then there exists an affine noetherian scheme S_0 such that G (resp. $\mathfrak{h}$) comes by base change from an adjoint reductive S_0 -group G_0 (resp. a submodule $\mathfrak{h}_0$ locally a direct summand of $\mathfrak{g}_0 = \underline{\mathrm{Lie}}(G_0/S_0)$). It suffices to show that there exists a subscheme Σ_0 of S_0 having the required properties (for then one will have $\Sigma = \Sigma_0 \times_{S_0} S$). Replacing S by S_0 , one may therefore assume S affine and noetherian (note that then every subscheme of S is of finite presentation over S). Finally, replacing S by a sufficiently small open subset, one may assume that $\mathfrak{g}$ is free of rank n and that $\mathfrak{h}$ is a direct summand, free of rank r .

One must first express that $\mathfrak{h}_{S'}$ is a Lie subalgebra of $\mathfrak{g}_{S'}$, i.e. that the morphism induced by the Lie bracket:

$$\varphi : \mathfrak{h} \otimes \mathfrak{h} \xrightarrow{[\cdot, \cdot]} \mathfrak{g}$$

factors through $\mathfrak{h}$. If $(e_1, \dots, e_n)$ is a basis of $\mathfrak{g}$ such that $(e_1, \dots, e_r)$ is a basis of $\mathfrak{h}$, then φ is given by sections a_k^{ij} of $\mathcal{O}_S$ (where $i, j = 1, \dots, r$ and $k = 1, \dots, n$), and the preceding condition is equivalent to saying that $S' \rightarrow S$ factors through the closed subscheme of S defined by the equations $a_k^{ij} = 0$ for $k = r+1, \dots, n$ and $i, j = 1, \dots, r$. Replace S by this closed subscheme.

Then, by 5.3.0, $N = \underline{\mathrm{Norm}}_G(\mathfrak{h})$ is a closed group subscheme of G , of finite presentation over G . One must now express (cf. 5.3.1) that $N_{S'}$ is smooth at every point of the identity section, of relative dimension $r = \mathrm{rank}(\mathfrak{h})$, and that the inclusion of $\mathfrak{h}_{S'}$ into $\underline{\mathrm{Lie}}(N_{S'}/S')$ is an equality.

⁽⁵¹⁾Since N is affine over S (being closed in G), the identity section $\varepsilon : S \rightarrow N$ is a closed immersion, so $\varepsilon(S)$ is defined by a quasi-coherent ideal $\mathcal{J}$ of $\mathcal{A}(N)$. Note that $\mathfrak{n}_{S/N} = \mathcal{J}/\mathcal{J}^2$ is identified with $\varepsilon^*(\Omega_{N/S}^1)$, hence its formation commutes with every base change $S' \rightarrow S$. By equivalence (c') $\Leftrightarrow$ (a) in EGA IV₄, 17.12.1 (applied to $f : N \rightarrow S$ and $j = \varepsilon$), $N_{S'} \rightarrow S'$ is smooth, of relative dimension r , at every point of $\varepsilon(S')$ if and only if $\mathfrak{n}_{S'/N'} = \mathfrak{n}_{S/N} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}$ is locally free of rank r and the morphism $\varphi_n(S') : \mathrm{Sym}^n(\mathfrak{n}_{S'/N'}) \rightarrow \mathcal{J}_{S'}^n/\mathcal{J}_{S'}^{n+1}$ is an isomorphism for every $n \geq 1$. Put $K_n(S') = \mathrm{Ker} \varphi_n(S')$. By TDTE I, Lemma 3.6, $\mathfrak{n}_{S/N} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}$ is locally free of rank r if and only if $S' \rightarrow S$ factors through a certain subscheme Z of S . Replacing S by Z , one may therefore assume that $\mathfrak{n}_{S/N} = \mathcal{J}/\mathcal{J}^2$ is locally free of rank r . Then, for every $S' \rightarrow S$, one has $(\mathcal{J}^2/\mathcal{J}^3) \otimes_{\mathcal{O}_S} \mathcal{O}_{S'} = \mathcal{J}_{S'}^2/\mathcal{J}_{S'}^3$, and hence $K_2(S') = K_2(S) \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}$. It follows that $\varphi_2(S')$ is an isomorphism if and only if $S' \rightarrow S$ factors through the closed subscheme S_2 of S defined by the ideal generated by the image of $\mathrm{Sym}^2(\mathfrak{n}_{S/N})^* \otimes K_2(S)$ in $\mathcal{O}_S$. Then, over S_2 , $\mathcal{J}^2/\mathcal{J}^3$ is isomorphic to $\mathrm{Sym}^2(\mathfrak{n}_{S/N})$, hence locally free, and the same argument shows that $\varphi_3(S')$ is an isomorphism if and only if $S' \rightarrow S$ factors through a certain closed subscheme S_3 of S , etc. One thus obtains that $N_{S'} \rightarrow S'$ is smooth, of relative dimension r , at every point of $\varepsilon(S')$ if and only if $S' \rightarrow S$ factors through the closed subscheme Z which is the intersection of the S_n . But then, for every $S' \rightarrow Z$, $\underline{\mathrm{Lie}}(N_{S'}/S')$ is locally a direct summand of rank r of $\mathfrak{g}_{S'}$, and hence the inclusion $\mathfrak{h}_{S'} \subset \underline{\mathrm{Lie}}(N_{S'}/S')$ is an equality. One then puts $H = N^0$.

⁽⁵⁰⁾**N.D.E.** cf. Exp. II 4.11.8.

⁽⁵¹⁾**N.D.E.** The original next indicated that, denoting by $\mathfrak{n}_{N/G}$ the inverse image of the conormal sheaf $\mathcal{N}_{N/G}$ by the identity section $S \rightarrow N$, the condition that $\mathfrak{n}_{N_{S'}/G_{S'}} \rightarrow \omega_{G_{S'}/S'}^1$ be universally injective is equivalent to the fact that $S' \rightarrow S$ factors through a certain open subscheme of S . We have not succeeded in justifying this point, owing to the fact that the formation of $\mathcal{N}_{N/G}$ does not commute with base change, and we have replaced this argument by the one which follows, indicated by O. Gabber.

Replacing S by Z , it only remains to express that $H_{s'}$ has the same reductive rank as $G_{s'}$ at every point $s' \in S'$, or, equivalently, that H_s has the same reductive rank as G_s at every point s of the (set-theoretic) image of S' in S . Now this condition defines an open subset of S (Exp. XIX 6.2).

Remark. In general, the scheme $\mathcal{H}$ is not smooth over S . It is however if 6 is invertible on S , or if there exists a prime number p such that $p \cdot 1_S = 0$ (i.e. if S has characteristic $p > 0$).

Corollary 5.8.2. *Let S be a scheme, G a reductive S -group, H a subgroup of type (R) of G . (Recall (5.3.10) that $\underline{\text{Norm}}_G(H)$ is representable by a closed group subscheme of G , smooth over S).*

Then the quotient sheaf $G/\underline{\text{Norm}}_G(H)$ is representable by a quasi-projective S -scheme, smooth and of finite presentation over S (which is in fact an open subset of $\mathcal{H}$).

Indeed, consider the morphism

$$f : G \longrightarrow \mathcal{H},$$

defined set-theoretically by $f(g) = \text{int}(g)H$. By 5.3.9, this morphism is smooth and of finite presentation, hence open. Let $V = f(G)$ be endowed with its structure of open subscheme of $\mathcal{H}$. The morphism $f : G \longrightarrow V$ is covering and has kernel $\underline{\text{Norm}}_G(H)$, which proves that $G/\underline{\text{Norm}}_G(H)$ is representable by V (cf. Exp. IV 4.6.5).

Corollary 5.8.3. *Let S be a scheme, G a reductive S -group. Consider the functors $\underline{\text{Tor}}(G)$, $\underline{\text{Bor}}(G)$, $\underline{\text{Kil}}(G)$ defined by*

$$\begin{aligned} \underline{\text{Tor}}(G)(S') &= \{\text{maximal tori of } G_{S'}\}, \\ \underline{\text{Bor}}(G)(S') &= \{\text{Borel subgroups of } G_{S'}\}, \\ \underline{\text{Kil}}(G)(S') &= \{\text{Killing pairs of } G_{S'} \text{ (cf. 5.3.13)}\}. \end{aligned}$$

- (i) $\underline{\text{Tor}}(G)$, $\underline{\text{Bor}}(G)$, $\underline{\text{Kil}}(G)$ are representable by smooth S -schemes of finite presentation, with integral geometric fibers, and respectively affine, projective, affine over S .
- (ii) The canonical morphism $\underline{\text{Kil}}(G) \longrightarrow \underline{\text{Tor}}(G)$ (resp. $\underline{\text{Kil}}(G) \longrightarrow \underline{\text{Bor}}(G)$) is finite étale surjective (resp. affine smooth surjective).
- (iii) Let T be a maximal torus of G (resp. B a Borel subgroup of G , resp. $B \supset T$ a Killing pair of G). The morphism

$$G \longrightarrow \underline{\text{Tor}}(G), \quad \text{resp. } G \longrightarrow \underline{\text{Bor}}(G), \quad \text{resp. } G \longrightarrow \underline{\text{Kil}}(G)$$

defined by

$$g \longmapsto \text{int}(g)T, \quad \text{resp. } g \longmapsto \text{int}(g)B, \quad \text{resp. } g \longmapsto (\text{int}(g)B, \text{int}(g)T)$$

induces an isomorphism

$$\begin{aligned} G/\underline{\text{Norm}}_G(T) &\xrightarrow{\sim} \underline{\text{Tor}}(G), & \text{resp. } G/B &\xrightarrow{\sim} \underline{\text{Bor}}(G), \\ & & \text{resp. } G/T &\xrightarrow{\sim} \underline{\text{Kil}}(G). \end{aligned}$$

One sees first that (iii) follows from the conjugacy theorem for maximal tori (resp. Borel subgroups, resp. Killing pairs) and the fact that

$$\underline{\text{Norm}}_G(B) = B, \quad \underline{\text{Norm}}_G(B) \cap \underline{\text{Norm}}_G(T) = T,$$

all results established previously (5.1.2, 5.3.12, 5.3.14, 5.6.1).

It follows first that the canonical morphisms

$$\underline{\text{Tor}}(G) \longrightarrow \mathcal{H}, \quad \underline{\text{Bor}}(G) \longrightarrow \mathcal{H}$$

are representable, locally for the étale topology, by open immersions (5.8.2 and 5.1.2 resp. 5.5.5), hence by descent that $\underline{\text{Tor}}(G)$ and $\underline{\text{Bor}}(G)$ are representable by open subsets of $\mathcal{H}$.

Likewise $\underline{\text{Kil}}(G)$ is locally (for the étale topology) representable by a scheme affine over the base (Exp. IX 2.3), hence representable by an affine S -scheme, by descent of affine schemes.⁽⁵²⁾

The assertions of (ii) follow at once from 5.5.5 (ii) and 5.6.13. It already follows that $\underline{\text{Tor}}(G)$ is affine over S (EGA II 6.7.1). It therefore only remains to prove the fact that $\underline{\text{Bor}}(G)$ is projective over S . One already knows that it is quasi-projective; it remains to prove that it is proper;⁽⁵³⁾ now it has connected fibers, hence, by EGA IV₃, 15.7.10, one is reduced to proving it on the geometric fibers; if S is the spectrum of an algebraically closed field, one has $\underline{\text{Bor}}(G) = G/B$ by (iii) and one concludes by *Bible*, § 6.4, th. 4 (or [Ch05], § 6.5, th. 5).

Remark 5.8.4. Under the conditions of 5.8.3, let Q be a central subgroup of multiplicative type of G . The obvious morphisms define isomorphisms

$$\begin{aligned}\underline{\text{Tor}}(G) &\simeq \underline{\text{Tor}}(G/Q), & \underline{\text{Bor}}(G) &\simeq \underline{\text{Bor}}(G/Q), \\ \underline{\text{Kil}}(G) &\simeq \underline{\text{Kil}}(G/Q).\end{aligned}$$

Corollary 5.8.5. *Let S be a scheme, G a reductive S -group, P a group subscheme of G , smooth and of finite presentation over S . The following conditions are equivalent:*

- (i) *For each $s \in S$, $P_{\bar{s}}$ is a parabolic subgroup of $G_{\bar{s}}$ (i.e. the quotient scheme $G_{\bar{s}}/P_{\bar{s}}$ is proper over $\bar{s}$, or $P_{\bar{s}}$ contains a Borel subgroup of $G_{\bar{s}}$, cf. *Bible*, § 6.4, th. 4 or [Ch05], § 6.5, th. 5).*
- (ii) *The quotient sheaf G/P is representable by a smooth S -scheme projective over S .*

Moreover, under these conditions, P is closed in G , has connected fibers and one has $P = \underline{\text{Norm}}_G(P)$.

One clearly has (ii) $\Rightarrow$ (i). If (i) holds, then $P(\bar{s}) = \text{Norm}_{G(\bar{s})}(P_{\bar{s}})$ and $P_{\bar{s}}$ is connected (for the first point, cf. *Bible*, § 12.3, lemma 4;⁽⁵⁴⁾ the second point follows from it, for $P' = P_{\bar{s}}^0$ is a parabolic subgroup of $G_{\bar{s}}$ normalized by $P_{\bar{s}}$, whence $P'(\bar{s}) = P(\bar{s})$ and hence $P' = P_{\bar{s}}$); it follows that P is of type (R), and that P equals $\underline{\text{Norm}}_G(P)$, hence is closed in G . By 5.8.2, $G/P = G/\underline{\text{Norm}}_G(P)$ is representable by a quasi-projective S -scheme. Its fibers are connected and proper, hence it is projective by the argument of 5.8.3.

Remark 5.8.6. Statements 5.8.1, 5.8.2, 5.8.5 hold for an S -group of type (RA), or for an S -group of type (RR) satisfying 5.1.8.⁽⁵⁵⁾

Remark 5.8.7. Through the inner automorphisms of G , one has canonical actions:

$$\begin{aligned}G &\longrightarrow \underline{\text{Aut}}_S(\underline{\text{Tor}}(G)), & G &\longrightarrow \underline{\text{Aut}}_S(\underline{\text{Bor}}(G)), \\ & & G &\longrightarrow \underline{\text{Aut}}_S(\underline{\text{Kil}}(G)),\end{aligned}$$

which, in the situation of 5.8.3 (iii), are identified with the canonical actions

$$\begin{aligned}G &\longrightarrow \underline{\text{Aut}}_S(G/\underline{\text{Norm}}_G(T)), & G &\longrightarrow \underline{\text{Aut}}_S(G/B), \\ & & G &\longrightarrow \underline{\text{Aut}}_S(G/T).\end{aligned}$$

One concludes in particular that

$$\begin{aligned}\text{Ker}(G \longrightarrow \underline{\text{Aut}}_S(\underline{\text{Tor}}(G))) &= \text{Ker}(G \longrightarrow \underline{\text{Aut}}_S(\underline{\text{Bor}}(G))) \\ &= \text{Ker}(G \longrightarrow \underline{\text{Aut}}_S(\underline{\text{Kil}}(G))) \\ &= \underline{\text{Centr}}(G).\end{aligned}$$

⁽⁵²⁾**N.D.E.** cf. SGA 1, VIII 2.1.

⁽⁵³⁾**N.D.E.** One may assume S affine and, since $\underline{\text{Bor}}(G)$ is of finite presentation over S by (i), reduce to the case where S is noetherian; one is then under the hypotheses of EGA IV₃, 15.7.10.

⁽⁵⁴⁾**N.D.E.** What follows has been expanded.

⁽⁵⁵⁾**N.D.E.** Indeed, the proof of 5.8.1 uses only 5.3.3 (valid for a group of type (RA)) and XIX 6.2 which, by XII 1.7 (b), also holds for groups of type (RR).

It is indeed clear that $\text{Centr}(G)$ acts trivially on each of the three schemes. Conversely, the kernel of $G \longrightarrow \underline{\text{Aut}}_S(\underline{\text{Kil}}(G))$ is “the intersection of the maximal tori of G ” in the sense of 4.1.7, hence equals $\text{Centr}(G)$ (*loc. cit.*). For $\text{Bor}(G)$, one notes that “the intersection of the Borel subgroups of G ” is also “the intersection of its maximal tori” (see the next number). For $\text{Tor}(G)$, one uses Exp. XII 4.11.

5.9. Properties peculiar to Borel subgroups

Most of these properties will be generalized in Exp. XXVI to parabolic subgroups.

Definition 5.9.1. Let S be a scheme, G a reductive S -group, B and B' two Borel subgroups of G . One says that B and B' are *in general position* (or that B' is in general position relative to B) if $B \cap B'$ is a torus (necessarily maximal) of G .

If T is a maximal torus of G contained in B and B' , one says that B and B' are *opposite* (relative to T) if $B \cap B' = T$.

Proposition 5.9.2. Let S be a scheme, G a reductive S -group, B a Borel subgroup of G , T a maximal torus of B . There exists a unique Borel subgroup B' of G , opposite to B relative to T .

If (G, T, M, R) is a splitting of G with respect to T and if $B = B_{R_+}$ (5.5.1), then $B' = B_{-R_+}$.

By faithfully flat descent, it suffices to prove the proposition in the split case, when $B = B_{R_+}$ (5.5.5 (iv)). Then B_{-R_+} is indeed opposite to B (4.1.2); let us show that it is the only Borel subgroup of G containing T which is opposite to B . If B' is a Borel subgroup of G containing T , then B' is locally over S of the form $B_{R'_+}$, where R'_+ is a second system of positive roots of R (5.5.5 (iii)). If $R'_+ \neq -R_+$, there exists $\alpha \in R'_+ \cap R_+$, hence such that $U_\alpha \subset B_{R_+} \cap B_{R'_+}$.

Proposition 5.9.3. Let S be a scheme, G a reductive S -group, B a Borel subgroup of G .

- (i) If B' is a Borel subgroup of G , the following conditions are equivalent:
 - (a) B' is in general position with respect to B (5.9.1).
 - (b) $B'^u \cap B^u = e$.
 - (b') $B'^u \cap B = e$.
 - (c) The product in G induces an open immersion $B'^u \times_S B \longrightarrow G$.
 - (c') The canonical morphism $B'^u \longrightarrow G/B$ is an open immersion.
- (ii) The functor $\underline{\text{Opp}}(B)$:

$$S' \longmapsto \left\{ \begin{array}{l} \text{Borel subgroups of } G_{S'} \text{ in} \\ \text{general position with respect to } B_{S'} \end{array} \right\}$$

is representable by an open subscheme of $\text{Bor}(G)$ (5.8.3). The morphism

$$\underline{\text{Opp}}(B) \longrightarrow \text{Tor}(B)$$

defined by $B' \longmapsto B \cap B'$ is an isomorphism. In particular (5.6.13), the inner automorphisms of B^u endow $\underline{\text{Opp}}(B)$ with a principal homogeneous bundle structure under B^u .

Let us first examine (i). One has (a) $\Rightarrow$ (c); indeed, (c) is local for the étale topology; by 5.5.5 (iv), one reduces to the case where G is split with respect to $B \cap B'$ and B has the form B_{R_+} ; by 5.9.2, one then has $B'^u = U_{-R_+}$ and one is reduced to 4.1.2.

One has trivially (c') $\Leftrightarrow$ (c) $\Rightarrow$ (b') $\Rightarrow$ (b). It therefore remains to prove (b) $\Rightarrow$ (a). Let us first prove (ii); the second assertion is a formal consequence of 5.9.2, the third follows at once by 5.6.13; let us then prove the first; it is local for the étale topology and one may therefore assume that B possesses a maximal torus T ; let B'_0 be the opposite of B relative to T (5.9.2).

By what precedes the morphism $B^u \rightarrow \underline{\text{Bor}}(G)$ induced by the canonical morphism $G \rightarrow G/B'_0 \rightarrow \underline{\text{Bor}}(G)$ (5.8.3) induces an isomorphism $B^u \xrightarrow{\sim} \underline{\text{Opp}}(B)$. One therefore has a commutative diagram

$$\begin{array}{ccc} B^u & \longrightarrow & G/B'_0 \\ \sim \downarrow & & \downarrow \sim \\ \underline{\text{Opp}}(B) & \longrightarrow & \underline{\text{Bor}}(G). \end{array}$$

Now the morphism $B^u \rightarrow G/B'_0$ is an open immersion (by (i) (a) $\Rightarrow$ (c')), which completes the proof of (ii). Note at once the corollary

Corollary 5.9.4. *Let G be a reductive S -group and B and B' two Borel subgroups of G . If $s \in S$ is such that B_s and B'_s are in general position, there exists an open subset V of S containing s such that B_V and B'_V are in general position.*

It therefore only remains to prove (b) $\Rightarrow$ (a). By the preceding corollary, it suffices to do so when S is the spectrum of an algebraically closed field k . One may assume G split with respect to a maximal torus T of B . Let B'_0 be the Borel subgroup opposite to B . Since the Borel subgroups of G are conjugate under $G(k)$, there exists $g \in G(k)$ such that $\text{int}(g)B'_0 = B'$. By Bruhat's theorem (5.7.4), one may write $g = bnb'$, with $b \in B(k)$, $b' \in B'_0(k)$, $n \in \underline{\text{Norm}}_G(T)(k)$. One therefore has

$$B' = \text{int}(b) \text{int}(n)B'_0$$

and $B' \cap B = \text{int}(b)(\text{int}(n)B'_0 \cap B)$. If $n \notin T(k)$, $\text{int}(n)B'_0 \cap B^u \neq e$ (cf. proof of 5.9.2); it follows that (b) implies $B' \cap B = \text{int}(b)(B'_0 \cap B) = \text{int}(b)T$. Q.E.D.

Proposition 5.9.5. *Let S be a scheme, G a reductive S -group, B a Borel subgroup of G , B^u its unipotent part. There exists a sequence of subgroups of B :*

$$U_0 = B^u \supset U_1 \supset \cdots \supset U_n \supset \cdots$$

having the following properties:

- (i) *Each U_i is smooth, with connected fibers, characteristic in B ; the inner automorphisms of B^u act trivially on the (sheaf) quotients U_i/U_{i+1} .*
- (ii) *For each $i \geq 0$, there exists a locally free $\mathcal{O}_S$ -module $\mathcal{E}_i$ and an isomorphism of S -sheaves of groups*

$$U_i/U_{i+1} \xrightarrow{\sim} \text{W}(\mathcal{E}_i).$$

- (iii) *For every $s \in S$, $(U_n)_s = e$ for $n \geq \dim(B_s^u)$.*

Assume first that there exist a splitting (G, T, M, R) of G and a system of positive roots R_+ of R such that $B = B_{R_+}$. One denotes by Δ the set of simple roots of R_+ ; for each $\alpha \in R_+$, one denotes by $\text{ord}(\alpha)$ the sum of the coefficients of α on the basis Δ of $\Gamma_0(R)$; this is the *order of α relative to R_+* . One has $\text{ord}(\alpha) \leq \text{Card}(R_+)$. For every $i > 0$, let $R^{(i)}$ be the set of roots of order $> i$; it is a closed set of positive roots, hence one may construct (5.6.5)

$$U_i = U_{R^{(i)}}.$$

If $\alpha \in R_+$ and $\beta \in R^{(i)}$, then $\alpha + \beta \in R^{(i+1)}$. It follows, by 5.5.2, that each U_i is an invariant subgroup of B and that the inner automorphisms of B^u act trivially on U_i/U_{i+1} . This group is moreover identified with

$$\prod_{\text{ord}(\alpha)=i+1} U_\alpha$$

and is therefore endowed with a vector structure.

If B has the form $B_{R'}$ for another splitting (G, T', M', R') of G , let us show that the groups U'_i constructed as above with the aid of the new splitting coincide with the U_i and that the vector structures on the successive quotients also coincide. By 5.6.13, there exists $b \in B^u(S)$ such that $T' = \text{int}(b)T$; the assertion to be proved is local over S and one may

therefore assume that the isomorphism $T \xrightarrow{\sim} T'$ induced by $\text{int}(b)$ comes by duality from an isomorphism of root data

$$h : (M', M'^*, R', R'^*) \xrightarrow{\sim} (M, M^*, R, R^*).$$

It is clear that the roots of R'_+ are the $\alpha \circ \text{int}(b) = h(\alpha)$, $\alpha \in R_+$, and that the simple roots of R'_+ are the $\alpha \circ \text{int}(b) = h(\alpha)$, $\alpha \in \Delta$, hence that $\text{ord}(h(\alpha)) = \text{ord}(\alpha)$ for $\alpha \in R_+$. On the other hand, it is clear by transport of structure that the vector groups $U'_{h(\alpha)}$ are none other than the $\text{int}(b)U_\alpha$. One therefore has $\text{int}(b)U_i = U'_i$, but since U_i is invariant, this gives $U_i = U'_i$.

Likewise the isomorphism of vector groups

$$\text{int}(b) : U_i/U_{i+1} \xrightarrow{\sim} U'_i/U'_{i+1}$$

is the identity, by what has already been proved.

Let us now treat the general case. There exists a covering family for the étale topology $\{S_i \rightarrow S\}$ and for each i a splitting (G_i, T_i, M_i, R_i) and a system of positive roots R_{i+} of R_i such that $B \times_S S_i = B_{R_{i+}}$ (5.5.5, (iii)). For each i , one therefore has a family

$$B_{S_i} = U_{i,0} \supset U_{i,1} \supset \cdots \supset U_{i,j} \supset \cdots$$

and vector structures on the $U_{i,j}/U_{i,j+1}$. By descent, it suffices to prove that for every pair (i, i') and every j , one has

$$U_{i,j} \times_{S_i} S_{ii'} = U_{i',j} \times_{S_{i'}} S_{ii'}$$

(one writes $S_{ii'} = S_i \times_S S_{i'}$) and that the vector structures on the quotients

$$(U_{i,j}/U_{i,j+1}) \times_{S_i} S_{ii'} \quad \text{and} \quad (U_{i',j}/U_{i',j+1}) \times_{S_{i'}} S_{ii'}$$

coincide. Now if $S_{ii'} = \emptyset$, this is trivial; if $S_{ii'} \neq \emptyset$, then one is in the situation studied previously: $B \times_S S_{ii'}$ is defined by the system of positive roots R_{i+} (resp. $R_{i'+}$) in the splitting $(G_{S_{ii'}}, T_i \times_{S_i} S_{ii'}, M_i, R_i)$ (resp. in the splitting $(G_{S_{ii'}}, T_{i'} \times_{S_{i'}} S_{ii'}, M_{i'}, R_{i'})$).

Corollary 5.9.6. *If S is affine, $H^1(S, B^u) = e$, i.e. every principal bundle under B^u possesses a section.*

Indeed, S decomposes as a direct sum of subschemes on each of which B^u has constant relative dimension. One may therefore, by 5.9.5 (iii), assume that there exists an n such that $U_n = e$. Since, by TDTE I, B.1.1,⁽⁵⁶⁾

$$H^1(S, U_i/U_{i+1}) = H^1(S, W(\mathcal{E}_i)) = 0,$$

one has $H^1(S, B^u) = 0$.

Corollary 5.9.7. *If S is affine, B possesses maximal tori. If T is a maximal torus of B , one has $H^1(S, T) = H^1(S, B)$.*

The first assertion follows at once from 5.9.6 and 5.6.13; the second is deduced from it in the standard way.⁽⁵⁷⁾

Corollary 5.9.8. *If G is a reductive S -group, the canonical morphism (cf. 5.8.3)*

$$\underline{\text{Kil}}(G) \rightarrow \underline{\text{Bor}}(G)$$

possesses sections over every affine open subset.

Corollary 5.9.9. *Under the conditions of 5.9.5, assume S affine; then there exists a locally free $\mathcal{O}_S$ -module $\mathcal{E}$ such that B^u is, as a scheme, S -isomorphic to $W(\mathcal{E})$.*

⁽⁵⁶⁾**N.D.E.** This is the corollary on page 18 of TDTE I.

⁽⁵⁷⁾**N.D.E.** Indeed, one has an exact sequence $H^1(S, B^u) \rightarrow H^1(S, B) \xrightarrow{\pi} H^1(S, B/B^u) = H^1(S, T)$, see [Se64], I § 5.5, Prop. 38 or [Gi71], III Prop. 3.3.1. Now $H^1(S, T) \rightarrow H^1(S, B)$ is a section of π , hence π is surjective; on the other hand $H^1(S, B^u) = 0$ by 5.9.6.

Let us show by induction on i that B^u/U_i is S -isomorphic to $W(\mathcal{E}_0 \oplus \cdots \oplus \mathcal{E}_{i-1})$. This is clear for $i = 0$; assume $i \geq 1$. Then B^u/U_i is a principal homogeneous bundle with base $X = B^u/U_{i-1}$ under the group $(U_{i-1}/U_i)_X$. Since B^u/U_{i-1} is affine, by the induction hypothesis, and since $U_{i-1}/U_i = W(\mathcal{E}_{i-1})$, this bundle is trivial. One therefore has (at least) an isomorphism of S -schemes

$$B^u/U_i \xrightarrow{\sim} (B^u/U_{i-1}) \times_S W(\mathcal{E}_{i-1}) = W(\mathcal{E}_0 \oplus \cdots \oplus \mathcal{E}_{i-1}).$$

One concludes at once by condition (iii) of 5.9.5.

Corollary 5.9.10. *Let S be a semilocal scheme, $\{s_i\}$ its closed points, B a Borel subgroup of the reductive S -group G . The canonical map*

$$B^u(S) \longrightarrow \prod_i B^u(\text{Spec } \kappa(s_i))$$

is surjective.

Indeed, if $S = \text{Spec}(A)$, $\kappa(s_i) = A/\mathfrak{p}_i$ and if $\mathcal{E}$ is given by the A -module E , one has

$$B^u(S) = E \otimes A, \quad B^u(\text{Spec } \kappa(s_i)) = E \otimes_A (A/\mathfrak{p}_i).$$

The assertion then follows at once from the fact that $A \longrightarrow \prod_i A/\mathfrak{p}_i$ is surjective.

5.10. Subgroups of type (R) with reductive fibers

Proposition 5.10.1. *Let (G, T, M, R) be a split S -group, R' a subset of R of type (R) (5.4.2), $H_{R'}$ the corresponding subgroup of G . The following conditions are equivalent:*

- (i) $H_{R'}$ is reductive (i.e. has reductive geometric fibers).
- (ii) One has $R' = -R'$, i.e. R' is symmetric.

Moreover, under these conditions, $(H_{R'}, T, M, R')$ is a splitting of $H_{R'}$; for every system of positive roots R_+ of R , $R'_+ = R' \cap R_+$ is a system of positive roots of R' and

$$B_{R_+} \cap H_{R'} = H_{R'_+}$$

is a Borel subgroup of $H_{R'}$, whose unipotent part is

$$U_{R_+} \cap H_{R'} = U_{R'_+}.$$

One clearly has (i) $\Rightarrow$ (ii) (it suffices to verify it fiber by fiber and R' is a system of roots of $H_{R'}$ with respect to T). To prove (ii) $\Rightarrow$ (i), one notes by 5.4.3 that

$$H_{R'} \cap Z_\alpha = \underline{\text{Centr}}_{H_{R'}}(T_\alpha) = Z_\alpha$$

for every $\alpha \in R'$ and one applies the criterion of Exp. XIX 1.12.

If R_+ is a system of positive roots of R , then $R'_+ = R_+ \cap R'$ is clearly a closed subset of R' such that $R'_+ \cup -R'_+ = R'$ and $R'_+ \cap -R'_+ = \emptyset$, hence a system of positive roots of R' . The two other assertions follow respectively from 5.6.1 (vi) and 5.6.7 (i).

Corollary 5.10.2. *Let S be a scheme, G a reductive S -group, H a reductive S -subgroup such that for every $s \in S$, $G_{\bar{s}}$ and $H_{\bar{s}}$ have the same reductive rank. Then H is closed in G , $\underline{\text{Norm}}_G(H)$ is smooth over S , $\underline{\text{Norm}}_G(H)/H$ is representable by a finite étale S -scheme.*

If T is a maximal torus of H and B a Borel subgroup of G containing T , then $B \cap H$ is a Borel subgroup of H , whose unipotent part is $(B \cap H)^u = B^u \cap H$.

The first assertions follow at once from 5.3.10 and 5.3.18, through the fact that the Weyl groups of G and of H are finite (Exp. XIX 2.5). The other assertions are local for the étale topology and reduce to the case studied in 5.10.1.

Proposition 5.10.3. *Let S be a scheme, G a reductive S -group.*

a) If Q is a torus of G , $\text{Centr}_G(Q)$ is a subgroup of type (R) of G with reductive fibers. If $Q \subset Q'$ are two tori of G , then $\text{Centr}_G(Q) \supset \text{Centr}_G(Q')$.

b) If H is a subgroup of type (R) of G with reductive fibers, then $\text{rad}(H)$ (4.3.6) is a torus of G . If $H \subset H'$ are two subgroups of type (R) of G with reductive fibers, then $\text{rad}(H) \supset \text{rad}(H')$.

c) If Q is a torus of G , one has

$$\begin{aligned} \text{rad}(\text{Centr}_G(Q)) &\supset Q \quad \text{and} \\ \text{Centr}_G(\text{rad}(\text{Centr}_G(Q))) &= \text{Centr}_G(Q). \end{aligned}$$

d) If H is a subgroup of type (R) of G with reductive fibers, one has

$$\text{Centr}_G(\text{rad}(H)) \supset H \quad \text{and} \quad \text{rad}(\text{Centr}_G(\text{rad}(H))) = \text{rad}(H).$$

Indeed, a) follows at once from Exp. XIX 2.8. To prove b), it suffices to note that $\text{rad}(H') \subset H$, for H contains (locally for (fpqc)) a maximal torus of G , hence of H' . The first assertion of c) (resp. d)) is trivial; the second follows by the usual argument.

This proposition leads to the following definition:

Definition 5.10.4. Let S be a scheme, G a reductive S -group, H a reductive subgroup of type (R) of G , and Q a subtorus of G .⁽⁵⁸⁾

- 1) One says that H is a *critical subgroup* if it is the centralizer of its radical.
- 2) One says that Q is a *C-critical torus* if it is the radical of its centralizer.

It then follows from proposition 5.10.3:

Corollary 5.10.5.

- (i) For every subtorus Q of G , $\text{Centr}_G(Q)$ is critical.
- (ii) For every subgroup of type (R) with reductive fibers H of G , $\text{rad}(H)$ is a C-critical torus of G .
- (iii) The maps

$$Q \longmapsto \text{Centr}_G(Q), \quad H \longmapsto \text{rad}(H)$$

are mutually inverse bijections between the set of C-critical tori of G and that of its critical reductive subgroups of type (R) .

- (iv) If Q is a torus of G , $\text{rad}(\text{Centr}_G(Q))$ is the smallest C-critical torus of G containing Q .
- (v) If H is a reductive subgroup of type (R) of G , $\text{Centr}_G(\text{rad}(H))$ is the smallest critical reductive subgroup of type (R) of G containing H .

Remark 5.10.5.1. ⁽⁵⁸⁾1) A torus T of G is a critical subgroup of G if and only if it is a maximal torus.

2) In what follows, “critical torus” means “C-critical torus”.

Proposition 5.10.6. Let (G, T, M, R) be a split S -group, R' a subset of R . The following conditions are equivalent:

- (i) R' is of type (R) , $H_{R'}$ is reductive and critical.
- (ii) There exists a system of simple roots Δ of R and a subset Δ' of Δ such that R' is the set of elements of R which are linear combinations of the elements of Δ' .
- (iii) R' is closed, symmetric, and every system of simple roots of R' is the intersection with R' of a system of simple roots of R .

Indeed, by Exp. XXI 3.4.8, (ii) and (iii) are equivalent and are also equivalent to the fact that R' is the intersection of R with a $\mathbf{Q}$ -vector subspace of $M \otimes \mathbf{Q}$. Now this last condition

⁽⁵⁸⁾**N.D.E.** The original has been modified by introducing the terminology “C-critical torus” instead of “critical torus”, in order to avoid confusion in later references (cf. Exp. XXVI, 3.9). Statement 5.10.5 has also been expanded and remark 5.10.5.1 added.

is implied by (i): if $H_{R'} = \underline{\text{Centr}}_G(Q)$, then R' is the set of elements of R vanishing on Q (Exp. II 5.2.3 (ii)). Finally, this condition implies (i), for $\text{rad}(H_{R'})$ is the maximal torus of $\bigcap_{\alpha \in R'} \text{Ker}(\alpha)$, hence $\underline{\text{Centr}}_G(\text{rad}(H_{R'}))$ is none other than $H_{R''}$ where R'' is the intersection of R with the vector subspace generated by R' .

5.10.7. — Let us collect some of the preceding results: let (G, T, M, R) be a split S -group, and let Δ be a system of simple roots of R and R_+ the corresponding system of positive roots; choose a subset Δ' of Δ , denote by R' the set of elements of R which are linear combinations of the elements of Δ' , and put $R'_+ = R' \cap R_+$. Let $T_{\Delta'}$ be the maximal torus of $\bigcap_{\alpha \in \Delta'} \text{Ker}(\alpha)$ and $Z_{\Delta'} = \underline{\text{Centr}}_G(T_{\Delta'})$.

Then $Z_{\Delta'}$ is a reductive subgroup of G , with radical $T_{\Delta'}$; $(Z_{\Delta'}, T, M, R')$ is a split S -group; $B_{R_+} \cap Z_{\Delta'}$ is the Borel subgroup of $Z_{\Delta'}$ defined by the system of positive roots R'_+ (or by the system of simple roots Δ'), and its unipotent part is $U_{R_+} \cap Z_{\Delta'} = U_{R'_+}$.

Remark 5.10.8. Under the conditions of 5.10.4, let Q be a critical torus of G , $L = \underline{\text{Centr}}_G(Q)$ its centralizer. Since $Q = \text{rad}(L)$, Q is a characteristic subgroup of L ; it follows at once that

$$\underline{\text{Norm}}_G(L) = \underline{\text{Norm}}_G(Q),$$

hence also

$$\underline{\text{Norm}}_G(L)/L = \underline{\text{Norm}}_G(Q)/\underline{\text{Centr}}_G(Q) = W_G(Q).$$

By 5.10.2, one deduces

Proposition 5.10.9. *Let S be a scheme, G a reductive S -group, Q a critical torus of G . The Weyl group $W_G(Q)$ is (étale) finite over S .*

Remark 5.10.10. Under the conditions of 5.10.7, one may make explicit

$$W_G(T_{\Delta'}) = \underline{\text{Norm}}_G(Z_{\Delta'})/Z_{\Delta'}.$$

It is the constant group associated with the quotient W_1/W_2 , where W_1 is the subgroup of W consisting of the elements which normalize the subgroup of M generated by Δ' and W_2 the subgroup of W generated by the s_α , $\alpha \in \Delta'$.

5.11. Subgroups of type (RC)

Definition 5.11.1. Let S be a scheme, G a reductive S -group. A group subscheme H of G is said to be of type (RC) if it is of type (R), i.e. (5.2.1) satisfies the following two conditions:

- (i) H is smooth over S , with connected fibers;
- (ii) for every $s \in S$, $H_{\bar{s}}$ contains a maximal torus of $G_{\bar{s}}$;

and if it satisfies in addition the following condition:

- (iii) for every $s \in S$ and every maximal torus T of $H_{\bar{s}}$, the set of roots of $H_{\bar{s}}$ with respect to T is a closed subset of the set of all roots of $G_{\bar{s}}$ with respect to T .

Remark 5.11.2. As we have already pointed out in 5.4.8, condition (iii) is a consequence of the others when 6 is invertible on S .⁽⁵⁹⁾

Lemma 5.11.3. *Let (G, T, M, R) be a split S -group and R' a closed subset of R . Let*

$$R_1 = \{\alpha \in R', -\alpha \in R'\} \quad \text{and} \quad R_2 = \{\alpha \in R', -\alpha \notin R'\}.$$

Then R_1 and R_2 are closed. Consider the groups $H_{R'}$, H_{R_1} and U_{R_2} (5.4.7 and 5.6.5), which are smooth and have connected fibers.

- (i) *The group U_{R_2} is invariant in $H_{R'}$ and $H_{R'}$ is the semidirect product of U_{R_2} by H_{R_1} .*

⁽⁵⁹⁾**N.D.E.** i.e. when each residue characteristic of S is > 3 .

- (ii) H_{R_1} is reductive, U_{R_2} has connected unipotent geometric fibers; every invariant subgroup of $H_{R'}$, smooth over S and with connected unipotent geometric fibers, is contained in U_{R_2} , and every reductive subgroup of $H_{R'}$ containing T is contained in H_{R_1} .
- (iii) One has $U_{R_2} \cap \underline{\text{Norm}}_G(H_{R_1}) = e$.

One first has (iii) by 5.6.7 (i). The first assertion of (i) follows from 5.6.7 (ii). Since $U_{R_2} \cap H_{R_1} = e$ by (iii), the semidirect product $H_{R_1} \cdot U_{R_2}$ is a subgroup of $H_{R'}$; but these are two subgroups of type (R) of G , containing T , and they have the same Lie algebra $\mathfrak{g}_{R'}$; they therefore coincide by 5.3.5, which completes the proof of (i).

Let us now prove (ii); the first two assertions are none other than 5.10.1 and 5.6.5. Let U be a group subscheme of $H_{R'}$, smooth and of finite presentation, invariant (hence normalized by T), with connected unipotent geometric fibers; by 5.6.12, one has, locally over S , $U = U_{R''}$, where R'' is a subset of R' such that $R'' \cap -R'' = \emptyset$. If $U \not\subset U_{R_2}$, then $R'' \not\subset R_2$, hence there exists $\alpha \in R''$ such that $-\alpha \in R'$. Then $Z_\alpha \subset H_{R'}$ (5.4.3), hence Z_α normalizes U . But U contains U_α and Z_α possesses a section w such that $\text{int}(w)U_\alpha = U_{-\alpha}$; this implies $-\alpha \in R''$, contradicting the hypothesis $R'' \cap -R'' = \emptyset$.

Finally, if L is a reductive subgroup of $H_{R'}$ containing T , one has locally over S , $L = H_{R'''}$, with R''' symmetric and contained in R' , hence contained in R_1 .

Proposition 5.11.4. *Let S be a scheme, G a reductive S -group, H a group subscheme of G of type (RC).*

- (i) H is closed in G , $\underline{\text{Norm}}_G(H)/H$ is representable by a finite étale S -group scheme.
- (ii) H possesses a largest smooth invariant group subscheme of finite presentation over S , with connected unipotent geometric fibers; one says that it is the unipotent radical of H and denotes it by $\text{rad}^u(H)$. The quotient sheaf $H/\text{rad}^u(H)$ is representable by a reductive S -group.
- (iii) If T is a maximal torus of H , H possesses a reductive subgroup L containing T , of type (RC), having the following two properties:
 - (a) Every reductive subgroup of H containing T is contained in L .
 - (b) H is the semidirect product $H = L \cdot \text{rad}^u(H)$, i.e. the canonical morphism $L \rightarrow H/\text{rad}^u(H)$ is an isomorphism.

Moreover, L is the unique reductive subgroup of H containing T and satisfying either of the two preceding conditions. Finally, one has the following equalities:

$$\underline{\text{Norm}}_H(L) = L, \quad \underline{\text{Norm}}_H(T) = \underline{\text{Norm}}_L(T), \quad W_H(T) = W_L(T),$$

in particular $W_H(T)$ is finite over S .

PROOF. Note first that (i) is local for the étale topology. Hence, by corollary 5.3.18, (i) is a consequence of the last assertion of (iii).

The assertions of (ii) are local for the étale topology. One may therefore assume one is in the situation of 5.11.3, where one concludes at once by (i) and (ii).

By the uniqueness assertions it contains, (iii) is also local for the étale topology and one may again reduce to the situation of 5.11.3, where properties (a) and (b) have been verified. The uniqueness of an L satisfying (a) is trivial; the uniqueness of an L satisfying (b) is obvious, in view of (a). The equality $\underline{\text{Norm}}_H(L) = L$ is none other than 5.11.3 (iii); if a section of H normalizes T , then it normalizes L , by uniqueness of L , hence is a section of L by what has just been proved, which proves the second equality; the third is then trivial. $\square$

Proposition 5.11.5. *Let S be a scheme, G a reductive S -group, $\mathcal{H}_c$ the functor of subgroups of type (RC) of G , which is a subfunctor of the functor $\mathcal{H}$ of 5.8.1.*

- (i) $\mathcal{H}_c$ is representable by an open subscheme of $\mathcal{H}$, smooth, quasi-projective and of finite presentation over S .

(ii) *There exists a finite étale S -scheme $C\ell_c$ and a morphism*

$$cl : \mathcal{H}_c \longrightarrow C\ell_c,$$

smooth, quasi-projective, of finite presentation, surjective and with connected geometric fibers, having the following property:

For every $S' \rightarrow S$ and all $H, H' \in \mathcal{H}_c(S')$, $cl(H) = cl(H')$ if and only if H and H' are conjugate in G locally for the étale topology (or, equivalently by 5.3.11, if for every $s \in S$, $H_{\bar{s}}$ and $H'_{\bar{s}}$ are conjugate by an element of $G(\bar{s})$).

- (iii) *$C\ell_c$ and cl are determined (up to a unique isomorphism) by the preceding conditions.*
- (iv) *If (G, T, M, R) is a splitting of G , let E be the set of conjugacy classes modulo W of closed subsets of R ; then there exists an isomorphism $C\ell_c \xrightarrow{\sim} E_S$ such that, for every closed subset R' of R , $cl(H_{R'})$ corresponds to the canonical image of R' in $E_S(S) = \text{Hom}_{\text{loc.const.}}(S, E)$.*

It is first clear that $\mathcal{H}_c$ is a sheaf for the étale topology and that (ii) implies that $C\ell_c$ is none other than the quotient sheaf of $\mathcal{H}_c$ by the equivalence relation defined by conjugacy.

This implies first (iii), as well as the fact that it suffices to verify (i) and (ii) locally for the étale topology. One therefore reduces to the situation of (iv); let us first construct a morphism

$$f : \mathcal{H}_c \longrightarrow E_S.$$

It suffices to construct a map $\mathcal{H}_c(S) \rightarrow E_S(S)$ functorial in S ; let therefore H be a subgroup of type (RC) of G ; since H possesses maximal tori locally for the étale topology, and since the maximal tori of G are conjugate locally for the étale topology, there exist a covering family $\{S_i \rightarrow S\}$ and for each i a $g_i \in G(S_i)$ and a closed subset R_i of R such that $\text{int}(g_i)(H \times_S S_i) = H_{R_i} \times_S S_i$; each R_i defines a section η_i of E_{S_i} , i.e. an element of $E_S(S_i)$; it now suffices to prove that the family (η_i) comes from a section $\eta = f(H)$ of E_S over S , and that the latter depends only on H .

To do this, one is reduced to proving that $H_{R'}$ and $H_{R''}$ are conjugate locally for the étale topology if and only if R' and R'' are conjugate by an element of the Weyl group W , which is trivial.

For every $\eta \in E$, there exists an $H_0 \in \mathcal{H}_c(S)$ such that $f(H_0) = \eta$: it suffices to take $H_0 = H_{R'}$ where R' is a closed subset of R whose image in E is η . If $H \in \mathcal{H}_0(S')$, $S' \rightarrow S$, H is conjugate to H_0 locally for the étale topology if and only if $f(H) = \eta$ (as one sees at once by the preceding argument), which shows that $f^{-1}(\eta)$ is identified with the quotient $G/\underline{\text{Norm}}_G(H_0)$, which by 5.8.2 is an open subset of $\mathcal{H}$, smooth, quasi-projective of finite presentation over S , with connected nonempty fibers. Since E_S is the sum of the open subschemes which are the images of the sections corresponding to the $\eta \in E$, $\mathcal{H}_c$ is identified with the sum of the $f^{-1}(\eta)$, $\eta \in E$, which proves (i) and (ii). Finally (iv) holds by construction.

Corollary 5.11.6. *If $u \in C\ell_c(S')$, $S' \rightarrow S$, $cl^{-1}(u)$ is a smooth quasi-projective S' -scheme of finite presentation with connected nonempty fibers; it is an open subset of $\mathcal{H}_c$ and a “homogeneous” scheme under $G_{S'}$ (by inner automorphisms). In particular, if $H \in cl^{-1}(u)(S')$, the morphism $G_{S'} \rightarrow (\mathcal{H}_c)_{S'}$ defined by $g \mapsto \text{int}(g)H$ identifies $G_{S'}/\underline{\text{Norm}}_{G_{S'}}(H)$ with $cl^{-1}(u)$.*

Examples 5.11.7. In particular, one has two canonical sections u_t, u_b of $C\ell_c$ corresponding respectively to maximal tori ($R' = \emptyset$) and to Borel subgroups ($R' =$ system of positive roots). The S -schemes $cl^{-1}(u_t)$ and $cl^{-1}(u_b)$ are none other than the S -schemes $\underline{\text{Tor}}(G)$ and $\underline{\text{Bor}}(G)$ introduced in 5.8.3. We will see other examples in Exp. XXVI.

Remark 5.11.8. One may construct an S -scheme $C\ell$, of finite presentation and unramified, and a smooth surjective morphism $\mathcal{H} \rightarrow C\ell$, with connected geometric fibers, enjoying properties analogous to 5.11.5 (ii) and (iii).

6. The derived group

6.1. Preliminaries

In this number, one fixes a scheme S , a split S -group (G, T, M, R) , a system of positive roots R_+ of R , and one writes

$$B = B_{R_+}, \quad B^- = B_{R_-}, \quad U = B^u, \quad U^- = (B^-)^u, \\ \Omega = \Omega_{R_+} = U^- \cdot T \cdot U.$$

6.1.1. — One denotes by T' the subtorus of T “image of the family α^* , $\alpha \in R$ ”; in other words T' is the image of the morphism of groups

$$\mathbf{G}_{m,S}^R \longrightarrow T$$

defined by $(z_\alpha)_{\alpha \in R} \mapsto \prod_{\alpha \in R} \alpha^*(z_\alpha)$. One sees at once that if Δ denotes the set of simple roots of R_+ , the morphism

$$\mathbf{G}_{m,S}^\Delta \longrightarrow T'$$

defined in the same way is surjective with finite kernel. If one identifies T with $D_S(M)$, then T' is identified with $D_S(M/N)$, where

$$N = M \cap \mathcal{V}(R^*)^\perp$$

(one denotes by $\mathcal{V}(R^*)^\perp$ the orthogonal of $\mathcal{V}(R^*)$ in the duality between V and V^*).

Lemma 6.1.2. *The morphism defined by the product in T*

$$\mathrm{rad}(G) \times_S T' \longrightarrow T$$

is an isogeny (cf. 4.2.9).

Indeed, the canonical morphism $\mathrm{rad}(T) \longrightarrow T/T'$ comes by duality from the morphism of commutative groups

$$M \cap \mathcal{V}(R^*)^\perp \longrightarrow M/M \cap \mathcal{V}(R),$$

which one sees at once to be injective with finite cokernel (cf. Exp. XXI 6.3).

Definition 6.1.3. One puts $\Omega' = U^- \cdot T' \cdot U$; it is a closed subscheme of $\Omega = U^- \cdot T \cdot U$.

Lemma 6.1.4. *Let α be a simple root and $w_\alpha \in \underline{\mathrm{Norm}}_G(T)(S)$ lifting s_α . One has*

$$\mathrm{int}(w_\alpha)\Omega' \cap \Omega \subset \Omega'.$$

It suffices to prove that if $g \in \Omega'(S)$ and if $\mathrm{int}(w_\alpha)g \in \Omega(S)$, then $\mathrm{int}(w_\alpha)g \in \Omega'(S)$. By 5.6.8, write

$$g = a \exp_{-\alpha}(Y) t \exp_\alpha(X) b,$$

with $a \in U_{-\hat{\alpha}}(S)$, $Y \in \Gamma(S, \mathfrak{g}^{-\alpha})$, $t \in T'(S)$, $X \in \Gamma(S, \mathfrak{g}^\alpha)$, $b \in U_{\hat{\alpha}}(S)$. One then has

$$\mathrm{int}(w_\alpha)g = \mathrm{int}(w_\alpha)a \cdot \mathrm{int}(w_\alpha)(\exp_{-\alpha}(Y) t \exp_\alpha(X)) \cdot \mathrm{int}(w_\alpha)b.$$

By 5.6.8 (iv), one has

$$\mathrm{int}(w_\alpha)a \in U_{-\hat{\alpha}}(S), \quad \mathrm{int}(w_\alpha)b \in U_{\hat{\alpha}}(S).$$

The following equivalences result (putting $h = \exp_{-\alpha}(Y) t \exp_\alpha(X)$):

$$\begin{aligned} \mathrm{int}(w_\alpha)g \in \Omega(S) &\iff \mathrm{int}(w_\alpha)h \in \Omega(S), \\ \mathrm{int}(w_\alpha)g \in \Omega'(S) &\iff \mathrm{int}(w_\alpha)h \in \Omega'(S). \end{aligned}$$

One is therefore reduced to the case where $g = h$. Since one has (4.1.12)

$$Z_\alpha \cap \Omega = U_{-\alpha} \cdot T \cdot U_\alpha, \quad Z_\alpha \cap \Omega' = U_{-\alpha} \cdot T' \cdot U_\alpha,$$

one is reduced to proving the following assertion:

$$\text{int}(w_\alpha)h \in (U_{-\alpha} \cdot T \cdot U_\alpha)(S) \implies \text{int}(w_\alpha)h \in (U_{-\alpha} \cdot T' \cdot U_\alpha)(S).$$

Now the latter follows at once from Exp. XX 3.12, which shows that the component on T of $\text{int}(w_\alpha)h$ has the form $t \cdot \alpha^*(z) \in T'(S)$.

Lemma 6.1.5. *For every $w \in \underline{\text{Norm}}_G(T)(S)$, there exists an open subset V_w of G , containing the identity section, such that*

$$\text{int}(w)\Omega' \cap V_w \subset \Omega'.$$

Choose for each simple root α an $n_\alpha \in \underline{\text{Norm}}_G(T)(S)$ lifting s_α . For every point $s \in S$, there exist an open subset V of S containing s , a $t \in T(V)$, and over V a relation

$$w = n_{\alpha_1} \cdots n_{\alpha_p} t, \quad \text{with the } \alpha_i \text{ simple.}$$

One may clearly content oneself with giving the proof for $V = S$; it is done by induction on p . If $p = 0$, then $w \in T(S)$ and one takes $V_w = G$; assume therefore $w = n_\alpha \cdot w'$, with w' satisfying the conclusion of the lemma; there therefore exists an open subset $V_{w'}$ of G , containing the identity section, such that $\text{int}(w')\Omega' \cap V_{w'} \subset \Omega'$. One may then write

$$\begin{aligned} \text{int}(w)\Omega' \cap \text{int}(n_\alpha)V_{w'} \cap \Omega &= \text{int}(n_\alpha)(\text{int}(w')\Omega' \cap V_{w'}) \cap \Omega \\ &\subset \text{int}(n_\alpha)\Omega' \cap \Omega \subset \Omega', \end{aligned}$$

by 6.1.4. One then takes $V_w = \text{int}(n_\alpha)V_{w'} \cap \Omega$ and one is done.

Lemma 6.1.6. *There exists an open subset V_0 of G , containing the identity section, such that for every $S' \rightarrow S$, one has*

$$U(S')U^-(S') \cap V_0(S') \subset \Omega'(S').$$

Let indeed n_0 be an element of $\underline{\text{Norm}}_G(T)(S)$ lifting the reflection w_0 of the Weyl group,⁽⁶⁰⁾ that is, such that $\text{int}(n_0)U = U^-$ (cf. Exp. XXI 3.6.14); then $n_0^2 \in T(S)$. Let us show that the open subset $V_0 = V_{n_0}$ of 6.1.5 answers the question. Indeed

$$\begin{aligned} U(S')U^-(S') &= \text{int}(n_0)(\text{int}(n_0)^{-1}U(S') \cdot \text{int}(n_0)^{-1}U^-(S')) \\ &= \text{int}(n_0)(U^-(S') \cdot U(S')) \subset \text{int}(n_0)\Omega'(S'). \end{aligned}$$

Whence

$$U(S')U^-(S') \cap V_0(S') \subset \text{int}(n_0)\Omega'(S') \cap V_0(S') \subset \Omega'(S').$$

Lemma 6.1.7. *Consider the morphism*

$$f : \Omega = U^- \cdot T \cdot U \rightarrow T/T'$$

composite of the second projection and the canonical morphism from T to T/T' . Then f is “generically multiplicative”: there exists an open subset V of $\Omega \times_S \Omega$, containing the identity section (and hence relatively schematically dense, Exp. XVIII 1.3), such that for every $S' \rightarrow S$ and every $(x, y) \in V(S')$, one has $xy \in \Omega(S')$ and $f(xy) = f(x)f(y)$.

Let indeed x and y be two sections of Ω over S' . Write

$$x = utv, \quad y = u't'v', \quad \text{with } u, u' \in U^-(S'), \quad t, t' \in T(S'), \quad v, v' \in U(S').$$

Let V_0 be the open subset of 6.1.6 and V the open subset of $\Omega \times_S \Omega$ defined by “ $vu' \in V_0(S')$ ” (it is the inverse image of V_0 by the morphism $\Omega \times_S \Omega$ written set-theoretically $(x, y) \mapsto vu'$). Then V answers the question. Indeed, for $(x, y) \in V(S')$, one has

$$xy = (utv)(u't'v') = (ut)(vu')(t'v').$$

But $vu' \in \Omega'(S')$, whence

$$xy \in U^-(S')t\Omega'(S')t'U(S') \subset U^-(S')tt'T'(S')U(S'),$$

⁽⁶⁰⁾**N.D.E.** The reflection w_0 is defined in XXI 3.6.14.

which shows that $xy \in \Omega(S')$ and that

$$f(xy) = f(tt') = f(t)f(t') = f(x)f(y).$$

Proposition 6.1.8. *There exists a morphism of groups*

$$f : G \longrightarrow T/T'$$

inducing on T the canonical projection. The kernel $\text{Ker}(f)$ of f is a closed group subscheme of G , smooth over S and with connected fibers. Every morphism of groups from G into a presheaf of commutative groups over S , separated for (fppf), vanishes on $\text{Ker}(f)$.

The first assertion follows at once from 4.1.11. One has immediately $\text{Ker}(f) \cap \Omega = \Omega'$, which proves that $\text{Ker}(f)$ is smooth over S at every point of the identity section.⁽⁶¹⁾ By 5.6.9 (ii), every morphism φ from G into a presheaf of commutative groups separated for (fppf) vanishes on U and U^- . By Exp. XX 2.7, φ therefore also vanishes on T' , hence on Ω' . Taking the notations of 5.7.10, one sees that the monoid U_1 is contained in $\text{Ker}(f)(S)$, which shows that

$$\text{Ker}(f) = \bigcup_{u \in U_1} u\Omega'.$$

It follows on the one hand that every φ as above vanishes on $\text{Ker}(f)$, and on the other hand that $\text{Ker}(f)$ has connected fibers, hence is smooth over S by Exp. VIB 3.10.

6.2. Derived group of a reductive group

Theorem 6.2.1. *Let S be a scheme, G a reductive S -group.*

- (i) $D_S(G) = \underline{\text{Hom}}_{S\text{-gr.}}(G, \mathbf{G}_{m,S})$ *is representable by a twisted constant S -group, whose type at $s \in S$ is $\mathbf{Z}^{\text{rgred}(G_s) - \text{rgss}(G_s)}$.*
- (ii) *Write $\text{corad}(G) = D_S(D_S(G))$, which is thus an S -torus. The biduality morphism (cf. Exp. VIII § 1)*

$$f_0 : G \longrightarrow \text{corad}(G)$$

is smooth and surjective.

- (iii) *The composite morphism*

$$\text{rad}(G) \longrightarrow G \longrightarrow \text{corad}(G)$$

is an isogeny (cf. 4.2.9).

- (iv) *The kernel of f_0 , denoted*

$$\text{dér}(G) = \text{Ker}(f_0)$$

is a closed group subscheme of G , semisimple over S , which one calls the derived group of G . If G is semisimple, one has $\text{dér}(G) = G$.

- (v) *Every morphism of groups from G into an S -presheaf of commutative groups, separated for (fppf), vanishes on $\text{dér}(G)$ and therefore factors through f_0 .*

PROOF. All assertions of the theorem are local for the étale topology; one may therefore reduce to the case where G is split over S . Consider then the morphism f of 6.1.8. By the last assertion of 6.1.8, one has at once an isomorphism

$$\underline{\text{Hom}}_{S\text{-gr.}}(G, \mathbf{G}_{m,S}) \xrightarrow{\sim} \underline{\text{Hom}}_{S\text{-gr.}}(T/T', \mathbf{G}_{m,S}),$$

⁽⁶¹⁾**N.D.E.** “At every point of the identity section” has been added, as well as the reference to VIB 3.10 at the end of the proof. On the other hand, in the following sentence “prescheme” has been replaced by “presheaf”.

which proves (i), then (ii), and gives a commutative diagram

$$\begin{array}{ccc} G & \xrightarrow{f_0} & \text{corad}(G) \\ & \searrow f & \downarrow \sim \\ & & T/T'. \end{array}$$

One then has (v) by 6.1.8, and (iii) by 6.1.2. One also has $\text{Ker}(f) = \text{Ker}(f_0)$, which by 6.1.8 implies that $\text{dér}(G)$ is smooth over S and has connected fibers; it remains to verify that its fibers are semisimple; now they are reductive by Exp. XIX 1.7, as invariant subgroups of reductive groups. By (iii), $\text{rad}(G) \cap \text{dér}(G)$ is finite, which indeed implies that the fibers of $\text{dér}(G)$ are semisimple. $\square$

Remark 6.2.2. a) By construction, in the case where G is split, $\text{dér}(G)$ is the (fppf) subsheaf of G generated by the U_α , $\alpha \in R$. (It even suffices to take the U_α , $\alpha \in \Delta \cup -\Delta$, where Δ is a basis of R).

b) ⁽⁶²⁾Let C be the presheaf of commutators of G , i.e. the S -group functor which associates with every $S' \rightarrow S$ the commutator group of $G(S')$ (i.e. the subgroup of $G(S')$ generated by the elements $xyx^{-1}y^{-1}$, for $x, y \in G(S')$), and let $\tilde{C}$ be the associated (fppf) sheaf. Since the quotient $G/\text{dér}(G) = T/T'$ is commutative, $\text{dér}(G)$ contains C and hence $\tilde{C}$ (cf. Exp. IV 4.3.12).

On the other hand, the quotient presheaf $G/\tilde{C}$ is separated (Exp. IV 4.4.8.1), and hence by (v) one has $\text{dér}(G) \subset \tilde{C}$, whence $\text{dér}(G) = \tilde{C}$, i.e. $\text{dér}(G)$ is the (fppf) sheaf of commutators of G .

Note finally that C , being a subpresheaf of G , is separated, but is not equal to $\text{dér}(G)$ in general: for example, $\text{dér}(\text{SL}_2) = \text{SL}_2$ but $\text{SL}_2(\mathbf{F}_2) \simeq \mathfrak{S}_3$ is not equal to its derived group.

c) When S is the spectrum of an algebraically closed field k , $\text{dér}(G)(k)$ is the commutator group of $G(k)$ (Exp. VIB 7.10).

6.2.3. — Consider now the two exact sequences

$$\begin{aligned} 1 &\longrightarrow \text{rad}(G) \longrightarrow G \longrightarrow \text{ss}(G) \longrightarrow 1, \\ 1 &\longrightarrow \text{dér}(G) \longrightarrow G \longrightarrow \text{corad}(G) \longrightarrow 1. \end{aligned}$$

Since $\text{rad}(G)$ is central in G , the product in G defines a morphism of groups

$$u : \text{rad}(G) \times_S \text{dér}(G) \longrightarrow G$$

which is covering by 6.2.1 (iii), hence surjective and flat (Exp. VIB 9.2 (xi)).⁽⁶³⁾ Its kernel is isomorphic to $\text{rad}(G) \cap \text{dér}(G)$, which is also the kernel of $\text{rad}(G) \rightarrow \text{corad}(G)$, hence is a finite subgroup of multiplicative type of $\text{rad}(G)$.

One argues in the same way for the morphism

$$G \longrightarrow \text{corad}(G) \times_S \text{ss}(G),$$

whose kernel is $\text{dér}(G) \cap \text{rad}(G)$. One therefore has the

Proposition 6.2.4. *Let G be a reductive S -group. The morphisms*

$$\begin{aligned} \text{rad}(G) \times_S \text{dér}(G) &\longrightarrow G, & G &\longrightarrow \text{corad}(G) \times_S \text{ss}(G), \\ \text{rad}(G) &\longrightarrow \text{corad}(G) \end{aligned}$$

are central isogenies, and their kernels are isomorphic.

⁽⁶²⁾**N.D.E.** In what follows, the original has been expanded, and the assertion that “ $\text{dér}(G)$ is the separated (fppf) presheaf of commutators of G ” has been deleted.

⁽⁶³⁾**N.D.E.** Indeed, G is the (fppf) quotient of $\text{rad}(G) \times_S \text{dér}(G)$ by $\text{Ker}(u)$, which is a group of multiplicative type, hence flat over S . Hence, by VIB 9.2 (xi), the morphism u is flat.

Corollary 6.2.5. *The following conditions are equivalent:*

- (i) G is the product of a semisimple group and a torus.
- (ii) $\mathrm{rad}(G) \times_S \mathrm{dér}(G) \xrightarrow{\sim} G$.
- (iii) $G \xrightarrow{\sim} \mathrm{corad}(G) \times_S \mathrm{ss}(G)$.
- (iv) $\mathrm{rad}(G) \cap \mathrm{dér}(G) = e$.

6.2.6. — Let us return temporarily to the case of a split group. Keep the notations of 6.1. Put $N = M \cap \mathcal{V}(R^*)^\perp$. One therefore has $T' = D_S(M/N)$. One has seen that $U^- \cdot T' \cdot U$ was an open neighborhood of the identity section of $\mathrm{dér}(G)$. One therefore has

$$\underline{\mathrm{Lie}}(\mathrm{dér}(G)/S) = \mathfrak{t}' \oplus \coprod_{\alpha \in R} \mathfrak{g}^\alpha.$$

Since the characters induced on T' by the $\alpha \in R$ are nonzero and distinct (cf. Exp. XXI 1.2.5—one has moreover already used this fact in 6.1.2), R is a system of roots of $\mathrm{dér}(G)$ with respect to T' . It is then immediate (for $U_\alpha \subset \mathrm{dér}(G)$) that the exp morphisms of $\mathrm{dér}(G)$ “are” those of G , and likewise for the coroots.

It follows:

Proposition 6.2.7. *In the preceding notations, $(\mathrm{dér}(G), T', M/N, R)$ is a split group with root datum $\mathrm{dér}(\mathcal{R}(G))$. The canonical morphism $\mathrm{dér}(G) \rightarrow G$ gives by functoriality the canonical morphism of root data $\mathcal{R}(G) \rightarrow \mathrm{dér}(\mathcal{R}(G))$ of Exp. XXI 6.5.*

N.B. The reader may as an exercise construct the diagram of split groups corresponding to the three left-hand columns of the diagram of root data of Exp. XXI 6.5.7.

Proposition 6.2.8. *Let S be a scheme, G a reductive S -group, $\mathrm{dér}(G)$ its derived group.*

- (i) *For every maximal torus T of G , $T \cap \mathrm{dér}(G)$ is a maximal torus of $\mathrm{dér}(G)$. For every maximal torus T' of $\mathrm{dér}(G)$, $\underline{\mathrm{Centr}}_G(T') = \mathrm{rad}(G) \cdot T'$ is a maximal torus of G . The two preceding constructions are mutually inverse and establish a bijective correspondence between maximal tori of G and of $\mathrm{dér}(G)$.*
- (ii) *For every Borel subgroup B of G , $B \cap \mathrm{dér}(G)$ is a Borel subgroup B' of $\mathrm{dér}(G)$. One has $B'^u = B^u$. For every Borel subgroup B' of $\mathrm{dér}(G)$, $\underline{\mathrm{Norm}}_G(B') = \mathrm{rad}(G) \cdot B'$ is a Borel subgroup of G . The preceding maps are mutually inverse and establish a bijective correspondence between Borel subgroups of G and of $\mathrm{dér}(G)$.*

By the local conjugacy theorem for maximal tori and the construction of the derived group, the only assertion which remains to be proved in (i) is the following: if T is a maximal torus of G , then

$$T = (T \cap \mathrm{dér}(G)) \cdot \mathrm{rad}(G) = \underline{\mathrm{Centr}}_G(T \cap \mathrm{dér}(G)).$$

The first equality is trivial (for one reduces to the split case); the second follows at once, for $\mathrm{rad}(G)$ is central in G , hence $T = \underline{\mathrm{Centr}}_G(T) = \underline{\mathrm{Centr}}_G(T \cap \mathrm{dér}(G))$. One argues in the same way for (ii).

6.3. Subgroups with commutative quotients

6.3.1. — Let G be a reductive S -group. If H is a subsheaf of groups of G , the following conditions are equivalent:

- H contains $\mathrm{dér}(G)$.
- H is invariant and G/H is commutative.

In this case, the canonical morphism $f_0 : G \rightarrow \mathrm{corad}(G)$ maps H onto a subsheaf $f_0(H)$ of $\mathrm{corad}(G)$; one has

$$\begin{aligned} G/H &\simeq \mathrm{corad}(G)/f_0(H), & H/\mathrm{dér}(H) &\simeq f_0(H), \\ \mathrm{dér}(G) &= \mathrm{dér}(H), & H &= f_0^{-1}(f_0(H)). \end{aligned}$$

Since $\text{dér}(G)$ is smooth over S and has connected fibers, then,⁽⁶⁴⁾ by Exp. IV, 5.3.1 and 6.3.1, and Exp. IVB 9.2, the map $H \mapsto f_0(H)$ establishes a bijective correspondence between group subschemes (resp. closed group subschemes) of G , containing $\text{dér}(G)$, smooth over S and with connected fibers, and group subschemes (resp. closed group subschemes) of $\text{corad}(G)$, smooth over S and with connected fibers.

Now, if H' is a group subscheme of $\text{corad}(G)$, smooth over S with connected fibers, then H' is of finite presentation over S (Exp. VIB 5.5) and its fibers are tori (since those of $\text{corad}(G)$ are), hence by Exp. X 8.2, H' is a subtorus of $\text{corad}(G)$, hence is closed in $\text{corad}(G)$ (Exp. IX 2.6).

Consequently, every subgroup of G , smooth with connected fibers and containing $\text{dér}(G)$, is closed in G .⁽⁶⁴⁾

6.3.2. — If H is a closed group subscheme of G , smooth over S , with connected fibers and invariant in G , then H is reductive. If moreover $H \supset \text{dér}(G)$, then $\text{dér}(H) = \text{dér}(G)$ and $f_0(H)$ is identified with $\text{corad}(H)$. One has therefore proved the

Proposition 6.3.3. *Let G be a reductive S -group. Every group subscheme H of G , invariant in G , with commutative quotient (i.e. containing $\text{dér}(G)$), smooth over S , with connected fibers,⁽⁶⁵⁾ is closed and reductive. One has $\text{dér}(H) = \text{dér}(G)$ and $f_0(H)$ is identified with $\text{corad}(H)$; one has*

$$G/H \simeq \text{corad}(G)/\text{corad}(H), \quad H = (H \cap \text{rad}(G)) \cdot \text{dér}(G).$$

Moreover, $H \mapsto f_0(H)$ defines a bijection between the set of subgroups H of G having the preceding properties and the set of subtori of $\text{corad}(H)$.

By a new application of Noether's theorem (Exp. IV, 5.3.1 and 6.3.1), one deduces the

Proposition 6.3.4. *Let S be a scheme, G a reductive S -group, T a maximal torus of G . For every subgroup H of G as above, $T \cap H$ is a maximal torus of G and one has*

$$G/H \simeq T/T \cap H, \quad H = (T \cap H) \cdot \text{dér}(G).$$

Moreover, $H \mapsto T \cap H$ is a bijection between the set of subgroups H of G as above and the set of subtori of T containing $T \cap \text{dér}(G)$.

⁽⁶⁴⁾**N.D.E.** The original has been expanded in what follows. In particular, the conclusion (implicit in the original) that every subgroup of G , smooth with connected fibers and containing $\text{dér}(G)$, is closed in G , has been added.

⁽⁶⁵⁾**N.D.E.** The hypothesis that H is retrocompact in G has been deleted, since it holds automatically: by VIB 5.5, G and H are separated and quasi-compact over S , hence $H \hookrightarrow G$ is quasi-compact by EGA IV₁, 1.1.2 (v).

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